GPW WATS 2.01 Native Order Gateway Specification

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CONTENTS

Contents **2**

1. Disclaimer **3**

2. Preface **4**

2.1. Target Audience 4

2.2. Document's Purpose 4

2.3. Associated Documents 4

3. Document History **6**

4. Service Description **18**

4.1. Native Service Description 18

4.2. Short Codes 20

4.3. Conditionally Required Fields 22

4.4. Security Identification 22

4.5. Numbers and Decimal Places 23

4.6. Order Management 23

4.7. Trade Capture 34

4.8. Quote Handling 36

4.9. Cancel on Disconnect 37

4.10. Mass Cancellation 37

4.11. Reject Messages 38

4.12. Timestamps and Dates 38

5. Connectivity **39**

5.1. Session Identification 39

5.2. Failover and Recovery 39

5.3. Message Throttling 43

5.4. Connectivity Policy 44

6. Session Layer **45**

6.1. Establishing a Session 45

6.2. Maintaining a Session 45

6.3. Terminating a Connection 46

7. Messages **47**

7.1. Header 48

7.2. BussinesBidOfferUpdate 48

7.3. ConnectionClose 49

7.4. GapFill 50

7.5. Heartbeat 50

7.6. Login 50

7.7. LoginResponse 50

7.8. Logout 51

7.9. LogoutResponse 51

7.10. MarketMakerCommand 52

7.11. MarketMakerCommandResponse 52

7.12. MassQuote 53

7.13. MassQuoteResponse 55

7.14. OrderAdd 57

7.15. OrderAddResponse 60

7.16. OrderCancel 62

7.17. OrderCancelResponse 63

7.18. OrderExecute 65

7.19. OrderMassCancel 66

7.20. OrderMassCancelResponse 67

7.21. OrderModify 68

7.22. OrderModifyResponse 70

7.23. Reject 71

7.24. RequestForExecution 71

7.25. TradeBust 72

7.26. TradeCaptureReportDual 72

7.27. TradeCaptureReportResponse 80

7.28. TradeCaptureReportSingle 80

7.29. TradingSessionStatus 85

7.30. Order Add Example 86

8. Message Kinematics **87**

8.1. Clob 88

8.2. Off-Book (Block or Cross) 106

1. DISCLAIMER

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In case of sections of documentation at a **High** level work progress according to the current version of *GPW WATS Advancement of Documentation*, Warsaw Stock Exchange will endeavor to limit changes to these sections of documents to those related to:

1. correcting errors in the documentation or in the software;
2. clarification of the documentation content or removing ambiguity;
3. implementation of approved change requests or;
4. regulatory changes.

2. PREFACE

This document has been prepared by Warsaw Stock Exchange in order to help in the implementation process of GPW WATS trading platform.

2.1. TARGET AUDIENCE

This document has been prepared to development staff, Independent Software Vendors who produce software integrated with GPW WATS, analysts, market participants and all clients who want to deepen their knowledge about GPW WATS.

2.2. DOCUMENT'S PURPOSE

The purpose of this document is to provide a full description of Native Order Gateway, which is part of GPW WATS.

2.3. ASSOCIATED DOCUMENTS

GPW WATS 2.01 Native Order Gateway Specification is a part of GPW WATS documentation set.

Please check the following documents to learn about the construction of Trading System.

- GPW WATS 1.01 Trading System.

Please check the documentation of the trading protocols supported by GPW WATS.

- **GPW WATS 2.01 Native Order Gateway Specification** (this document),
- GPW WATS 2.02 FIX Order Gateway Specification.

Please check the description of the communication with Data Distribution Service.

- GPW WATS 3.01 Market Data Protocol.

Please check the description of the communication with Internet Data Distribution System.

- GPW WATS 3.02 Internet Data Distribution System,
- GPW WATS 3.03 Streaming Messages for IDDS,
- GPW WATS 3.04 Rest API Messages for IDDS.

Please check the additional documentation, which explains other services provided within GPW WATS.

- GPW WATS 4.01 Drop Copy Gateway,
- GPW WATS 4.02 Post Trade Gateway,
- GPW WATS 5.01 Risk Management Gateway.

Please check the additional documentation describing the following:

- GPW WATS 2.03 Rejection Codes,
- GPW WATS 2.04 BenDec Message Definition Format,
- GPW WATS 4.03 Contract Notes,
- GPW WATS 6.01 Connectivity,
- GPW WATS 6.02 (ENG) Short Code Record Keeping,
- GPW WATS 6.02 (PL) Mapowanie Short Code,
- GPW WATS 6.03 Short-Long Mapper User Guide.

It is recommended to read ***GPW WATS 1.01 Trading System*** document first.

Gateway messages are available in the Repository shared with ISV.

The contract is based on the BenDec library which is publicly available, see Bendec library.

3. Document History

Version	Date	Description
0.51	29.06.2023	The initial publication of the documentation.
0.52	26.07.2023	Messages that have been added or updated in version 0.52: **Added new messages:** • MassQuote • MassQuoteResponse • RiskLimitDefinition • RiskLimitDefinitionResponse • TradeBust **Modified** • OrderAddResponse - field source has new Value 7 - IcebergRefill • OrderCancelResponse - field source has new Value 7 - IcebergRefill • RiskLimitBreach - field limitType has new values (301-326) RiskLimitBreach - new fields: • riskWarningLevelAction • riskLimitUtilizationAmount • riskLimitUtilizationPercent. **Cancelled:** • OrderAddResponde field memo was deleated.
0.53	16.08.2023	Messages that have been updated accordingly to changes in messages v0.53: **MassQuoteResponse**: OrderBookRebuild value added in responses.source field. **OrderAddResponse**: OrderBookRebuild value added in source field. **OrderCancelResponse**: OrderBookRebuild value added in source field.
0.54	06.09.2023	Messages that have been updated accordingly to changes in messages v0.54: Changes in **MassQuote** messgage fields: • mifidFields.client.qualifier • mifidFields.executingTrader.qualifier **MassQuoteResponse** message field: • responses. **OrderAdd** • mifidFields.client.qualifier (values) • mifidFields.executingTrader.qualifier (values) • mifidFields.investmentDecisionMaker.qualifier (values) **OrderAddResponse** • reason (values) • source (values Activated added) **OrderCancelResponse** • reason (values) • source (values Activated added) **OrderModifyResponse** • reason (values) **RiskLimitDefinition** deleted: • elementType, side field added:

Version	Date	Description
		• marketSegmentId, capacity field **RiskLimitDefinitionResponse** • reason (value MissingMicCode) **TradeCaptureReportDual** • algorithmicTradeIndicator (values) • tcrPartyBuy.mifidFields.client.qualifier (values) • tcrPartyBuy.mifidFields.executingTrader.qualifier (values) • tcrPartyBuy.mifidFields.investmentDecisionMaker.qualifier (values) • tcrPartyBuy.mifidFields.investmentDecisionMaker.qualifier (values) • tcrPartySell.mifidFields.client.qualifier • tcrPartySell.mifidFields.executingTrader.qualifier • tcrPartySell.mifidFields.investmentDecisionMaker.qualifier **TradeCaptureReportResponse** • reason (values) **TradeCaptureReportSingle** • algorithmicTradeIndicator • tcrParty.mifidFields.client.qualifier • tcrParty.mifidFields.executingTrader.qualifier • tcrParty.mifidFields.investmentDecisionMaker.qualifier
0.55	16.10.2023	Messages that have been updated accordingly to changes in messages v0.55: Changes in **OrderAdd** message Listed values added in **reason** field: • InvalidPartyRoleForCLOB (1073) • InvalidPartyRoleQualifierForPartyId (1075) • RiskLimitNotDefined (7000) • RiskMaximumOrderVolumeExceeded (7001) • RiskMaximumOrderValueExceeded (7002) • RiskOrderPriceCollarExceeded (7003). **Reject** message Listed values added in **rejectReason** field: • NA (0) • InvalidSettlementDate (4). **RiskLimitDefinition** message **riskLimitId** field added. Modified values in **clientRoleQualifier** field: • NA (1) • Algorithm (2) • FirmOrLegalEntity (3) • NaturalPerson (4). **TradeCaptureReportDual** message Deleted fields: • tradeRequestId • lastParPx • tcrPartyBuy.side • tcrPartyBuy.counterpartyId • tcrPartySell.side • tcrPartySell.counterpartyId. **TradeCaptureReportResponse** message

Version	Date	Description
		Added values in **reason** field: • SettlementDateCannotBeEarlierThanMinimumSettlementDate (2015) • SettlementDateCannotBeLaterThanMaximumSettlementDate (2016). **TradeCaptureReportSingle** message Deleted fields: • tradeRequestId • lastParPx. **tcrParty.counterpartyId** changed into **counterpartyId**
0.56	09.11.2023	Messages that have been updated accordingly to changes in messages v0.56: New message **TestEvent** has been added. Changes in **Header** message: **msgType** length value changed from 1 to 2 bytes.
0.57	30.11.2023	Messages that have been updated accordingly to changes in messages v0.57: • Unused messages removed: RiskLimitDefinition, RiskLimitDefinitionResponse, Test and TestEvent. • Rejection codes updated. • Added the value 'Stp' to the enum 'OrderSource'. Cross transactions added in sections 4.1.2 and 4.7. Order source filed values updated – section 4.6.5. New chapter 4.8 describing quotes handling.
0.58	15.12.2023	Publication of v0.58.
0.59	25.01.2024	**RegularTrade** value has been removed from **TradeType** filed from **TradeCaptureReport** messages. New values have been added for **responses** (OrderRejectReason) field: • InvalidBidAskSpread – 1208, • RequestNotAllowedForBlockInstrument – 2026, • RequestNotAllowedForCrossInstrument – 2028. New value has been added for **source** (OrderSource) filed: • CorporateAction - 11. New values have been added for **rejectReason** field: • SettlementDateRequired – 5, • TradeReportIdRequired – 6, • MissingReportIdSecondaryTradeReportIdOrTradeReportRefId – 7. **MassQuoteResponse** message Fields **status** (MassQuoteStatus) and **reason** (MassQuoteRejectionReason) have been added. **TradeCaptureReportDual** message Values for **matchStatus** filed have been changed to: • NA – 0, • Matched – 1, • Unmatched – 2. Values for **reason** (TcrRejectionReason) field have changed to: Deleated: • CannotFindMatchingFirstLeg – 2025. Added: • InvalidMatchStatus – 2029, • CrossNotAllowedOutsideOfClobInstrumentSpread – 2030, • CrossPriceNotEqualToTheReferencePrice – 2031,

Version	Date	Description
		• CrossNotAllowedDuringClobInstrumentAuctionOrSuspension – 2032, • ForbiddenSecondaryTradereportId – 2033, • UnknownSecondaryTradereportId – 2034. **TradeCaptureReportSingle** message Added fields: • secondaryTradeReportId Values for **matchStatus** filed have been changed to: • NA – 0, • Matched – 1, • Unmatched – 2. Correction of the throttling mechanism description. Division into business and technical limits.
0.62	25.03.2024	The document has been updated with: • modifications of orders in terms of GTD validity (4.4.3.3), • enhancement of the description in section 4.5 Trade Capture for Cross and Block trades, • new section 4.8 Mass Cancellation, The following messages in section 7 have been updated: The list of messages is now described not by type, but by Id. New messages: • InitiateState • InitianeStateResponse • OrderMassCancel • OrderMassCancelResponse • RequestForExecution **MassQuote:** • **stpId** length has changed to 1 • **count** field name has changed to **quotes.count** • quotes field name has changed to **quotes.item;** its type has changed to QuotesArray • **count** field name has changed to **responses.count** • **responses** field name has changed to **responses.item;** its type has changed to QuoteOrderResponseArray; new or changed values for **reason** field: o 1028 – Description amended to "lower", o 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, o 1039 – Name changed to InvalidTimeInForceForOrderType, o 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, o 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, o 1043 – New value, o 1044 – New value, o 1046 – New value, o 1069 – New value, o 1201 – New value, o 1203 – New value, o 1301 – New value, o 1401 – New value. • **reason** field – 1210 – New value added. **OrderAdd:** • **stpId** length has changed to 1 **OrderAddResponse**

Version	Date	Description
		• New or changed values for **reason** field: o 1028 – Description amended to "lower", o 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, o 1039 – Name changed to InvalidTimeInForceForOrderType, o 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, o 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, o 1043 – New value, o 1044 – New value, o 1046 – New value, o 1201 – New value, o 1203 – New value, o 1301 – New value, o 1401 – New value. • **source** field – new value added: 12 MassCancel **OrderModify** • New field added: **expire**. **OrderModifyResponse** • New or changed values for **reason** field: o 1028 – Description amended to "lower", o 1032 – Name changed to InvalidOrdTypeForSelectedMarketMode, o 1039 – Name changed to InvalidTimeInForceForOrderType, o 1040 – Name changed to InvalidTimeInForceForCurrentMarketPhase, o 1041 – Name changed to InvalidTimeInForceForSelectedMarketModel, o 1043 – New value, o 1044 – New value, o 1046 – New value, o 1069 – New value, o 1201 – New value, o 1203 – New value, o 1301 – New value, o 1401 – New value. **TradeCaptureReportResponse:** • **secondaryTradeReportId** field added, • New or changed values for **reason** field: o 1028 – Description amended to "lower", o 2024- New Value, • **counterpartyICode** field and its type amended.
1.0	30.04.2024	Changes propagating across multiple chapters: • Harmonized order statuses with FIX Order Gateway • Introduced a field indicating the execution of an order, execTypeReason, replacing the source field. 2.3. Associated Documents Updated the list of related documents. 4.1 Native Service Description /4.1.2 Business layer Enhanced the functional scope to include "mass quotes" and mass "cancellation" Figure 2 - updated the main scenario of message exchange with Gateway about currently used status fields and execTypeReason (the successor to the source field). 4.4.5 Order Status Harmonized order statuses with FIX Order Gateway. The previously used status Acknowledged corresponds to New (similarly as in FIX).

Version	Date	Description
		Included statuses Partially filled and Expired. There is no Modification status (Replaced in FIX), which in FIX 5.0 was also treated as deprecated. Information about order execution in the execTypeReason field. The new execTypeReason field contains information similar to the ExecType/ExecTypeReason field in FIX Order Gateway. 4.4.7. Modification Specified fields that can be subject to modification. Extended the scope of modifications e.g., the expire field. 4.5.1. Block Market Model Updated information on the execType field value for executing a Trade. 4.6. Quote Handling Indicated the possibility of using Mass Quote in the Hybrid market model 4.7 Cancel on Disconnect Changed the operation of the CoD mechanism to be fully manageable by Exchange Members (added a dedicated field execInst to orders) 5.2. Failover and Recovery Expanded the description of the section by separating Failover and Recovery modes. 7. Messages Added the BidOfferUpdate message related to IPO and Tender offer Added messages for Hybrid market model: • MarketMakerCommand • MarketMakerCommandResponse • RequestForExecution RiskLimitBreach message deleted (only for RMA purpose) Removed the AlgorithmicTrade flag from mifidFields.flags (a qualifier is used to indicate the use of an algorithm) Removed BIC code from ClearingIdentifier - LEI and Custom codes are used. Adopted new values for the OrderStatus field. Introduced a field indicating the execution of an order, execTypeReason, replacing the source field. Updated codes for OrderRejectionReason and TcrRejectionReason. In the OrderAdd message, added fields: • execInst - for the purpose of the CoD mechanism • feeStructureId - fee scheme for billing purposes. In the OrderAddResponse message, added fields displayQty and filled. In the OrderCancel message, added fields mifidFields and status. In the OrderMassCancel message, added field executingTrader. In the OrderModify message, expanded the scope of modifications e.g., the expire field. In the OrderModifyResponse message, added field filled. In the TradeCaptureReportDual and TradeCaptureReportSingle, corrected the execType and tradeReportType fields, added the feeStructureId field, removed the matchStatus field (using execType = Trade), added information about the clearingMember. Updated the kinematics for the gateway (Chapter 8)
1.1	28.06.2024	Changes propagating across multiple chapters Enums TimeInForce - Indicates the order's time in force (e.g. GTC). FOK corrected to: A Fill or Kill order must be immediately fully filled or canceled. VFA corrected to: Valid For Auction. VFC corrected to: Valid For Closing. Affected messages: OrderAdd. ClearingIdentifier- Clearing member identifier type.

Version	Date	Description
		BIC - Business Identifier Code, value removed. Affected messages: MassQuote, OrderAdd, TradeCaptureReportDual, TradeCaptureReportSingle. ExecType - Type of execution being reported. Uses subset of ExecType for trade capture reports. New – Value removed. Affected messages: TradeCaptureReportDual, TradeCaptureReportSingle From TradeCaptureReportDual message has been removed feeStructuredId field. Two fields: tcrPartyBuy.feeStructureId and tcrPartySell.feeStructureId have been added. From TradeCaptureReportSingle message has been removed feeStructuredId field. The field tcrParty.feeStructureId has been added.
1.1.2	9.08.2024	**ConnectionClose** message: • New value added in reason field: CloseOps. **4.4.7 Order Modification** – MiFID Fields modification are no longer included in the list of modifications. Short Codes modification has been added to the list.
1.2	18.09.2024	TradeReportRefId and TradeReportId field lengths have been increased to 21 characters.
1.3	17.10.2024	**GapFill** New message added. **OrderAddResponse** message New values addes in exectTypeReason field: • CancelonBuyOnlyStateEntry • CancelonKnockedOutStateEntry • CancelByRiskManagement **OrderCancelResponse** message New values addes in exectTypeReason field: • CancelonBuyOnlyStateEntry • CancelonKnockedOutStateEntry • CancelByRiskManagement **Reject** message New values added in rejectReason field: • InvalidTradeReportId • InvalidGapFillSeqNum
1.4	6.12.2024	Unpublished version. All changes in this version have been documented in v1.5.
1.5	3.02.2025	total size = number of Lots price = price per unit (smallest portion) of instrument **ConnectionClose** message • reason field: new value added: 8 - Disconnect **MassQuote** message • length changed to 1200 new fields added: • feeStructureId • interestedParty • quoteId onBehalOf field has been deleted. **MassQuoteResponse** message new field added: • feeStructureId

Version	Date	Description
		OrderAdd message • length changed to 167 new field added: • InterestedParty • clientOrderId onBehalOf field has been deleted. **TradeCaptureReportDual** message • length changed to 271 new fields added: • tcrPartyBuy.interestedParty • tcrPartySell.interestedParty **TradeCapture ReportSingle** message • length changed to 206 new field added: • tcrParty.interestedParty **Dual Sided - Accepted** kinematics has been updated
1.5.4	30.04.2025	**MassQuote** message: • stpId kind changed to Alias (u8) **MassQuote** message: quotes.item field: • Kind of bid.price and ask.price is changed to i64 **TradeCaptureReportSingle** message • lastPX field: kind changed to Alias(i64) **Single Sided - Declined by Counterparty and Single Sided – Cancelled by Initiator** diagrams have been amended: **Dual Sided – Accepted, Dual Sided – Rejected, Single Sided Accepted, Single Sided – Declined by Counterparty, Single Sided – Cancelled by Initiator, Single Sided – Rejected:** • Change from execType = NA (0) to execType = NA (1) in Off-Book kinematics. • Editorial changes in Kinematics (name and structure amends).
1.6	26.05.2025	**OrderAdd message** • stpId field - type changed from u16 to u8 **OrderAddResponse message** execTypeReason: • value added: 19 - CancelOnDisconnect **OrderCancelResponse message** execTypeReason: • value added: 19 - CancelOnDisconnect **Trade message** • price field - kind changed from Number to i64. **TradeCaptureReportDual message** • secondaryTradeReportID - new field • algorithmicTradeIndicator field - values corrected: 2 - NonAlgorithmicTrade; 3 - AlgorithmicTrade • lastPx field - kind changed from Number to i64. **TradeCaptureReportResponse message** • tradeReportId field - description changed **TradeCaptureReportSingle message** • tradeReportId field - description changed • counterpartyCode field - Type changed from CcpCode to ParticipantCode

Version	Date	Description
1.6.5	18.06.2025	**Header message** • seqNum - description updated • timestamp - description updated
1.6.6	10.07.2025	Publication of v1.6.6. No changes in the document.
1.6.7	7.08.2025	4.2 Short Codes – new section added. 6.2.1 Sequence Number: message Reject deleted from the list of the messages with seqNum=0. Certain fields have added **Optional** requirement in their description: • stpID (MassQuote), • account (MassQuote), • memo (MassQuote), • clearingMemberCode (MassQuote), • feeStructureId (MassQuote), • interestedParty (MassQuote), • quoteId (MassQuote), • stpID (OrderAdd), • triggerPrice (OrderAdd), • displayQty (OrderAdd), • account (OrderAdd), • expire (OrderAdd), • memo (OrderAdd), • clientOrderId (OrderAdd), • clearingMemberCode (OrderAdd), • feeStructureId (OrderAdd), • interestedParty (OrderAdd), • targetPartyId (OrderMassCancel), • marketSegmentId (OrderMassCancel), • instrumentId (OrderMassCancel), • triggerPrice (OrderModify), • displayQty (OrderModify), • expire (OrderModify),
1.6.8	14.08.2025	Publication of v1.6.8. No changes in the document.
1.6.12	12.09.2025	4.5.1.3 TVTIC – new section added. Certain fields have updated their description: • id (Trade) • tradeId (TradeCaptureReportDual) • tradeId (TradeCaptureReportSingle) 4.6.2 Cross Trade • settlementDate field updated 4.8 Cancel of Disconect – section updated
1.6.15	30.09.2025	MassQuoteResponse Responses.items.status field updated: Cancelled, Flled, PartiallyFilled, Expired statuses deteleted.as these won't be used in the future .Additionally the field description has been updated.
1.6.16	24.10.2025	4.7 Quote Handling – Quotation rules have been updated 7. Messages • 7.7 Login Response – Result field has been updated

Version	Date	Description
1.7.1	18.11.2025	4.5 Numbers and Decimal Places section has been added. It describes a number of decimal places in Number types 4.6.1.2 clientOrderId section has been added. It is an optional identifier to be used by Exchange Member 4.8 Cancel on Disconnest • For functionality mass quotes elaboration (orders from mass quotes by default subject of CoD). 7. Messages chapter: Length of messages updated The field MifidFlags changed to OrderFlags and extracted from Mifid fields for messages: • MassQuote, • OrderAdd, • TradeCaptureReportSingle, • TradeCaptureReport. MifidFlags fields have been deleted from messages: • OrderModify • OrderCancel Trade message changed to OrderExecute.message. OrderExecute message has been changed: additional attributes or attributes names changes: • id field changed to tradeId • new fields have been added: o instrumentId, o liquidityIndicator, o currency, o side, o clientOrderId. Field clientOrderId hass benn added to the messages: • OrderAddResponse, • OrderCancel, • OrderCancelResponse, • OrderModify, • OrderModifyResponse, • OrderMassCancel, • OrderMassCancelResponse. The fields OrderRestrictions and orderOrigination have been deleted from the messages: • TradeCaptureReportDual • TradeCaptureReportSingle. Messages OrderAddResponse and Order Cancel Response: • To the field execTypeReason the new values have been added: o VfaActivated o HybridActivated Message Header: • sourceTimestamp field has been added • timestamp field definition has been updated 8. Kinematics 8.1 Clob - the section has been updated
1.7.1.1	12.12.2025	4.2 Short Codes – section has been updated 7. Messages

Version	Date	Description
		• 7.15 OrderAddResponse ○ execTypeReason – description for values 4 has been updated • 7.18 OrderExecute ○ liquidityIndicator – value NotApplicable has been updated • 7.26 TradeCaptureReportDual - certain fields have added **Optional** requirement in their description ○ settlementDate ○ tcr.Party.Buy.clearingMemberCode ○ tcr.Party.Buy.clearingMemberIdentifier ○ tcr.Party.Buy.account ○ tcr.Party.Buy.feeStructuredId ○ tcr.Party.Buy.interestedParty ○ tcr.Party.Buy.memo ○ tcr.Party.Sell.clearingMemberCode ○ tcr.Party.Sell.clearingMemberIdentifier ○ tcr.Party.Sell.account ○ tcr.Party.Sell.feeStructuredId ○ tcr.Party.Sell.interestedParty ○ tcr.Party.Sell.memo • 7.28 TradeCaptureReportSingle - certain fields have added **Optional** requirement in their description ○ secondaryTradeReportId ○ tradeReportRefId ○ tcrParty.clearingMemberCode ○ tcrParty.clearingMemberIdentifier ○ tcrParty.account ○ tcrParty.feeStructuredId ○ tcrParty.interestedParty ○ tcrParty.memo
1.7.2-1.7.4		No changes in the document.
1.7.5	06.03.2026	Section 6.2.1 Sequence Number has been updated Section 7 Messages • 7.17 OrderCanelResponse – status and exectTypeReason fields have been updated. This is a clarification of possible values for OrderCancelResponse message in the status and execTypeReason fields, • 7.18 OrderExecute – LiquidityIndicator field – description has been updated • 7.26 TradeCaptureReportDual – ExecType field – description has been updated • 7.28 TradeCaptureReportSingle – ExecType field – description has been updated • 7.29 TradingSessionStatus – date field has been moved in the message as editorial correction
1.7.6-1.7.9		No changes in the document.
1.7.10	03.06.2026	Section 5.2.2 Recovery mode has been updated Section 5.2.3 Recovery Mode (DR Scenario / BCP Plan) has been added Section 5.3 Message Throttling – the measurement period has been updated
1.7.11-1.7.12		No changes in the document.
1.7.13	15.07.2026	Section 4.6.5 Order Status – the table has been added with possible combinations ExectTypeReason vs Status. Section 4.6.9 Text Fields - ASCII characters range adjustment(range 32 – 126)

Version	Date	Description
		New chapter 4.6.12 Clearing Member Identification has been added with recommendation how to fill ClearingMemberData Section 6.1 Establishing a Session - Additional remark of how logging to secondary Gateway works. Section 6.2.1 Sequence Number - Additional remarks of heartbeat being sent by Order Gateway with non zero sequence and recommendation for heartbeat message to be sent with seqNum=0 by Exchange member.
1.7.17	27.08.2026	**All updates are editorial in nature and do not alter existing system logic.** The following chapters have been corrected: • 4.5 Numbers and Decimal Places - description of little-endian byte used in GPW WATS • 4.12 Timestamps and Dates - description has been updated • 7.1 Header - the description has been updated for sourceTimestamp field • 7.7 LoginResponse - AccountLocked has been added to result field • 7.22 OrderModifyResponse - unused statuses have been removed • 7.26 TradeCaptureReportDual - description has been updated for tcrPartyBuy.flags field Minor editorial changes: • 4.3 Conditionally Required Fields • 4.6.1.4 TVTIC • 4.6.3.4 Good Till Time (GTT) • 4.6.10 Order Book Restatement • 4.7.2 Cross Trades • 8.1 Message Kinematics for CLOB
1.7.6.20	17.09.2026	**All updates are editorial in nature and do not alter existing system logic.** Minor editorial changes: 8.1 Message Kinematics (CLOB) - Stop Loss Order – Accepted, Triggered and Cancelled

4. SERVICE DESCRIPTION

4.1. NATIVE SERVICE DESCRIPTION

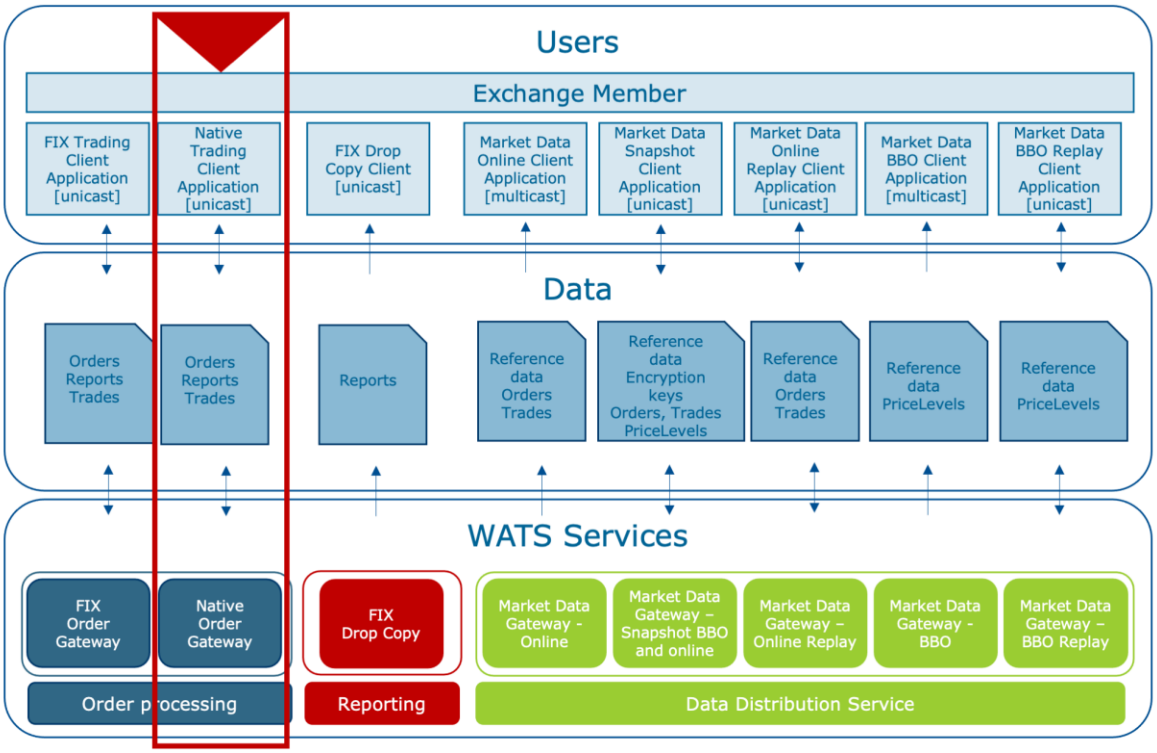
Native Order Gateway service provides a low-latency trading interface based on a proprietary binary protocol.

The main service functionalities:

- order submission,
- order modification,
- order cancellation and mass cancellation,
- transaction submission (block market),
- mass quotes.

Native Order Gateway is a point-to-point service based on the TCP-IP protocol.

Figure 1. GPW WATS Services



The following description is a brief introduction to the operation of the service, divided into the session and business layer. Detailed information can be found in the subsequent chapters of the document.

4.1.1. SESSION LAYER

To use the service, log in by sending a Login message. The message contains access parameters such as a token and a connection identifier. System confirms successful login with a LoginResponse message which contains the result field set to "OK".

After logging in, a potential synchronization of missing messages takes place based on the sequential numbers exchanged during the login process (also called gap filling).

If no messages are sent within a specified time period, each party should send a Heartbeat message.

Participant can terminate the session at any time by sending a Logout message and closing the network connection. The Service responds to the Logout message with a LogoutResponse message.

Gateway can terminate the session by sending a ConnectionClose message and closes the network connection. The ConnectionClose message can be sent in a positive scenario similar to closing trading day, as well as in the case of significant session-level errors, e.g. communication protocol errors.

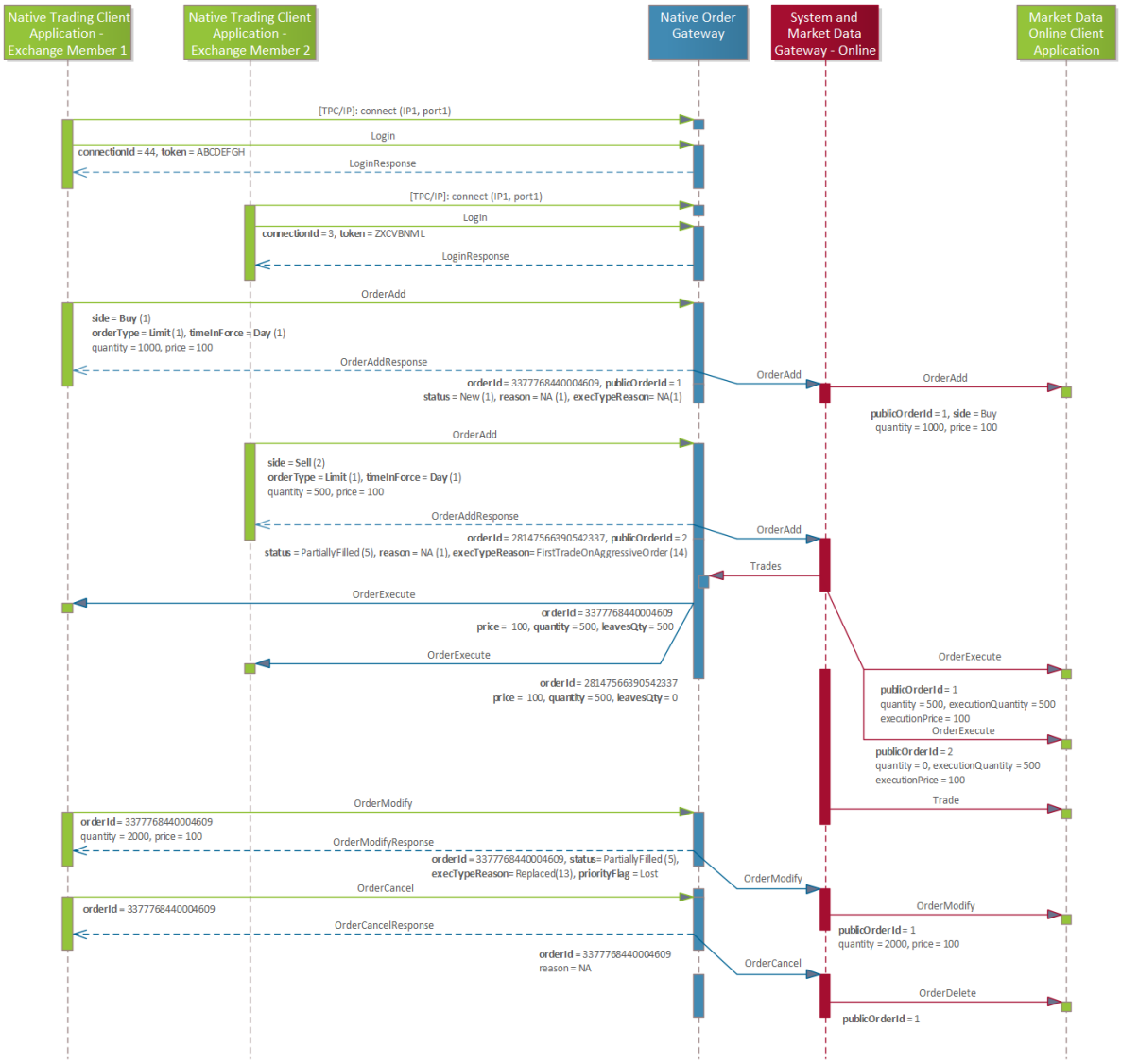
4.1.2. BUSINESS LAYER

Business communication is primarily based on the request - response pattern. We have the following main communication scenarios:

- Exchange Member places an order by sending an OrderAdd message. Gateway responds with an OrderAddResponse message, which confirms the successful submission of the order or otherwise rejects it,
- Exchange Member modifies an order by sending an OrderModify message. Gateway responds with an OrderModifyResponse message to confirm the accuracy of the modification or otherwise rejects it,
- Exchange Member cancels an order by sending an OrderCancel message. Gateway responds with an OrderCancelResponse message to confirm the accuracy of the cancellation or subsequently rejects it,
- Exchange Member reports a block transaction conducted outside Market by sending a TradeCaptureReportSingle or TradeCaptureReportDual message. Gateway responds with a TradeCaptureReportResponse message and continues to handle the Blocked transaction,
- Exchange Member reports a Cross transaction using trade capture reports - similar to block transactions.
- Mass cancellation of orders carried out through the OrderMassCancel message.

In addition to the above request – response pattern, where Exchange Member initiates communication, it is also possible to receive ad-hoc messages from Gateway concerning the status of orders, transactions and order cancellations performed by System. Changes in the order status are sent to Exchange Member in the OrderAddResponse message. Information about transactions based on the submitted orders is provided in the OrderExecute message. With regards to order cancellations performed by System, the OrderCancelResponse message is used. Below is a diagram illustrating the usage of key business messages.

Figure 2. Scenario of messages exchange via Native Order Gateway



The diagram depicts the actions of two Exchange Members. Both log into System and then submit crossed orders (an OrderAdd message). Gateway notifies about the transaction being executed by sending a OrderExecute message. Next, Exchange Member 1 modifies the order by increasing the quantity to 2000 and cancels it. The above actions of Exchange Members are shown in the context of provided market data. It demonstrates the orders submission, execution, modification and cancellation.

4.2. Short Codes

Members use short codes to safely exchange data about parties required by MiFID2/MiFIR for reporting purposes. Participants must additionally provide real data referred by the short codes, in a separate document.

A short code ranges from 4 to 4.294.967.295 (2^32 – 1). The values 0, 1, 2, and 3 are reserved for special purposes. In BIN messages mifidFields.client.shortCode; mifidFields.executingTrader.shortCode; mifidFields.investmentDecisionMaker.shortCode fields contains short codes.

- Client of the Member (mifidFields.client.shortCode):

  provides information about short codes which identify a natural person or firm or legal entity. Client short code is mandatory if the order is submitted with capacity = 1 (Agency) or 3 (Riskless Principal). A Client short code is linked to mifidFields.client.qualifier equal to 3 (FirmOrLegalEntity) or 4 (NaturalPerson). In case of aggregated orders or pending allocations the mifidFields.client.shortCode field is set to 1 or 2 and mifidFields.client.qualifier should be set to 1 (N/A).

- Investment Decision Maker (mifidFields.investmentDecisionMaker.shortCode),

  An Investment decision maker party role provides information about short codes which identify a natural person or algorithm responsible for the investment decision. An Investment decision maker identifier is mandatory if an order is submitted with capacity = 2 (Principal).

  The mifidFields.investmentDecisionMaker.shortCode for Investment decision maker value can range from 4 to 4.294.967.295. An Investment decision maker short code is linked to mifidFields.investmentDecisionMaker.qualifier equal to 4 (NaturalPerson) or 2 (Algorithm). mifidFields.investmentDecisionMaker.qualifier =4 is used when a natural person within the member is responsible for the investment decision. mifidFields.investmentDecisionMaker.qualifier = 2 is used when an algorithm is responsible for the investment decision.

- Executing Trader (mifidFields.executingTrader.shortCode),

  An Executing trader party role provides information on short codes which identify a natural person or algorithm responsible for an order execution. The Executing trader short code is mandatory if capacity is equal to Agency (1), Principal (2) and Riskless Principal (3). mifidFields.executingTrader.shortCode field value ranges from 4 to 4.294.967.295. Executing trader short code is linked to mifidFields.executingTrader.qualifier equal to NaturalPerson (4) or Algorithm (2). mifidFields.executingTrader.qualifier = 4 is used when a natural person within the member is responsible for order execution. mifidFields.executingTrader.qualifier = 2 is used when an algorithm is responsible for order execution. When the time and location of order execution are determined by the client of the participant, then the mifidFields.executingTrader.shortCode has the reserved code =3 (NONE) and mifidFields.executingTrader.qualifier should be set to N/A (1).

Fields mifidFields.client.qualifier; mifidFields.executingTrader.qualifier; mifidFields.investmentDecisionMaker.qualifier provides further qualification of respectively: mifidFields.client.shortCode; mifidFields.executingTrader.shortCode; mifidFields.investmentDecisionMaker.shortCode:

- 1 = NA
- 2 = Algorithm (applicable to mifidFields.executingTrader.shortCode; mifidFields.investmentDecisionMaker.shortCode),
- 3 = Firm or legal entity (LEI) (applicable to mifidFields.client.shortCode),
- 4 = Natural person (applicable to mifidFields.client.shortCode; mifidFields.executingTrader.shortCode; mifidFields.investmentDecisionMaker.shortCode).

The table below provides the summary of the combination of MiFiD fields group used to specify short codes.

Short Code	BIN Fields
Client – Natural person (National ID)	mifidFields.client.shortCode = short code value

Short Code	BIN Fields
	mifidFields.client.qualifier = 4
Client – Legal entity (LEI)	mifidFields.client.shortCode = short code value mifidFields.client.qualifier = 3
Client's aggregated orders	mifidFields.client.shortCode = 1 (AGGR) mifidFields.client.qualifier = 1
Client's pending allocation	mifidFields.client.shortCode = 2 (PNAL) mifidFields.client.qualifier = 1
Investment Decision Maker - Natural person (National ID)	mifidFields.investmentDecisionMaker.shortCode = short code value mifidFields.investmentDecisionMaker.qualifier = 4
Investment Decision Maker - Algorithm	mifidFields.investmentDecisionMaker.shortCode = short code value mifidFields.investmentDecisionMaker.qualifier = 2
Executing Trader - Natural person (National ID)	mifidFields.executingTrader.shortCode = short code value mifidFields.executingTrader.qualifier = 4
Executing Trader - Algorithm	mifidFields.executingTrader.shortCode = short code value mifidFields.executingTrader.qualifier = 2
Executing Trader - Order execution is determined by the client	mifidFields.executingTrader.shortCode = 3 (Client) mifidFields.executingTrader.qualifier = 1.

Depending on the capacity in the message, the following instances of short codes are mandatory or not:

	capacity		
Short Code	**1 (Agency)**	**2 (Principal)**	**3 (Riskless Principal)**
Client ID	R	O	R
Executing Trader	R	R	R
Investment Decision Maker	O	R	O

where :
R – short code required
O – short code optional

4.3. Conditionally Required Fields

Conditional fields are ignored unless they are necessary in the message data (e.g. triggerPrice is not allowed in order types other than Stop Loss and Stop Limit).

One sets unused conditional and optional numeric fields should to zero. One sets unused conditional and optional enumerable fields to a value which indicates their unrequired use (e.g. not applicable).

4.4. Security Identification

System assigns the instrument identifier (e.i. instrumentId). On the basis of reference data, one obtains other identifiers, e.g. ISIN number.

4.5. Numbers and Decimal Places

Numbers are encoded differently depending on type. All attributes with type Number have 8 decimal places. From a mathematical perspective to correctly interpret attributes with Type Number values should be divided by 10^8. The most common alias of the Number type is the Price type.

In GPW WATS, messages are formatted in little-endian byte order.

4.6. Order Management

The chapter describes important information about orders, including their identifiers, types, and processing rules.

4.6.1. ORDER IDENTIFIERS

The following order identification fields are used in messages:

- orderId,
- publicOrderId.

4.6.1.1. orderId

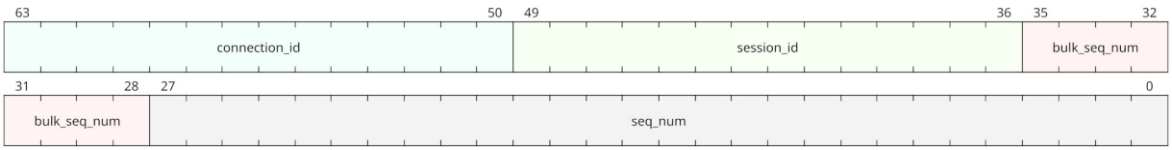
Trading System assigns a unique identifier for an order. The orderId field is unique across all trading days, all order books and all Participants. It is persistent for the entire life of an order (i.e. it does not change after an order modification). orderId is known only for Exchange Member who submits the order, Exchange and possibly for supervisory entities (e.g. Polish Financial Supervision Authority - KNF). Other Exchange Members do not know that orderId field.

An orderId field is generated for each order as a concatenation of connection id, session id, bulk sequence number and sequence number of the OrderAdd message sent by Exchange Member:

- connection Id – a connection number - connectivity parameter provided by GPW,
- session Id – a session number received during the login process,
- bulk sequence number – zero value for single orders (e.g. Market, Limit) and the value of subsequent orders for messages with multiple orders or quotes (e.g. MassQuote),
- sequence number - a sequence number of OrderAdd message.

All the above parts of the orderId field are known at the time of sending the OrderAdd message.

Figure3. orderId field construction.



Example

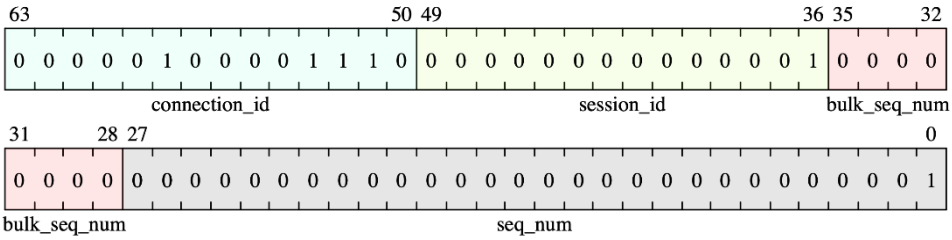
1. OderAddResponse message with orderId (highlighted in bold blue):

   1b 00 05 01 00 00 00 **01 00 00 00 10 00 38 04** 01 00 00 00 00 00 00 00 02 01 00 01,

5. Raw bytes of order id: 01 00 00 00 10 00 38 04,

6. Order id value 0x0438001000000001 (little endian byte order) and parts value:
    a) connection id = 0b00_0001_0000_1110 = 0x10E = 270,
    b) session id = 0b00_0000_0000_0001 = 1,
    c) bulk = 0b0000_0000 = 0,
    d) sequence number = 0b0000_0000_0000_0000_0000_0000_0001 = 1.

Figure 4. orderId =0x0438001000000001



Despite the facility to calculate orderId during order sending, Gateway sends back the orderId value in response to the received order (see the OrderAddResponse message specification).

Then orderId is used across multiple messages throughout the entire life of the order.

orderId can be used by Exchange Member when requesting an order modification (the OrderModify message) or order cancellation (the OrderCancel message) to identify the submitted order.

4.6.1.2. clientOrderId

clientOrderId is an optional identifier which can be used by Exchange Meber to identify or tag orders.

Gateway doesn't check uniqueness of received clientOrderId.

It can be assigned or reassigned to order in OrderAdd, OrderModify, OrderCancel.

Last assigned clientOrderId is returned in OrderAddResponse, OrderModifyResponse, OrderCancelResponse and populated on Drop Copy Gateway.

clientOrderId can also be used to identify or tag MassCancel operation but in such a case it's populated only into MassCancelResponse on both Order Gateway or Drop Copy Gateway.

4.6.1.3. publicOrderId

System assigns the order identifier published in Market Data (public information) and it corresponds to the publicOrderId field in Market Data OrderAdd (9), OrderModify (10), OrderCancel (11), OrderExecute (12) messages.

publicOrderId is unique across all order books and all Exchange Members, but only within a single trading day. On the next trading day, the numbering is reset and starts afresh(i.e. publicOrderId is not consistent for the entire life of an order). Moreover it also changes with each refill of an iceberg order or applies to transfer orders (e.g. orders with time in force set to Good Till Cancel or Good Till Date).

Gateway provides publicOrderId for an order only in the OrderAddResponse message. publicOrderId can be used to relate submitted orders to Market Data, however orders are modified or cancelled only using the orderId field.

4.6.1.4. TVTIC

Trading Venue Transaction Identification Code is a number generated by the WATS trading system for transactions reporting purposes in accordance with Article 12 of Commission Delegated Regulation (EU) 2017/580.

The TVTIC numbers are unique systemwide and assigned across all market segments and all financial instruments covering both book and off-book trades.

At the start of the trading day TVTIC is reset and starts from 1.

TVTIC is provided in the message "OrderExecute" - field "tradeId" (Type "TradeID") and for Off-book trades in the message "Trade Capture Report" – field: "tradeId" (Type "TradeID")

4.6.2. ORDER TYPES

Exchange Members may submit the following order types in the OrderAdd message via Gateway:

- Limit Order,
- Market Order,
- Market To Limit Order,
- Iceberg Order,
- Stop Limit Order,
- Stop Loss Order.

4.6.2.1. Limit Order

Limit Order is an order to buy or sell at a stipulated limit price or a better one. Limit Order can be executed at a price equal to or better than its limit price, which means for buy order at its limit or lower price and for sell order at its limit or higher price. Unexecuted remainder of a Limit Order is added to the order book, unless order's validity attribute timeInForce specifies otherwise.

Relevant message fields:

- orderType = 1 (Limit),
- price = Limit Price.

4.6.2.2. Market Order

Market Order is an order to buy or sell without specified price. Market Order is executed at the best opposite prices with one or more counterpart resting orders.

During Continuous Trading Market Phase, Market Orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. timeInForce = 3, 4, 5, 7.

During Auction Market Phase, Market Orders can only be submitted with VFA and VFC validity attributes, i.e. timeInForce = 5, 7.

Relevant message fields:

- orderType = 2 (Market),
- timeInForce = 3 (Immediate Or Cancel), 4 (Fill Or Kill), 5 (Valid For Auction), 7 (Valid For Closing).

4.6.2.3. Market To Limit

Market To Limit Order is an order to buy or sell without a specified price. Market To Limit Order is executed at the best opposite price level with one or more counterpart resting orders.

During Continuous Trading Market Phase, Market To Limit Orders can only be submitted with IOC, FOK, VFA, VFC validity attributes, i.e. timeInForce = 3, 4, 5, 7.

During Auction Market Phase, Market To Limit Orders can only be submitted with VFA and VFC validity attributes i.e. timeInForce = 5, 7.

Relevant message fields:

- ordType = 3 (Market To Limit),
- timeInForce = 3 (Immediate Or Cancel), 4 (Fill Or Kill), 5 (Valid For Auction), 7 (Valid For Closing).

4.6.2.4. Iceberg

Iceberg Order is an order, for which only a part of the total quantity is disclosed to Market via Market Data. The rest of the order is hidden (invisible) to the general market. Exchange Member submitting an Iceberg Order must specify additionally the quantity that is going to be displayed initially, i.e. displayQty, which:

- must be lower than the total order quantity,
- must be greater or equal to the Minimum Display Quantity set in Trading System configuration,
- may be randomized by Trading System within a specified range.

Once displayed quantity is completely filled, a new portion of the order is disclosed to Market according to the displayQty requested at the order entry (with possible randomization).

The value of each Iceberg Order (i.e. price x orderQty) at the moment of entry or during any modification must be equal or greater than the Minimum Iceberg Size set in Trading System configuration.
Iceberg Orders can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.
Relevant message fields:

- ordType = 4 (Iceberg),
- price = Limit Price,
- displayQty = Initially Displayed Quantity,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.6.2.5. Stop Limit

Stop Limit is an order to buy or sell at a stipulated limit price or a better one, which is activated and inserted into the order book upon reaching or exceeding pre-defined price level (called Trigger Price) by Last Traded Price on Market. Until activation, Stop Limit is hidden and cannot interact with the order book. Once activated, Stop Limit is processed the same way as a Limit Order with the same validity attribute that the original Stop Order had before its activation (nonetheless maintains its original Order Type = 5).

When submitting or modifying a Stop Limit order, the following conditions must be met:

- for buy order: Price >= Trigger Price > Last Traded Price (LTP),
- for sell order: Price <= Trigger Price < Last Traded Price (LTP).

Stop limit can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.

The Trigger Price may be modified for inactive Stop Order, but once the order is activated, Trigger Price cannot be modified anymore.

Relevant message fields:

- ordType = 5 (Stop Limit),
- price = Limit Price,
- triggerPrice = Activation Price,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.6.2.6. Stop Loss

Stop Loss is an order to buy or sell without specified price, which is activated and inserted into the order book upon reaching or exceeding pre-defined price level (called Trigger Price) by the Last Traded Price on Market. Until activation, Stop Loss is hidden and cannot interact with the order book. Once activated, Stop Loss is processed the same way as Market Order with Immediate Or Cancel (IOC) validity attribute (nonetheless maintains its original Order Type = 6).

When submitting or modifying a Stop Loss Order, the following conditions must be met:

- for buy order: Trigger Price > Last Traded Price (LTP),
- for sell order: Trigger Price < Last Traded Price (LTP).

Stop Loss can only be submitted with DAY, GTC, GTD, GTT validity attributes, i.e. timeInForce = 1, 2, 6, 8.

The Trigger price may be modified for inactive Stop order, but once the order is activated, Trigger Price cannot be modified anymore.

Relevant message fields:

- ordType = 6 (Stop Loss),
- triggerPrice = Activation Price,
- timeInForce = 1 (Day), 2 (Good Till Cancel), 6 (Good Till Date), 8 (Good Till Time).

4.6.3. ORDER VALIDITY TYPES

Exchange Members may submit orders with the following validity attributes (timeInForce field):

- Day (DAY),
- Good Till Cancel (GTC),
- Good Till Date (GTD),
- Good Till Time (GTT),
- Immediate Or Cancel (IOC),
- Fill Or Kill (FOK),
- Valid For Auction (VFA).
- Valid For Closing (VFC).

Not all validity attributes may be used with every order type throughout various Market Phases.

4.6.3.1. Day (DAY)

Day order is valid until the end of the current trading day only. Unexecuted Day orders are expired by Trading System at the end of the current trading day.

Day validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 1 (Day).

4.6.3.2. Good Till Cancel (GTC)

Good Till Cancel order is valid until it gets fully executed or canceled by the submitter or market operations (whichever comes first).

There is a maximum number of days threshold set up in Trading System configuration, that defines for how long a Good Till Cancel order may rest in the order book. After that time elapses, GTC order gets expired as well.

Good Till Cancel validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 2 (Good Till Cancel).

4.6.3.3. Good Till Date (GTD)

Good Till Date order is valid until the end of the specified day.

If the expiry date specified for an order is not a trading day, the order gets expired at the end of the previous day.

The specified date cannot exceed the maximum number of days threshold defined by Market Operator (ranging from 1 to 365 days).

Good Till Date validity attribute may be used with all order types, except for Market Order and Market To Limit.

It is possible to modify orders with GTD validity type. Modification of the date is possible from the current date to the limit for GTD (otherwise the modification is rejected). Modified orders do not change priority.

Relevant message fields:

- timeInForce = 6 (Good Till Date),
- expire = Expiry Date.

4.6.3.4. Good Till Time (GTT)

Good Till Time order is valid until the specified time of the current trading day.

If the expiry time specified for an order is later than the end of the last Market Phase, the order gets expired at the end of the current trading day.

Good Till Time validity attribute must be used with the expiry field set to the current date and time in the future.

Good Till Time validity attribute may be used with all order types, except for Market Order and Market To Limit.

Relevant message fields:

- timeInForce = 8 (Good Till Time),
- expire = Expiry Time (UTC timestamp with date 1970-01-01).

4.6.3.5. Immediate Or Cancel (IOC)

Immediate Or Cancel order must be executed immediately, in full or partially, upon order entry. IOC orders may be matched against one or more opposite orders (with the price limits within Trade Price Collars). Any unexecuted part of an IOC order is automatically canceled by Trading System.

Immediate Or Cancel orders may be submitted during Continuous Trading only.

Immediate Or Cancel validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 3 (Immediate Or Cancel).

4.6.3.6. Fill Or Kill (FOK)

Fill Or Kill order must be executed in full, immediately upon order entry. FOK orders may be matched against one or more opposite orders (with the price limits within Trade Price Collars). An unexecuted FOK order is automatically canceled by Trading System.

Fill Or Kill orders may be submitted during Continuous Trading Market Phase only.

Fill Or Kill validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 4 (Fill Or Kill).

4.6.3.7. Valid For Auction (VFA)

Valid For Auction order is activated and inserted into the order book at the beginning of the next Auction Market Phase (whether scheduled or unscheduled). VFA order submitted in Continuous Trading is hidden until its activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFA order is valid until the end of Auction and gets expired by Trading System immediately after uncrossing (if left unexecuted). VFA orders are valid for the current trading day only. Activated VFA orders retain their original priority timestamp from the moment of submission.

Valid For Auction validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 5 (Valid For Auction).

4.6.3.8. Valid For Closing (VFC)

Valid For Closing order is activated and inserted into the order book at the beginning of the Closing Auction Market Phase only. VFC order submitted in Continuous Trading or Auction other than Closing, is hidden until its activation (i.e. it is not published via Market Data) and cannot interact with the order book. Once activated, VFC order is valid until the end of the Closing Auction and gets expired by Trading System immediately after the uncrossing (if left unexecuted). VFC orders are valid for the current trading day only. Activated VFC orders retain their original priority timestamp from the moment of submission.

Valid For Closing validity attribute may be used with all order types, except for Stop Limit, Stop Loss and Iceberg.

Relevant message fields:

- timeInForce = 7 (Valid For Closing).

4.6.4. ORDER CAPACITY

Order capacity defines the capacity of the company, which submits an order. The capacity field values:

- 1 - Agency mapped as AOTC (Any other capacity),
- 2 - Principal mapped as DEAL (Dealing on own account),
- 3 - Riskless Principal mapped as MTCH (Matched principal).

It is required to set the correct capacity field value by Exchange Member.

4.6.5. ORDER STATUS

The order status informs on the state in which it is in GPW WATS. With regards to orders submitted by Exchange Member, the status may take the values acknowledged, cancelled, rejected or filled. Successful modification of the order by Exchange Member results in the modified status. Similarly, a successful cancellation changes the status of the order to be cancelled.

Exchange Member may receive the following status for orders from Gateway:

- New (1) – status in which an order is accepted into the order book until there is any execution on it (trade) or until an order is cancelled or expired. In case aggressive order was partially or fully filled there is no New status assigned to order. To help Exchange Member to identify such a situation, the field execTypeReason is set to value FirstTradeOnAggressiveOrder (First trade on aggressive order),
- Partially filled (5) – status in which there was at least one execution on an order until the order is fully filled (there is no remaining quantity on it) or until the order is cancelled or expired,
- Filled (4) - status in which an order is fully executed (there is no remaining quantity on an order),
- Cancelled (2) - status in which an order is canceled by Exchange Member, Trading System (including situation when an incoming IOC/FOK order cannot be fully executed upon arrival) or by the Market Operations. Additional information about the reason for unsolicited cancellation is provided in the execTypeReason field,
- Rejected (3) - status in which an order is rejected by Trading System upon entry due to various business validations,
- Expired (6) – status in which an order is automatically removed from the order book due to its validity condition specified in timeInForce field (including removal of unexecuted part of VFA/VFC order after the Auction).

Information about the status of orders is sent in OrderAddResponse, OrderCancelResponse and OrderModifyResponse messages.

The reason for the order status change may be an order modification or cancelation by the Exchange Member, orders matching, System actions (e.g. cancelation via Cancel on Disconnect mechanism) and session supervision team operations.

More information concerning the reason for the order state change can be found in the execTypeReason field description (see execTypeReason field description in OrderAddResponse message). The field informs why an order was executed and about order lifecycle events. It takes the following values:

- CancelOnDisconnect – cancelled order by Cancel on Disconnect mechanism,

- Expired - cancelled order by System in the case of expiry,
- Triggered - notification of Stop or VFA/VFC order activation,
- CancelOnSuspension - cancelled order by Trading System in the case of instrument suspension,
- OrderRestatement - reinstated order,
- IcebergOrderRefill - iceberg order refill,
- CancelByStp – cancelled order due to self-trade prevention,
- CancelByCorporateAction- cancelled order due to submitted Corporate Action,
- CancelByMassCancel – an order cancelled due to mass cancel,
- CancelIocFokOrder – an order cancelled due to IOC/FOK,
- CancelByMarketOperations – an order cancelled due to market operations,
- Replaced – order modified,
- FirstTradeOnAggressiveOrder – new and aggressive order was filled or partially filled,
- Rejected - order rejected.
- VfaActivated –

The below table presents the possible combinations ExectTypeReason vs Status.

		Status					
		New	Cancelled	Rejected	Filled	Partially Filled	Expired
execTypeReason	NA						
	CancelOnDisconnect		X				
	Expired						X
	Triggered	X			X	X	
	CancelOnSuspension		X				
	OrderRestatement	X				X	
	IcebergOrderRefill	X			X	X	
	CancelByStp		X				
	CancelByCorporateAction		X				
	CancelByMassCancel		X				
	CancelIocFokOrder		X				
	CancelByMarketOperations		X				
	Replaced	X				X	
	FirstTradeOnAggressiveOrder				X	X	
	Rejected			X			
	CancelonBuyOnlyStateEntry		X				
	CancelonKnockedOutStateEntry		X				
	CancelByRiskManagement		X				
	CancelOnDcDisconnect		X				

	VfaActivated	X					
	HybridActivated	X			X	X	

4.6.6. CANCELLATION

Cancellation of a previously submitted order by Exchange Member is only possible on the same connection id on which the order was submitted. For example, an order submitted on the connection id = 123 cannot be removed from the connection id = 242.

The order is cancelled by sending the OrderCancel message with the appropriate orderId field (i.e. the orderId received in the OrderAddResponse message)

Connection allocation management is the responsibility of the Exchange Member.

Deleting orders is subject to business validation, in particular, it is not possible to delete an order which has already been fully executed.

More information about connections is presented in the Connectivity chapter.

4.6.7. MODIFICATION

Modification of the order, similarly to its cancellation, is only possible on the connection on which the order was previously sent (i.e. the same connection id).

The order is modified by sending the OrderModify message with the appropriate orderId field (i.e. the orderId received in the OrderAddResponse message).

Exchange Members are allowed to modify the following order characteristics:

- total quantity,
- initial displayed quantity (in case of Iceberg Orders only),
- price,
- trigger price (in case of Stop Orders only),
- expire (in case of timeInForce = GTD (i.e. Good Till Date),
- Short Code.

MiFID fields shall be provided as in a OrderAdd message or modified. MiFID fields will be validated in the same way as in the case of the OrderAdd message.

4.6.8. ACCOUNT STRUCTURE

There are the following types of accounts (i.e. accoutType field):

- Customer - account is carried on the customer side of the books,
- House – house trader.

If there is no account placed in the order, the accountType field should be set to "Missing" and the account field should be filled with zeros.

4.6.9. TEXT FIELDS

A character in the text field is treated as an unsigned byte type. In the trading_port.json contract, the above type is called AnsiChar. The string is created as an AnsiChar array. Examples of text fields are account, token, micCode etc.

If the length of the text is less than the length of the field, the empty space should be filled with zeros.

ASCII characters, in the range 32 – 126, should be used. Messages containing text fields with characters outside the above range is rejected.

4.6.10. ORDER BOOK RESTATEMENT

At the beginning of each trading day, Exchange Members receive incoming orders from Gateway which were placed during the previous session and are still valid (e.g., orders with a Time In Force equal to GTD or GTC), known as restated orders.

More information can be found in the document *GPW WATS 1.01 Trading System*, chapter Trading Schedules.

Restated orders are sent using an OrderAddResponse message with fields set to the values from the end of the previous day. An order status is also set to the last status from the previous day (e.g. new, partially filled). The field execTypeReason is set to an orderRestatement value.

Restated orders are not sent publicly in Market Data, only to Exchange Members which submitted them.

4.6.11. EXPIRED ORDERS ARE NOT TRANSFERRED – RESTATED (E.G. ORDER WITH TIF = DAY, OR GOOD TILL DATE ORDERS WITH A PAST EXPIRY DATE).UNSOLICITED ORDER CANCELLATIONS GENERATED BY TRADING SYSTEM

Unsolicited order cancellations generated by System is carried out using OrderCancelResponse messages with the following characteristics:

1. order execTypeReason (i.e. execTypeReason field) takes a different value than submitted. This value explains the source of the order cancellation e.g. Cancel on Disconnect mechanism,
2. reason – a reason for cancelation if available.

4.6.12. CLEARING MEMBER IDENTIFICATION

Clearing Member is a financial institution or a firm that acts as an intermediary between buyers and sellers to facilitate the settlement of transactions. Each Exchange Members connection should have clearing member set to be used for settlement purposes, however Exchange Members may indicate different clearing member on the level of submitted order or BLOCK/CROSS transaction.

Clearing member can be submitted by Exchange Member using two optional attributes:

- clearingMemberClearingIdentifier
- clearingMemberCode

Recommendation: For GPW WATS system version V1, the use of KDPW CCP participant codes is recommended (clearingMemberClearingIdentifier: Custom).If an **LEI** value is provided and is not recognized by the system, the settlement will be processed using the default clearing account in accordance with its applicable KDPW CCP procedure.

4.7. Trade Capture

There are several situations where transaction capture reports are exchanged:

- Block transactions (field tradeType - BlockTrade):
    - Exchange Member may send a dual transaction capture report in a situation where a transaction involving clients of a given Exchange Member was made on the block market. The TradeCaptureReportDual message is used by Exchange Member as a dual transaction capture report. Gateway response should be the TradeCaptureReportResponse message.
    - Exchange Member may send a single transaction capture report which requires confirmation by the other party. System ensures correct message exchange between the transaction parties. The TradeCaptureReportSingle message is used by Exchange Member as a single transaction capture report. Gateway response should be the TradeCaptureReportResponse message.
- Cross transactions (field tradeType - PrivatelyNegotiatedTrade):
    - Exchange Member may send a dual transaction capture report (a TradeCaptureReportDual message) in a situation where a cross transaction is submitted.

FIX Order Gateway equivalent of the message about transaction capture reports are TradeCaptureReport (AE)/ TradeCaptureReportAck (AR) messages.

4.7.1. BLOCK MARKET MODEL

Block trading is a market model in which trades are concluded bilaterally in the process of the negotiations between the counterparties (beneficiary clients). The beneficiary clients may belong to one or a few Exchange Members. The negotiations usually take place beyond Trading System infrastructure. Block trading is concluded when the respective trade entry is recorded in Trading System based on the Trade Capture Report message/messages submitted by Exchange Members.

Block Trade facility enables Exchange Members to report pre-arranged large trades through Market Operator Trading System.

The value of Block Trades must not be lower than Large In Scale value. The price of the Block Trade must be within thresholds set by the exchange. Price thresholds can be set on different levels depending on the session Phase of the same instrument in the Central Limit Order Book. Only instruments traded in the Central Limit Order Book are subject to Block Trades.

Block Trade is reported to the exchange by entering "one-party" or "two-party" Trade Capture Report. In the "one-party for pass-through" Block Trade the beneficiary clients belong to different Exchange Members. The initiator who enters one-party Block Trade requires confirmation by the counterparty to achieve the Block Trade. In "two-party" Block Trade both beneficiary clients belong to the same Exchange Member. "Two-party" Block Trade is used to report internal trade where the trade can be concluded and

reported between two clients of Exchange Member (Order Capacity; Agency vs Agency) or between the firm and the client (order Capacity: Principal vs Agency).

Block Trade features:

- Despite the fact that MiFID 2 tick size regime does not apply to Block Trades, simplified tick size tables for block instruments have been adopted,
- Both parties to the Block Trade must be the Exchange Member,
- Block Trades are reported by using the Trade Capture Reports messages.
- Block Trades are not pre-transparent thus not matched Trade Capture Reports (i.e. the execType field is not set to the value *Trade*) are not disseminated in Market data stream,
- Block Trades are post-trade transparent thus matched Block Trades (i.e. the execType field is set to the value *Trade*) are immediately disseminated in Market Data stream,
- Block Trades do not update the Last Trade Price.
- Block Trades are not taken into account when calculating the opening price or the closing price.

4.7.1.1. Trade Capture Report Dual for Block Trades

TradeCaptureReportDual is applied when both beneficiary parties to the trade belong to the reporting firm (Exchange Member). The TradeCaptureReportDual is accepted by Gateway by sending back TradeCaptureReportResponse with the field status = 2 (Accepted). If TradeCaptureReportDual is rejected, then Exchange Member receives the TradeCaptureReportResponse with status = 3 (Rejected) with the respective information about the reason for rejection provided in the **reason** field.

4.7.1.2. Trade Capture Report Single for Block Trades

TradeCaptureReportSingle is applied when the beneficiary parties to the trade belong to different Exchange Members. The applied one-sided model is the "one-party report for pass-through". The initiator submits a report (first leg to the trade), which is accepted by the counterparty to the trade.

This is the basic TradeCaptureReportSingle flow enabling Exchange Members to conclude the block trade:

- The initiator submits the first leg to the trade by sending TradeCaptureReportSingle with the values:
    a) tradeReportTransType = 1 (New),
    b) tradeReportType = 1 (Submit).
- The first leg is accepted by System by sending TradeCaptureReportResponse with the status = 2 (Accepted).
- System forwards the TradeCaptureReportSingle to the counterparty by sending notification with the values:
    a) tradeReportTransType = 1 (New),
    b) tradeReportType = 2 (Alleged).
- The counterparty accepts the first leg by sending TradeCaptureReportSingle with the values:
    a) tradeReportTransType = 3 (Replace),
    b) tradeReportType = 3 (Accept).

- System sends the TradeCaptureReportSingle informing the initiator and the counterparty about the concluded trade with the values:
    a) execType = 15 (Trade),

4.7.2. CROSS TRADES

For a given instrumentId, Cross trades can only be submitted if the corresponding instrumentId with the same ISIN code in the Order Book is in the Continuous Trading Phase (variable price and fixed price), i.e. in:

- Continuous price and time (ContinuousPriceTime),
- Continuous time at reference price (ContinuousRefPriceTime),
- Trade at last (ContinuousLastAuctionTime).

Cross trades do not change Last Trade Price for an instrument with the same ISIN code in the central order book.

Cross trades are transmitted to System via Trade Capture Report message. This is a dual-sided TradeCaptureReportDual.

The content of the TradeCaptureReportDual message for Cross trades is analogous to that for bilateral Block trades, if the following tags have the following fixed values:

- tradeType = 22 (PrivatelyNegotiatedTrade),
- settlementDate = 0(settlement date is not applicable for cross trades) .

The values in tradeReportType have been reduced to the requirements of a Cross transaction to:

- 1 = Submit,
- 7= Trade Report Cancel.

Cross transaction scenario:

1. Transmission of a TCR Cross to System with a tradeReportType field with the value 1 = Submit.
2. For a valid TCR Cross, System sends a TradeCaptureReportResponse with an Accepted status (status = 2 (Accepted)).
3. For an invalid TCR Cross, System sends a TradeCaptureReportResponse with status Rejected (status = 3 (Rejected)).

4.8. QUOTE HANDLING

Market makers send quotes into a market using a MassQuote message. For a given instrument, it is possible to send one quotation. The MassQuote message enables the sending of up to 30 quotations.

The Trading Platform sends a confirmation of the accepted and rejected quotations by transmitting a MassQuoteResponse message.

Quotations rules:

- Two-sided quotes are allowed,
- Only one market maker quotation is allowed per instrument,

- MassQuoteResponse message includes all submitted quotations along with their acceptance status (e.g. Accepted, Rejected),
- Modification of submitted quotations involves placing another MassQuote message with changed values for price or quantity,
- Cancellation of submitted quotations involves placing another MassQuote message with the quantity set to 0 for one(Buy or Sell) or both sides of quotation for the same Instrument
- Orders created from mass quotes are by default subject to CoD(will be canceled when CoD service is set).
- Mass Quote messages can be used on CLOB and Hybrid market model.

4.9. Cancel on Disconnect

Cancel on Disconnect (CoD) mechanism prevents orders from remaining outside the Exchange Member's control after an incorrect disconnection from System, i.e. without logging out.

Disconnection from Gateway is understood as breaking the TCP/IP connection or lack of communication from the Client Application, including the lack of Heartbeat messages for a period of 12 seconds (so-called grace period).

The CoD is a sub-service of Gateway service and can be enabled for any Exchange Member connection. Elimination of orders follows as per the CoD mechanism rules on a given trading connection. If the user has more than one connection, then the mechanism can be activated on each of these connections or only on chosen connections. The GPW is responsible for the configuration of the service and performs it at Exchange Member request.

Orders in which the Exchange Member has set the CancelOnConnectionLoss flag in the execInst field are exclusively subject to the CoD mechanism. Orders created from mass quotes are by default subject to CoD(there is no execInst field in mass quote, orders will be canceled when CoD service is set).

In case CoD service is not set for connection all new orders (sent on this connection) with execInst flag set (CancelOnConnectionLoss) will be rejected.to avoid misunderstanding.

4.10. Mass Cancellation

Mass Cancel functionality allows Exchange Member to cancel all active orders according to the selection criteria they have specified. In the case of a previously partially executed order, only the remaining unexecuted part of the order is cancelled. This means that Mass Cancel doesn't affect the already executed volume of the order. Mass Cancel functionality works only for CLOB market model (i.e. the Mass Cancel functionality does not include the cancellation of unmatched Trade Capture Reports).

This functionality can only be used by Exchange Member in respect of orders that they have sent themselves. Sending a Mass Cancel message is not possible via the DEA (Direct Electronic Access). Exchange Member must indicate the Executing Trader performing the operation (the executingTrader field) and provide shortcode.

Exchange Member performs Mass Cancel by sending the OrderMassCancel message. The OrderMassCancel message allows the cancellation of all active orders placed on all connections (ConnectionID) of a Participant. OrderMassCancel message contains fields that define the selection criteria for cancelled orders. These fields can be divided into two groups:

- General criteria specified in the massCancelRequestType field,

- Narrowing criteria contained in the body of the of the message (e.g. targetPartyId).

Selection Criteria

Three main selection criteria are available for the mass cancellation of orders. These main criteria are disjoint and are contained in the required massCancelRequestType field:

- 1 = CancelForSecurity,
- 7 = CancelAllOrders,
- 9= CancelOrdersForMarketSegment,

In addition, a number of narrowing criteria is available that only work in combination with one of the above three main criteria for Mass Cancel. It is allowed to select only one narrowing criterion from the targetPartyRole, which automatically means that the value of the targetPartyId field should correspond to the selected role:

- None and LiquidityProvider – targetPartyId equals 0,
- SenderLocation - targetPartyId is set to connection_id,
- ClientID, ExecutingTrader and InvestmentDecisionMaker - targetPartyId is set to the short code of a party.

4.11. Reject Messages

Messages sent by Exchange Member can be rejected at the session level and at the application level.

The Reject message informs on the message rejection at the session level i.e. in a situation where the message cannot be considered valid and can no longer be processed. Reason for rejection are given in the rejectReason field (e.g. Invalid message type).

Session-level valid messages related to submitting, cancelling and modifying orders may be rejected at the application level. Relevant information about the rejection is provided in the status and reason fields of the message which are a Response to the Exchange Member's message.

4.12. Timestamps and Dates

The time and date are stored on the Timestamp type, which is based on a 64-bit unsigned integer. On the Timestamp type, the time is given within nano second accuracy.

Below are the fields in which the time is given and the rules for filling them in:

- Expire – in the case of GTT orders, only the time is filled in, the date component of expire field must be set to 1970-01-01; in the case of GTD orders, only the date is filled in, the time component should be filled in with 0.

5. CONNECTIVITY

This chapter provides information regarding: logon authentication, network parameters, failover and recovery.

5.1. SESSION IDENTIFICATION

5.1.1. LOGIN PARAMETERS

Establishing a session with GPW WATS requires logging into System. To do this, it is necessary to have a token associated with a unique connection ID. The token and connection ID are fields of the login message. Both parameters are obtained from GPW. You cannot log in with the same connection ID multiple times.

5.1.2. NETWORK CONNECTION

IP addresses and ports for Gateway are published in a separate configuration document. There are 2 IP addresses available for each connection. The primary address for typical use and secondary address used in the case of failure.

5.1.3. TECHNICAL SESSION IDENTIFICATION

The technical session begins when System starts up and ends when it is turned off. Thus, it typically covers a trading day. In special cases, such as an emergency system restart, another technical session is started. The technical session ID is published in the StartOfTechnicalSession market data message as a field sessionId. It is also possible to obtain the current session id when logging in. Then the session id is given in response to the login message.

The session ID is used to create an order ID (see section ***Order Identifiers***).

5.2. FAILOVER AND RECOVERY

Due to a reduction to operational risk within GPW WATS, the failover and recovery mechanism has been designed.

Failover and recovery modes:

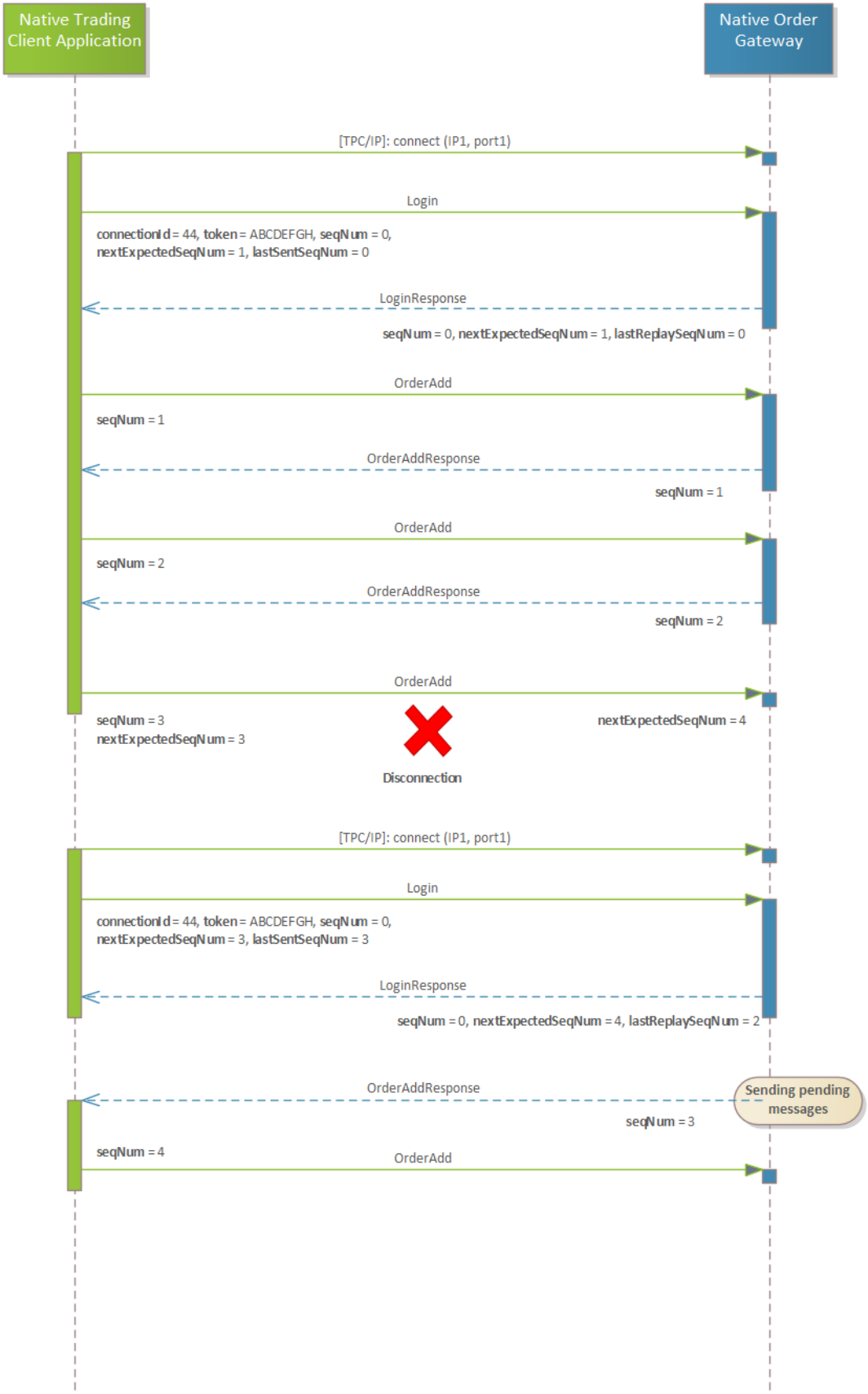
1. **Failover Mode**: This mode is activated in response to temporary connectivity losses. The system is designed to allow reconnection attempts within a timeframe of several seconds, up to approximately fifteen seconds. If the connection cannot be re-established after multiple attempts, users will be required to transition to Recovery Mode .
2. **Recovery Mode**: This mode is implemented when there is a need to shift sessions to an alternative gateway. An alternative gateway may be located at a different SITE.

5.2.1. FAILOVER MODE

In the case of temporary and short unavailability of Gateway, Exchange Member should attempt to re-establish the connection and login to Gateway. Subsequently the messages are resynchronized between Exchange Member and Gateway based on the sequence numbers included in the Login and LoginResponse messages. There are nextExpectedSeqNum and lastSentSeqNum in the Login message.

The nextExpectedSeqNum field means the next expected message sequence number to be received and the lastSentSeqNum field means the last sent sequence number. The LoginResponse message also contains the nextExpectedSeqNum field and, additionally, the lastReplaySeqNum field indicating the number of the last sent message or the last message currently waiting to be sent, if any. Any gap in the messages is filled by sending missing messages by each party. The diagram below shows such a situation.

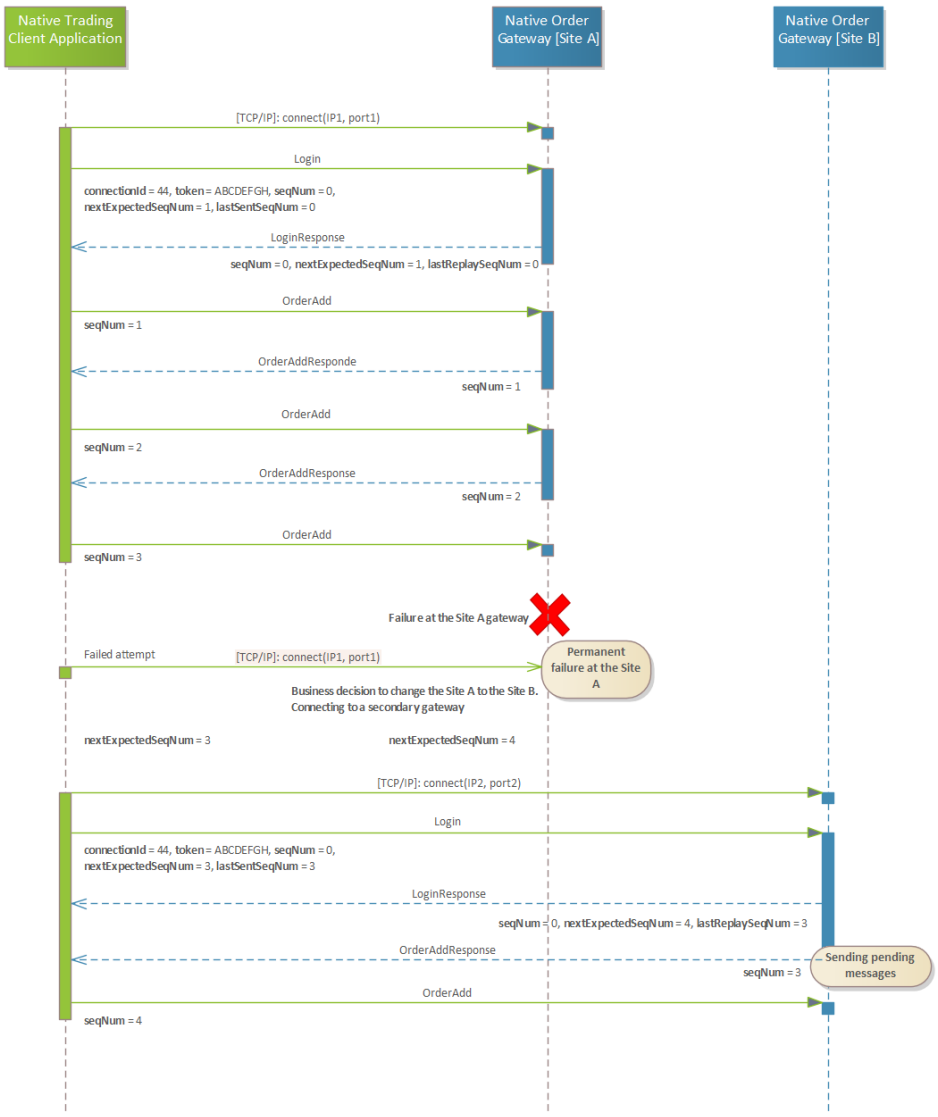
Figure 5. The synchronization after disconnection scenario.



5.2.2. RECOVERY MODE

When the primary Gateway is unavailable for a long period (minimum 15 seconds), the procedure to switch to a secondary could be launched. It is a business decision of Exchange Member. The connection is established with the same login parameters but on a new IP address and a port number. After logging in, business messages are synchronized in the same manner as in the first case (i.e. failover mode). In case, during logging to site B, there is still open session remaining in Site A it will be closed and subsequently session in site B will be established. The diagram below shows such a situation. Exchange Member should number subsequent application messages in accordance with the sequence number received in the LoginResponse (the nextExpectedSeqNum field).

Figure 6. Switching gateway location in the case of disconnection.



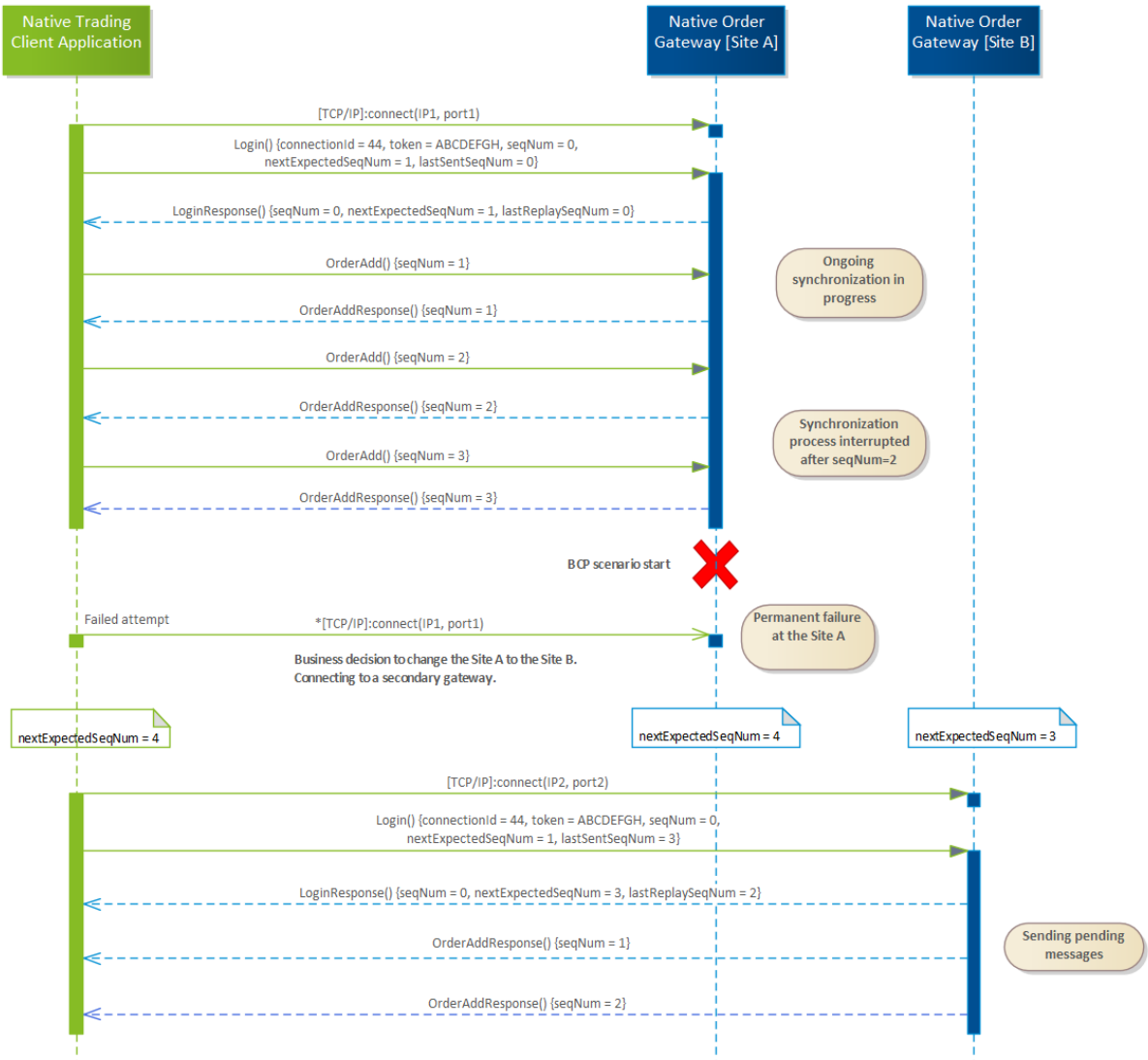
In the event that a full restoration of the message state is necessary on the side of the Exchange Member, the nextExpectedSeqNum field in the Login message will be set to 1, which will enable the recovery of the entire state of messages received from System.

In case of further connection issues, please contact the System Operator.

Please note that when using the CoD mechanism and disconnecting without a logout message, which is natural for most failures, orders are cancelled (see section Cancel on Disconnect).

5.2.3. RECOVERY MODE (DR SCENARIO / BCP PLAN)

In the event of DR scenario, announced by GPW, the full restoration of the message state is necessary on the side of the Exchange Member. The nextExpectedSeqNum field in the Login message should be set to 1, which will enable the recovery of the entire state of messages received from WATS System.



5.3. MESSAGE THROTTLING

The throttling mechanism protects Gateway against overloading with an excessive number of input messages.

There are 2 limits set in Gateway:

1. business throttling - after exceeding the business limit level, consecutive messages are rejected using Reject messages with the reason set to MaxThroughputExceeded,
2. technical throttling - after exceeding the technical limit level, the connection is closed by sending a ConnectionClose message with the reason set to AntiFloodingThresholdExceeded and then the socket is closed.

Limit calculations take into account all messages. However, Logout and OrderCancel messages will not be rejected.

The MassQuote messages are treated differently. When calculating limits for MassQuote, each side of each quotation is counted as a separate message. For example, a MassQuote message containing 8 quotations is counted as 16 messages.

The above limits are calculated over a period of time configurable in Gateway (i.e., measurement period).

The throttling parameters are configurable at Gateway service as follows:

- The business limit level is configured for each connection and depends on the level of service provision agreed upon between System Operator and the Exchange Member. The measurement period - last 2 seconds.
- The technical limit level is configured as an internal gateway parameter and is set to 5000 messages per second. The measurement period - last 100 milliseconds.

In addition, after exceeding the technical limit level, it is not possible to log into System for 10 seconds.

Note: the measurement period means the period of time during which messages are counted and the limits which have possibly been exceeded are determined.

5.4. Connectivity Policy

Connectivity policy is described in the document *GPW WATS 6.01 Connectivity*.

6. Session Layer

6.1. Establishing a Session

Starting a session between the Native Trading Client Application and Gateway begins with establishing a TCP/IP connection and sending the correct Login message. The message should contain the connection parameters presented in the Connectivity chapter. In particular, the Login message should contain a token and a connection identifier. Both of the above parameters are provided by System Operator.

When successfully logging in, the client receives a LoginResponse message with the result field equals "OK". When there is a business level error, a reason is given in the result field, e.g. invalid token.

When there are protocol errors or it has not been possible to establish a session, Gateway sends the ConnectionClose message which contains the reason and then the TCP/IP connection would be closed.

Each user can only be logged in once with the same credentials. When the already logged user tries login again during another established session, Gateway sends the LoginResponse message with the result field set to "AlreadyLoggedIn". In the recovery mode scenario, when member logs in to secondary Gateway, session established with primary Gateway is terminated. Such an approach guarantees only one session is established per connection and, in case the state of primary Gateway is invalid, new session can be created with secondary Gateway.

If Participant exceeds the number of login attempts, Gateway blocks the account and the LoginResponse message is set to "AccountLocked". The limit of failed login attempts is set to 5.

The maximum limit of the login attempts with Gateway is set up as once every 3 seconds. In the situation of a failed login attempt, Exchange Member is advised to contact the GPW service desk for further assistance.

6.2. Maintaining a Session

6.2.1. SEQUENCE NUMBER

Sequence numbers are only assigned to business messages within a trading day. Each trading day is associated with a session id available after logging in (see. a LoginResponse message and the sessionId field) or in market data (see. StartOfTechnicalSession and EndOfTechnicalSession messages). The sequence numbers start with 1 and are incremented by 1 in each sent business message. Session messages have a sequence number set to 0.

The following is a list of session message types (i.e. messages with a sequence number of zero):

- Login,
- LoginResponse,
- Logout,
- LogoutResponse,
- ConnectionClose,
- Heartbeat (beeing sent with sequence of last bussiness message by WATS Native Gateway, recomended to be sent with seqNum=0 by Exchange Member)

Message types other than those listed above belong to the business message set and require setting a sequence number In case sequence number is not iterated correctly (last sent sequence number +1)

Gateway disconnects connection. In such a situation, Client Application should reestablish connection and resynchronize sequences. There is a possibility for Client Application to skip sequences by using GapFill message. Client Application can send GapFill with header:seqNum set to new_seqnum (it must be higher than seqNum which was expected) and continue starting from seqNum set to new_seqNum+1 in subsequent messages.

6.2.2. HEARTBEATS

The heartbeat message used by Gateway and Client Application to provide information about the communication line during time gaps of inactivity and to verify that interfaces at each end are accessible. The length of heartbeat interval is 4 seconds. The heartbeat send time (i.e. interval) is counted from the time of the last s) message.

The Logout procedure and closing the TCP/IP connection is done once the server noticed an inactivity for 3 heartbeat intervals.

6.2.3. REJECTIONS

Client Application message is rejected at the session level by sending a "Reject" message. The reason for rejecting is presented in the “rejectReason” field and includes:

- exceeding throughput limits (MaxThroughputExceeded),
- sending the wrong type of a message (InvalidMsgType),
- wrong expire field format (InvalidExpireTimePrecision).
- invalid settlement date in trade capture report (InvalidSettlementDate).

In the case of communication protocol errors, instead of rejecting the message, the connection is terminated (see next section).

6.3. Terminating a Connection

Client Application should terminate all used connections at the end of trading day by sending a Logout message. Then Gateway service closes TCP/IP connection.

If Client Application does not terminate a connection, Gateway does it automatically once it shuts down (a ConnectionClose message is sent).

During the trading day, in extraordinary cases, Gateway may initiate the termination of connection by sending the ConnectionClose message.

7. MESSAGES

Name	ID	Length	Description
BidOfferUpdate	31	53	Message send during the IPO to the sell side or during the Tender Offer to the buy side.
ConnectionClose	12	25	The ConnectionClose message confirms the termination of a session through the trading port service.
GapFill	34	24	Create a SeqNum gap between current SeqNum and header::seqNum
Heartbeat	13	24	Heartbeat message
Login	2	44	The login message authenticates a user establishing a connection to the trading port service. The login message must be the first message sent by the client application to request the initiation of a trading port session.
LoginResponse	3	35	The login response message. The result field describes the login status, indicating whether the login was successful or not (i.e. successful login).
Logout	11	24	The logout message is sent from the client application to terminate the communication session with the trading port.
LogoutResponse	14	24	The logout response message confirms the client logout message.
MarketMakerCommand	32	29	Market Maker command request.
MarketMakerCommandResponse	33	31	The response to the Market Maker command.
MassQuote	24	1208	Mass Quote
MassQuoteResponse	25	727	The response to a MassQuote message.
OrderAdd	4	175	Message used to add new orders to System.
OrderAddResponse	5	80	The message is a response to an OrderAdd message and includes the order execution status.
OrderCancel	6	67	Message used to cancel the previously submitted order.
OrderCancelResponse	7	56	The message is a response to an order cancel request and contains information about its execution, in particular whether the order to cancel was found or not.
OrderExecute	10	88	The message used to report trades between counterparties (i.e. generated when two or more orders are matched).
OrderMassCancel	29	63	Message used to cancel multiple existing orders.
OrderMassCancelResponse	30	72	Response to message used to cancel multiple existing orders.
OrderModify	8	107	Message used to modify the submitted order.

Name	ID	Length	Description
OrderModifyResponse	9	64	The response message for an OrderModify.
Reject	15	29	The reject message is sent by the trading port service when receiving an erroneous message that cannot be further processed.
RequestForExecution	28	29	The information for the MM that one of the quotes has been crossed.
TradeBust	23	28	Message used to inform about cancellation of previously accepted Trade
TradeCaptureReportDual	19	271	Trade Capture Report - dual sided.
TradeCaptureReportResponse	20	64	The message is a response to an Trade Capture Report message, containing the state of TCR execution.
TradeCaptureReportSingle	18	206	Trade Capture Report - single side.
TradingSessionStatus	35	33	The Trading Session Status provides information on the status of a trading session.

7.1. Header

Name	Type	Kind	Length	Description
length	MsgLength	Alias (u16)	2	Total length of the message.
msgType	MsgType	Enum	2	Type of the message (e.g. Login).
seqNum	SeqNum	Alias (u32)	4	Sequence number of the message added by the sender. Session messages such as Heartbeat, Login, LoginResponse, Logout, LogoutResponse, ConnectionClose should have seqNum set to 0. Oher (business) messages should have subsequent values in range <1 , 2^28-1>.
timestamp	Timestamp	Alias (u64)	8	Sending time.
sourceTimestamp	Timestamp	Alias (u64)	8	Time of the business event that triggered or generated the message

7.2. BussinesBidOfferUpdate

Name	Data type	Kind	Length	Description
header	Header	Struct	24	Message header.

Name	Data type	Kind	Length	Description
instrumentId	ElementId	Alias (u32)	4	Instrument ID.
updateType	BidOfferUpdateType	Enum	1	Indicates a type of the BidOfferUpdate message. **Ipo** (1): IPO. **TenderOffer** (2): Tender Offer.
totalBidSize	Quantity	Alias (u64)	8	Specifies the total bid size (number of Lots).
totalOfferSize	Quantity	Alias (u64)	8	Specifies the total offer size (number of Lots).
bidOrders	u32	Primitive	4	Number of bid orders.
offerOrders	u32	Primitive	4	Number of offer orders.

7.3. ConnectionClose

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
reason	ConnectionCloseReason	Enum	1	Connection close reason. **ProtocolError** (1): Invalid message or frame length. **InvalidSeqNum** (2): Message came with an incorrect sequence number. **EndOfDay** (3): The session day has come to an end. **SyncFail** (4): The synchronization of messages has failed. **AntiFloodingThresholdExceeded** (5): The second level of the throttling limit has been exceeded. **ConnectionConfigChanged** (6): Connection configuration has changed **CloseOps** (7): Service closed the connection because of an exchange operation (e.g. supervision operation). **Disconnect** (8): Service closed the connection because of an internal operation (e.g. grace period end).

7.4. GapFill

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.

7.5. Heartbeat

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.

7.6. Login

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
version	MsgVersion	Alias (u16)	2	Indicates the version of the protocol in which the message is defined.
token	Token	Array (AnsiChar)	8	The security data required for authentication, which is a token received by Exchange Member during the registration process.
connectionId	ConnectionId	Alias (u16)	2	ID of the connection.
nextExpectedSeqNum	SeqNum	Alias (u32)	4	Next expected message sequence number value to be received.
lastSentSeqNum	SeqNum	Alias (u32)	4	Last sent sequence number.

7.7. LoginResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
result	LoginResult	Enum	1	Login response status code.

Name	Type	Kind	Length	Description
				Name / **Value** / **Description** **Ok** / 1 / Successful login. **NotFound** / 2 / User not found. **InvalidToken** / 3 / Authorization failure. **AlreadyLoggedIn** / 4 / Already logged in. **AccountLocked** / 5 / Account is locked **LoginNotAllowed** / 6 / Login is currently unavailable due to reasons such as service unavailability. **ThrottlingTemporaryLock** / 8 / The login attempt failed due to exceeding the anti-flooding threshold.
nextExpectedSeqNum	SeqNum	Alias (u32)	4	Next expected message sequence number value to be received.
lastReplaySeqNum	SeqNum	Alias (u32)	4	Last replay sequence number.
sessionId	SessionId	Alias (u16)	2	ID of the session.

7.8. Logout

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.

7.9. LogoutResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.

7.10. MarketMakerCommand

Name	Data type	Kind	Length	Description
header	Header	Struct	24	Message header.
instrumentId	ElementId	Alias (u32)	4	Instrument ID.
action	CommandAction	Enum	1	The action for the Market Maker command request. **Name** / **Value** / **Description** **ChangeToHybridBuyOnly** / 1 / Transition of the instrument from the Hybrid phase to the Hybrid Buy Only phase. **ChangeToKnockOut** / 2 / Change of the instrument's phase to KnockOut. **RevokeKnockOut** / 3 / Cancellation of the KnockOut phase. New orders can be placed again

7.11. MarketMakerCommandResponse

Name	Data type	Kind	Length	Description
header	Header	Struct	24	Message header.
refId	u32	Primitive	4	A reference id to the Market Maker command request message.
result	CommandResult	Enum	1	Confirmation of achieving the intended action. If the action (e.g., Buy Only state) has already been achieved before the command, the result will be positive. **Name** / **Value** / **Description** **Success** / 1 / **Failure** / 2 /
rejectionCode	CommandRejectionCode	Enum	2	Please, see ***GPW WATS 2.03 Rejection Codes*** for a full list of rejection codes.

7.12. MassQuote

Name	Type	Kind	Length	Description
header	Header	Struct	24	Message header.
stpId	STPId	Alias (u8)	1	An ID assigned by the client used in Self Match Prevention mechanism. **Optional:** Never required, can be 0 to disable stp.
capacity	Capacity	Enum	1	Capacity of the party making the order (either principal or agency). **Agency** (1): Agency (mapped to AOTC). **Principal** (2): Principal (mapped to DEAL). **RisklessPrincipal** (3): Riskless Principal (mapped to MTCH)
account	Account	Array (AnsiChar)	16	Account mnemonic as agreed between buy and sell sides. **Optional:** Required for AccountType different than Missing, otherwise filled with 0x0.
accountType	AccountType	Enum	1	Type of account associated with the order. **Missing** (1): Account is missing. Account is expected to be filled with 0x00. **Customer** (2): Account is carried on customer side of the books. **House** (3): House trader.
flags	OrderFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive. **None** (0b0000) **LiquidityProvisionActivity** (0b0001): Order submitted by any firm providing liquidity to the market. **DirectOrSponsoredAccess** (0b0010): Order submitted by Sponsored client using Sponsored Access connectivity. **MarketMakerOrSpecialist** (0b0100): Order submitted as part of Liquidity Provision Activity by firm which have an agreement with Market.
mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant.

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not Applicable.</td></tr>
<tr><td>Algorithm</td><td>2</td><td>Short code of the algorithm used when submitting an order.</td></tr>
<tr><td>FirmOrLegalEntity</td><td>3</td><td>Qualifier if an order is submitted by a client being legal entity.</td></tr>
<tr><td>NaturalPerson</td><td>4</td><td>Qualifier if an order is submitted by a client being natural person</td></tr>
</table>
</td></tr>
<tr><td>mifidFields.executingTrader.shortCode</td><td>ShortCode</td><td>Alias (u32)</td><td>4</td><td>Short code of MiFID participant.</td></tr>
<tr><td>mifidFields.executingTrader.qualifier</td><td>PartyRoleQualifier</td><td>Enum</td><td>1</td><td>Qualifier of MiFID participant.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not Applicable.</td></tr>
<tr><td>Algorithm</td><td>2</td><td>Short code of the algorithm used when submitting an order.</td></tr>
<tr><td>FirmOrLegalEntity</td><td>3</td><td>Qualifier if an order is submitted by a client being legal entity.</td></tr>
<tr><td>NaturalPerson</td><td>4</td><td>Qualifier if an order is submitted by a client being natural person</td></tr>
</table>
</td></tr>
<tr><td>mifidFields.investmentDecisionMaker.shortCode</td><td>ShortCode</td><td>Alias (u32)</td><td>4</td><td>Short code of MiFID participant.</td></tr>
<tr><td>mifidFields.investmentDecisionMaker.qualifier</td><td>PartyRoleQualifier</td><td>Enum</td><td>1</td><td>Qualifier of MiFID participant.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not Applicable.</td></tr>
<tr><td>Algorithm</td><td>2</td><td>Short code of the algorithm used when submitting an order.</td></tr>
<tr><td>FirmOrLegalEntity</td><td>3</td><td>Qualifier if an order is submitted by a client being legal entity</td></tr>
<tr><td>NaturalPerson</td><td>4</td><td>Qualifier if an order is submitted by a client being natural person</td></tr>
</table>
</td></tr>
<tr><td>memo</td><td>Memo</td><td>Array (AnsiChar)</td><td>18</td><td>Free text.
Optional: Never required, if unused filled with 0x0.</td></tr>
<tr><td>clearingMemberCode</td><td>ClearingCode</td><td>Array (AnsiChar)</td><td>20</td><td>Clearing member code.
Optional: Never required, if unused filled with 0x0.</td></tr>
<tr><td>clearingMemberClearingIdentifier</td><td>ClearingIdentifier</td><td>Enum</td><td>1</td><td>Clearing member's clearing identifier.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NotApplicable</td><td>33</td><td>Not Applicable.</td></tr>
<tr><td>Lei</td><td>78</td><td>Legal Entity Identifier.</td></tr>
<tr><td>Custom</td><td>68</td><td>Custom clearing identifier.</td></tr>
</table>
</td></tr>
</table>

Name	Type	Kind	Length	Description
quotes.count	u8	Primitive	1	The number of quotes this message contains.
quotes.items	QuotesArray	Array (Quote)	1080	The array of quotes. Quote: **Name** / **Type** / **Kind** / **Length** / **Description** **instrumentId** / ElementId / Alias (u32) / 4 / ID of the instrument being traded. **bid.price** / Price / Alias (i64) / 8 / Indicates the price of the given order. **bid.quantity** / Quantity / Alias (u64) / 8 / Indicates the quantity of the instrument included in the order (number of Lots). **ask.price** / Price / Alias (i64) / 8 / Indicates the price of the given order. **ask.quantity** / Quantity / Alias (u64) / 8 / Indicates the quantity of the instrument included in the order (number of Lots).
feeStructureId	u8	Primitive	1	Optional identifier of a fee scheme for billing purposes. 1 to 99 - for internal purposes for Members 100 to 255 - for internal purposes for WSE (GPW). **Optional:** Never required, if unused 0
interestedParty	InterestedParty	Array (AnsiChar)	8	3rd party interested in this order or trade. **Optional:** Never required, if unused filled with 0x0.
quoteId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the mass quote. **Optional:** Never required, if unused filled with 0x0.

7.13. MassQuoteResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Message header.
massQuoteId	OrderId	Alias (u64)	8	Quote id.
responses.count	u8	Primitive	1	The number of quote order responses in this message.
responses.items	QuoteOrderResponseArray	Array (QuoteOrderResponse)	690	The array of quote order responses. QuoteOrderResponse:

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Data type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>instrumentId</td><td>ElementId</td><td>Alias (u32)</td><td>4</td><td>ID of the instrument being traded.</td></tr>
<tr><td>bidOrderId</td><td>OrderId</td><td>Alias (u64)</td><td>8</td><td>Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.</td></tr>
<tr><td>askOrderId</td><td>OrderId</td><td>Alias (u64)</td><td>8</td><td>Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.</td></tr>
<tr><td>status</td><td>OrderStatus</td><td>Enum</td><td>1</td><td>Status of the requested action..
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>New</td><td>1</td><td>Operation accepted (New order, Cancellation, Aggressive order execution)</td></tr>
<tr><td>Rejected</td><td>3</td><td>Operation rejected.</td></tr>
</table></td></tr>
<tr><td>reason</td><td>OrderRejectionReason</td><td>Enum</td><td>2</td><td>Reason for rejecting the given order. Please, see GPW WATS 2.03 Rejection Codes for a full list of rejection codes.</td></tr>
</table>
</td></tr>
<tr><td>status</td><td>MassQuoteStatus</td><td>Enum</td><td>1</td><td>Status of the given mass quote order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Accepted</td><td>1</td><td>Mass quote acknowledged by system.</td></tr>
<tr><td>Rejected</td><td>2</td><td>Mass quote rejected.</td></tr>
</table></td></tr>
<tr><td>reason</td><td>MassQuoteRejectionReason</td><td>Enum</td><td>2</td><td>Reason for rejecting the given mass quote order. Please, see GPW WATS 2.03 Rejection Codes for a full list of rejection codes.</td></tr>
<tr><td>feeStructureId</td><td>u8</td><td>Primitive</td><td>1</td><td>Optional identifier of a fee scheme for billing purposes.
1 to 99 - for internal purposes for Members
100 to 255 - for internal purposes for WSE (GPW)</td></tr>
</table>

7.14. OrderAdd

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
stpId	STPId	Alias (u8)	1	ID assigned by the client used in the Self Trade Prevention mechanism. **Optional:** Never required, can be 0 to disable stp
instrumentId	ElementId	Alias (u32)	4	ID of the instrument being traded.
orderType	OrderType	Enum	1	Indicates the order type. **Limit** = 1 — Limit order type. **Market** = 2 — Market order type. **MarketToLimit** = 3 — Market to limit order type. **Iceberg** = 4 — Iceberg order type. **StopLimit** = 5 — Stop limit order type. **StopLoss** = 6 — Stop loss order type.
timeInForce	TimeInForce	Enum	1	Indicates the order's time in force (e.g. GTC). **Day** = 1 — A day order is valid until the end of the trading day. **GTC** = 2 — A GTC order is good till canceled. **IOC** = 3 — An Immediate or Cancel order must be filled immediately or canceled. **FOK** = 4 — A Fill or Kill order must be immediately fully filled or canceled. **VFA** = 5 — Valid For Auction. **GTD** = 6 — A Good Till Date order must be filled before timestamp provided in `Expire` field or canceled. **VFC** = 7 — Valid For Closing. **GTT** = 8 — A Good Till Time order must be filled before timestamp provided in `Expire` field or canceled within the day of submission.

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>side</td><td>OrderSide</td><td>Enum</td><td>1</td><td>Indicates the order's side (buy or sell).
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Buy</td><td>1</td><td>Indicates a buy-side order.</td></tr>
<tr><td>Sell</td><td>2</td><td>Indicates a sell-side order.</td></tr>
</table></td></tr>
<tr><td>price</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Indicates the price per unit (smallest portion) of instrument.</td></tr>
<tr><td>triggerPrice</td><td>Price</td><td>Alias (Number)</td><td>8</td><td>Indicates the trigger price (Last Trade Price - LTP) after which the order should be added to the order book (per unit(smallest portion) of instrument).
Optional: Required for OrderType::StopLimit and OrderType::StopLoss, otherwise 0.</td></tr>
<tr><td>quantity</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Indicates the quantity of the instrument included in the order (number of Lots).</td></tr>
<tr><td>displayQty</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Used only for iceberg order. The quantity to be displayed (number of Lots).
Optional: Required for OrderType::Iceberg, otherwise 0.</td></tr>
<tr><td>capacity</td><td>Capacity</td><td>Enum</td><td>1</td><td>Capacity of the party making the order (either principal or agency).
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Agency</td><td>1</td><td>Agency (mapped to AOTC).</td></tr>
<tr><td>Principal</td><td>2</td><td>Principal (mapped to DEAL).</td></tr>
<tr><td>RisklessPrincipal</td><td>3</td><td>Riskless Principal (mapped to MTCH)</td></tr>
</table></td></tr>
<tr><td>account</td><td>Account</td><td>Array (AnsiChar)</td><td>16</td><td>Account number.
Optional: Required for AccountType different than Missing, otherwise filled with 0x0.</td></tr>
<tr><td>accountType</td><td>AccountType</td><td>Enum</td><td>1</td><td>Type of account associated with the order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Missing</td><td>1</td><td>Account is missing. Account is expected to be filled with 0x00.</td></tr>
<tr><td>Customer</td><td>2</td><td>Account is carried on customer side of the books.</td></tr>
<tr><td>House</td><td>3</td><td>House trader.</td></tr>
</table></td></tr>
<tr><td>flags</td><td>OrderFlags</td><td>Enum</td><td>1</td><td>Flags raised on an order in compliance with the MiFID directive.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b0000</td><td></td></tr>
<tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td>Order submitted by any firm providing liquidity to the market.</td></tr>
</table></td></tr>
</table>

Name	Type	Kind	Length	Description
				DirectOrSponsoredAccess 0b0010 Order submitted by Sponsored client using Sponsored Access connectivity. **MarketMakerOrSpecialist** 0b0100 Order submitted as part of Liquidity Provision Activity by firm which have an agreement with Market.
mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
expire	Timestamp	Alias (u64)	8	Expiration time indicating the validity of the order - relevant only when timeInForce is set to GTD (Good Till Date) or GTT (Good Till Time). **Optional:** Required for TimeInForce::GTD or TimeInForce::GTT, otherwise 0.
memo	Memo	Array (AnsiChar)	18	Free text. **Optional:** Never required, if unused filled with 0x0.

Name	Type	Kind	Length	Description
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.
clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code. **Optional:** Never required, if unused filled with 0x0.
clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier. **Name** / **Value** / **Description** **NotApplicable** / 33 / Not Applicable. **Lei** / 78 / Legal Entity Identifier. **Custom** / 68 / Custom clearing identifier.
execInst	ExecInst	Enum	1	Instructions for order handling on exchange trading floor. **Name** / **Value** / **Description** **None** / 0b00000000 / **CancelOnConnectionLoss** / 0b00000001 / Cancel on connection loss
feeStructureId	u8	Primitive	1	Optional identifier of a fee scheme for billing purposes. 1 to 99 - for internal purposes for Members. 100 to 255 - for internal purposes for WSE (GPW). **Optional:** Never required, if unused 0.
interestedParty	InterestedParty	Array (AnsiChar)	8	3rd party interested in this order or trade. **Optional:** Never required, if unused filled with 0x0.

7.15. OrderAddResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
publicOrderId	PublicOrderId	Alias (u64)	8	Unique for single trading day order identifier assigned by the trading system and shared publicly in market data (public information).

Name	Type	Kind	Length	Description
displayQty	Quantity	Alias (u64)	8	Used only for iceberg order. The quantity to be displayed (number of Lots).
filled	Quantity	Alias (u64)	8	Indicates the quantity of the order which has already been filled (number of Lots).
status	OrderStatus	Enum	1	Status of the given order. **New** (1): New order. **Cancelled** (2): Order canceled. **Rejected** (3): Order rejected. **Filled** (4): Order filled. **PartiallyFilled** (5): Order partially filled. **Expired** (6): Order expired.
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order. Please, see ***GPW WATS 2.03 Rejection Codes*** for a full list of rejection codes.
execTypeReason	ExecTypeReason	Enum	1	Describes why an order was executed and the events related to its lifecycle. **NA** (1): Not applicable. **CancelOnDisconnect** (2): Order cancelled by the canceller on client disconnect. **Expired** (3): Order cancelled by the canceller due to expiration. **Triggered** (4): Notification of Stop order activation. **CancelOnSuspension** (5): Order cancelled by canceller due to instrument suspension. **OrderRestatement** (6): Order reinstated. **IcebergOrderRefill** (7): Iceberg order refill. **CancelByStp** (8): Order cancelled due to self-trade prevention. **CancelByCorporateAction** (9): Order cancelled due to submitted Corporate Action. **CancelByMassCancel** (10): Order cancelled due to mass cancel

Name	Type	Kind	Length	Description		
				CancelIocFokOrder	11	Order cancelled due to IOC/FOK.
				CancelByMarketOperations	12	Order cancelled due to market operations.
				Replaced	13	Order replaced.
				FirstTradeOnAggressiveOrder	14	First trade on aggressive order.
				Rejected	15	Order rejected.
				CancelonBuyOnlyStateEntry	16	Cancel on BuyOnly state entry
				CancelonKnockedOutStateEntry	17	Cancel on KnockedOut state entry
				CancelByRiskManagement	18	Order cancelled due to exceeding RMA post trade limit.
				CancelOnDcDisconnect	19	Order cancelled due to drop copy disconnect
				VfaActivated	20	Notification of VFA/VFC order activation.
				HybridActivated	21	Notification of Hybrid order activation.
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.		

7.16. OrderCancel

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** (1) – Not Applicable. **Algorithm** (2) – Short code of the algorithm used when submitting an order.

Name	Type	Kind	Length	Description
				FirmOrLegalEntity 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.

7.17. OrderCancelResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>status</td><td>OrderStatus</td><td>Enum</td><td>1</td><td>Status of the given order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Cancelled</td><td>2</td><td>Order canceled.</td></tr>
<tr><td>Rejected</td><td>3</td><td>Order rejected.</td></tr>
<tr><td>Expired</td><td>6</td><td>Order expired.</td></tr>
</table></td></tr>
<tr><td>reason</td><td>OrderRejectionReason</td><td>Enum</td><td>2</td><td>Reason for rejecting the given order. Please, see GPW WATS 2.03 Rejection Codes for a full list of rejection codes.</td></tr>
<tr><td>execTypeReason</td><td>ExecTypeReason</td><td>Enum</td><td>1</td><td>Describes why an order was executed and the events related to its lifecycle.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>CancelOnDisconnect</td><td>2</td><td>Order cancelled by the canceller on client disconnect.</td></tr>
<tr><td>Expired</td><td>3</td><td>Order cancelled by the canceller due to expiration.</td></tr>
<tr><td>CancelOnSuspension</td><td>5</td><td>Order cancelled by canceller due to instrument suspension.</td></tr>
<tr><td>OrderRestatement</td><td>6</td><td>Order reinstated.</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>CancelByStp</td><td>8</td><td>Order cancelled due to self-trade prevention.</td></tr>
<tr><td>CancelByCorporateAction</td><td>9</td><td>Order cancelled due to submitted Corporate Action.</td></tr>
<tr><td>CancelByMassCancel</td><td>10</td><td>Order cancelled due to mass cancel</td></tr>
<tr><td>CancelByMarketOperations</td><td>12</td><td>Order cancelled due to market operations.</td></tr>
<tr><td>Rejected</td><td>15</td><td>Order rejected.</td></tr>
<tr><td>CancelonBuyOnlyStateEntry</td><td>16</td><td>Cancel on BuyOnly state entry</td></tr>
<tr><td>CancelonKnockedOutStateEntry</td><td>17</td><td>Cancel on KnockedOut state entry</td></tr>
<tr><td>CancelByRiskManagement</td><td>18</td><td>Order cancelled due to exceeding RMA post trade limit.</td></tr>
<tr><td>CancelOnDcDisconnect</td><td>19</td><td>Order cancelled due to drop copy disconnect</td></tr>
</table></td></tr>
</table>

Name	Type	Kind	Length	Description
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.

7.18. OrderExecute

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument being traded.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
tradeId	TradeId	Alias (u32)	4	ID of the trade (TVTIC).
price	Price	Alias (i64)	8	Price per unit (smallest portion) of the given trade .
quantity	Quantity	Alias (u64)	8	Quantity of the instrument involved in the given trade (number of Lots).
leavesQty	Quantity	Alias (u64)	8	How much of the given security is left on Market after the trade is concluded (number of Lots).
liquidityIndicator	LiquidityIndicator	Enum	1	Liquidity indicator **Name** / **Value** / **Description** **BuyOrderExecution** / 1 / Buy order execution (for buy order it means 'removed liquidity' and for sell order it means 'added liquidity) **SellOrderExecution** / 2 / Sell order execution (for buy order it means 'added liquidity' and for sell order it means 'removed liquidity') **AuctionExecution** / 3 / Auction execution (for buy and sell orders it means execution during an auction) **NotApplicable (not used)** / 4 / Not applicable (value not used)
currency	Currency	Enum	2	Currency - ISO 4217 code (e.g. PLN)
side	OrderSide	Enum	1	Indicates the order's side (buy or sell). **Name** / **Value** / **Description**

Name	Type	Kind	Length	Description
				Buy 1 Indicates a buy-side order. **Sell** 2 Indicates a sell-side order.
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.

7.19. OrderMassCancel

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
massCancelRequestType	MassCancelRequestType	Enum	1	Mass cancel request type. **CancelForSecurity** 1 Cancel orders for a security. **CancelAllOrders** 7 Cancel all orders. **CancelOrdersForMarketSegment** 9 Cancel orders for a market segment.
targetPartyRole	TargetPartyRole	Enum	1	Target party role filter selection field. **None** 0 No filtering. **LiquidityProvider** 35 Liquidity provider. **SenderLocation** 54 Sender location. **MarketMaker** 66 Market maker. **ClientID** 3 Client ID. **ExecutingTrader** 12 Executing trader. **InvestmentDecisionMaker** 122 Investment decision maker.
targetPartyId	u32	Primitive	4	Used to identify the party targeted for the action specified in the message.

Name	Type	Kind	Length	Description
				Optional: : Required for TargetPartyRole::SenderLocation/ClientID/ExecutingTrader/InvestmentDecisionMaker, otherwise 0.
marketSegmentId	ElementId	Alias (u32)	4	Identifies Market segment for request type CancelOrdersForMarketSegment. **Optional :** Required for MassCancelRequestType::CancelOrdersForMarketSegment, otherwise 0.
instrumentId	ElementId	Alias (u32)	4	**Optional :** Required for MassCancelRequestType::CancelForSecurity, otherwise 0.
executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **Name** / **Value** / **Description** **NA** / 1 / Not Applicable. **Algorithm** / 2 / Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** / 3 / Qualifier if an order is submitted by a client being legal entity **NaturalPerson** / 4 / Qualifier if an order is submitted by a client being natural person
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the mass cancel. **Optional :** Never required, if unused filled with 0x0.

7.20. OrderMassCancelResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
totalAffectedOrders	u64	Primitive	8	Total number of orders affected by the mass cancel request.
massCancelRequestType	MassCancelRequestType	Enum	1	Mass cancel request type. **Name** / **Value** / **Description** **CancelForSecurity** / 1 / Cancel orders for a security. **CancelAllOrders** / 7 / Cancel all orders. **CancelOrdersForMarketSegment** / 9 / Cancel orders for a market segment.
massCancelId	OrderId	Alias (u64)	8	Mass cancel request ID.

Name	Type	Kind	Length	Description
targetPartyRole	TargetPartyRole	Enum	1	Target party role filter selection field. **Name** / **Value** / **Description** **None** / 0 / No filtering. **LiquidityProvider** / 35 / Liquidity provider. **SenderLocation** / 54 / Sender location. **MarketMaker** / 66 / Market maker. **ClientID** / 3 / Client ID. **ExecutingTrader** / 12 / Executing trader. **InvestmentDecisionMaker** / 122 / Investment decision maker.
targetPartyId	u32	Primitive	4	Used to identify the party targeted for the action specified in the message.
marketSegmentId	ElementId	Alias (u32)	4	Identifies Market segment for request type CancelOrdersForMarketSegment.
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given mass order cancel request. Please, see ***GPW WATS 2.03 Rejection Codes*** for a full list of rejection codes.
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the mass cancel. **Optional :** Never required, if unused filled with 0x0.

7.21. OrderModify

	Type	Kind	Length	Description
header	Header	Struct	24	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
price	Price	Alias (Number)	8	Indicates the price per unit (smallest portion) of the given order
triggerPrice	Price	Alias (Number)	8	Indicates the trigger price (Last Trade Price - LTP) after which the order should be added to the order book (per unit (smallest portion) of instrument). **Optional:** Required for OrderType::StopLimit and OrderType::StopLoss before activation, otherwise 0.

	Type	Kind	Length	Description
quantity	Quantity	Alias (u64)	8	Indicates the quantity of the instrument included in the order (number of Lots).
displayQty	Quantity	Alias (u64)	8	Used only for iceberg order. The quantity to be displayed (number of Lots). **Optional:** Required for OrderType::Iceberg, otherwise 0.
expire	Timestamp	Alias (u64)	8	Expiration time indicating the validity of the order - relevant only when timeInForce is set to GTD (Good Till Date). **Optional:** Required for TimeInForce::GTD, otherwise 0
mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **Name** / **Value** / **Description** **NA** / 1 / Not Applicable. **Algorithm** / 2 / Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** / 3 / Qualifier if an order is submitted by a client being legal entity **NaturalPerson** / 4 / Qualifier if an order is submitted by a client being natural person
mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **Name** / **Value** / **Description** **NA** / 1 / Not Applicable. **Algorithm** / 2 / Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** / 3 / Qualifier if an order is submitted by a client being legal entity **NaturalPerson** / 4 / Qualifier if an order is submitted by a client being natural person
mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **Name** / **Value** / **Description** **NA** / 1 / Not Applicable.

	Type	Kind	Length	Description		
				Algorithm	2	Short code of the algorithm used when submitting an order.
				FirmOrLegalEntity	3	Qualifier if an order is submitted by a client being legal entity
				NaturalPerson	4	Qualifier if an order is submitted by a client being natural person
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.		

7.22. OrderModifyResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
orderId	OrderId	Alias (u64)	8	Unique for each trading day order identifier based on the sequence number of order message, bulk sequence number, session ID and connection ID.
filled	Quantity	Alias (u64)	8	Indicates the quantity of the order which has already been filled (number of Lots).
status	OrderStatus	Enum	1	Status of the given order. **Name** / **Value** / **Description** **New** / 1 / New order. **Rejected** / 3 / Order rejected. **PartiallyFilled** / 5 / Order partially filled.
priorityFlag	PriorityFlag	Enum	1	Indicates whether the priority flag is lost or retained after modification. **Name** / **Value** / **Description** **Lost** / 1 / The priority flag was lost. **Retained** / 2 / The priority flag was retained.
reason	OrderRejectionReason	Enum	2	Reason for rejecting the given order. Please, see ***GPW WATS 2.03 Rejection Codes*** for a full list of rejection codes.
clientOrderId	ClientOrderId	Array (AnsiChar)	20	Arbitrary user provided value associated with the order. **Optional:** Never required, if unused filled with 0x0.

7.23. REJECT

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
refSeqNum	SeqNum	Alias (u32)	4	Sequence number of the rejected message.
rejectReason	RejectReason	Enum	1	Reject reason.

Name	Value	Description
NA	0	Not applicable
MaxThroughputExceeded	1	Max throughput exceeded.
InvalidMsgType	2	Invalid message type.
InvalidExpireTimePrecision	3	Invalid expire time precision.
InvalidSettlementDate	4	Invalid settlement date.
SettlementDateRequired	5	Settlement date required.
TradeReportIdRequired	6	Trade report ID required.
MissingReportIdSecondaryTradeReportIdOrTradeReportRefId	7	Missing report id (SecondaryTradeReportId or TradeReportRefId).
InvalidTradeId	8	Invalid trade ID.
InvalidAlgorithmicTradeIndicator	9	Invalid algorithmic trade indicator.
InvalidTradeReportId	10	Invalid Trade report ID.
InvalidGapFillSeqNum	11	GapFill seqNum should be higher than current seqNum

7.24. REQUESTFOREXECUTION

Name	Type	Kind	Length	Description
header	Header	Struct	24	Message header.
instrumentId	ElementId	Alias (u32)	4	Instrument ID
reason	RequestForExecutionReason	Enum	1	The reason for this RFE.

Name	Type	Kind	Length	Description
				Name / **Value** / **Description** **PassiveQuote** / 1 / **AggressiveQuote** / 2 /

7.25. TradeBust

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.

7.26. TradeCaptureReportDual

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.
tradeReportId	TradeReportId	Array (AnsiChar)	21	Unique identifier of trade capture report. Last character is reserved for exchange-assigned identifiers.
secondaryTradeReportId	OrderId	Alias (u64)	8	ID of the trade capture report.
tradeId	TradeId	Alias	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty (TVTIC).

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td>(u32)</td><td></td><td></td></tr>
<tr><td>tradeReportTransType</td><td>TradeReportTransType</td><td>Enum</td><td>1</td><td>Identifies Trade Report message transaction type.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>New</td><td>1</td><td>New.</td></tr>
<tr><td>Cancel</td><td>2</td><td>Cancel.</td></tr>
<tr><td>Replace</td><td>3</td><td>Replace.</td></tr>
</table></td></tr>
<tr><td>tradeReportType</td><td>TradeReportType</td><td>Enum</td><td>1</td><td>Type of Trade Report.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Submit</td><td>1</td><td>Submit</td></tr>
<tr><td>Alleged</td><td>2</td><td>Alleged</td></tr>
<tr><td>Accept</td><td>3</td><td>Accept</td></tr>
<tr><td>Decline</td><td>4</td><td>Decline</td></tr>
<tr><td>TradeReportCancel</td><td>7</td><td>Trade Report Cancel</td></tr>
<tr><td>TradeBreak</td><td>8</td><td>Trade Break</td></tr>
</table></td></tr>
<tr><td>tradeType</td><td>TradeType</td><td>Enum</td><td>1</td><td>Type of trade.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>PrivatelyNegotiatedTrade</td><td>22</td><td>Privately negotiated trade.</td></tr>
<tr><td>BlockTrade</td><td>38</td><td>Block trade.</td></tr>
</table></td></tr>
<tr><td>algorithmicTradeIndicator</td><td>AlgorithmicTradeIndicator</td><td>Enum</td><td>1</td><td>Indicates algorithmic trade.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>NonAlgorithmicTrade</td><td>2</td><td>Non-algorithmic trade.</td></tr>
<tr><td>AlgorithmicTrade</td><td>3</td><td>Algorithmic trade.</td></tr>
</table></td></tr>
</table>

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td>execType</td><td>ExecType</td><td>Enum</td><td>1</td><td>Type of execution being reported. Uses subset of ExecType for trade capture reports.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>Rejected</td><td>8</td><td>Rejected.</td></tr>
<tr><td>Expired</td><td>12</td><td>Expire</td></tr>
<tr><td>Trade</td><td>15</td><td>Trade.</td></tr>
<tr><td>TradeCorrect (for future use)</td><td>16</td><td>Trade Correct.</td></tr>
<tr><td>TradeCancel</td><td>17</td><td>Trade Cancel.</td></tr>
</table></td></tr>
<tr><td>tradeReportRefId</td><td>TradeReportRefID</td><td>Array (AnsiChar)</td><td>21</td><td>Reference identifier used with Cancel and Replace transaction types. The TradeReportID that is being referenced for trade correction or cancelation.</td></tr>
<tr><td>lastQty</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Quantity (e.g. shares) bought/sold on this (last) fill.</td></tr>
<tr><td>lastPx</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Price per unit(smallest portion) of instrument of this (last) fill.</td></tr>
<tr><td>settlementDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Settlement date of the trade is equal to current date plus actual settlement offset calendar days.
Optional: required if tradeType is BlockTrade, required to be 0 if tradeType is PrivatelyNegotiatedTrade</td></tr>
<tr><td>tcrPartyBuy.flags</td><td>OrderFlags</td><td>Enum</td><td>1</td><td>Flags raised on an order in compliance with the MiFID directive.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b0000</td><td>Order submitted by any firm providing liquidity to the market.</td></tr>
<tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td>Order submitted by Sponsored client using Sponsored Access connectivity.</td></tr>
<tr><td>DirectOrSponsoredAccess</td><td>0b0010</td><td>Order submitted as part of Liquidity Provision Activity by firm which have an agreement with Market.</td></tr>
</table></td></tr>
</table>

Name	Type	Kind	Length	Description
				MarketMakerOrSpecialist 0b0100 Order submitted by any firm providing liquidity to the market.
tcrPartyBuy.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartyBuy.mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrPartyBuy.mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartyBuy.mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrPartyBuy.mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartyBuy.mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable.

Name	Type	Kind	Length	Description
				Algorithm 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrPartyBuy.clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code. **Optional:** required if clearingMemberClearingIdentifier is different than NotApplicable
tcrPartyBuy.clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier. **Optional:** required if clearingMemberCode is present **NotApplicable** 33 Not Applicable. **Lei** 78 Legal Entity Identifier. **Custom** 68 Custom clearing identifier.
tcrPartyBuy.account	Account	Array (AnsiChar)	16	Account number. **Optional**
tcrPartyBuy.accountType	AccountType	Enum	1	Type of account associated with the order. **Missing** 1 Account is missing. Account is expected to be filled with 0x00. **Customer** 2 Account is carried on customer side of the books. **House** 3 House trader.
tcrPartyBuy.orderCapacity	Capacity	Enum	1	Designates the capacity of the firm placing the order. **Agency** 1 Agency (mapped to AOTC). **Principal** 2 Principal (mapped to DEAL).

Name	Type	Kind	Length	Description
				RisklessPrincipal | 3 | Riskless Principal (mapped to MTCH)
tcrPartyBuy.feeStructureId	u8	Primitive	1	Optional identifier of a fee scheme for billing purposes. **Optional**
tcrPartyBuy.interestedParty	InterestedParty	Array (AnsiChar)	8	3rd party interested in this order or trade. **Optional**
tcrPartyBuy.memo	Memo	Array (AnsiChar)	18	Free text. **Optional**
tcrPartySell.flags	OrderFlags	Enum	1	Flags raised on an order in compliance with the MiFID directive. **None** | 0b0000 | Order submitted by any firm providing liquidity to the market. **LiquidityProvisionActivity** | 0b0001 | Order submitted by Sponsored client using Sponsored Access connectivity. **DirectOrSponsoredAccess** | 0b0010 | Order submitted as part of Liquidity Provision Activity by firm which have an agreement with Market. **MarketMakerOrSpecialist** | 0b0100 | Order submitted by any firm providing liquidity to the market.
tcrPartySell.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartySell.mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** | 1 | Not Applicable. **Algorithm** | 2 | Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** | 3 | Qualifier if an order is submitted by a client being legal entity **NaturalPerson** | 4 | Qualifier if an order is submitted by a client being natural person

Name	Type	Kind	Length	Description
tcrPartySell.mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartySell.mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** (1) – Not Applicable. **Algorithm** (2) – Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** (3) – Qualifier if an order is submitted by a client being legal entity **NaturalPerson** (4) – Qualifier if an order is submitted by a client being natural person
tcrPartySell.mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrPartySell.mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** (1) – Not Applicable. **Algorithm** (2) – Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** (3) – Qualifier if an order is submitted by a client being legal entity **NaturalPerson** (4) – Qualifier if an order is submitted by a client being natural person
tcrPartySell.clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code. **Optional:** required if clearingMemberClearingIdentifier is different than NotApplicable
tcrPartySell.clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier. **Optional:** required if clearingMemberCode is present Name / Value / Description

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><td>NotApplicable</td><td>33</td><td>Not Applicable.</td></tr>
<tr><td>Lei</td><td>78</td><td>Legal Entity Identifier.</td></tr>
<tr><td>Custom</td><td>68</td><td>Custom clearing identifier.</td></tr>
</table>
</td></tr>
<tr><td>tcrPartySell.account</td><td>Account</td><td>Array (AnsiChar)</td><td>16</td><td>Account number.
Optional</td></tr>
<tr><td>tcrPartySell.accountType</td><td>AccountType</td><td>Enum</td><td>1</td><td>Type of account associated with the order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Missing</td><td>1</td><td>Account is missing. Account is expected to be filled with 0x00.</td></tr>
<tr><td>Customer</td><td>2</td><td>Account is carried on customer side of the books.</td></tr>
<tr><td>House</td><td>3</td><td>House trader.</td></tr>
</table>
</td></tr>
<tr><td>tcrPartySell.orderCapacity</td><td>Capacity</td><td>Enum</td><td>1</td><td>Designates the capacity of the firm placing the order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Agency</td><td>1</td><td>Agency (mapped to AOTC).</td></tr>
<tr><td>Principal</td><td>2</td><td>Principal (mapped to DEAL).</td></tr>
<tr><td>RisklessPrincipal</td><td>3</td><td>Riskless Principal (mapped to MTCH)</td></tr>
</table>
</td></tr>
<tr><td>tcrPartySell.feeStructureId</td><td>u8</td><td>Primitive</td><td>1</td><td>Optional identifier of a fee scheme for billing purposes.
Optional</td></tr>
<tr><td>tcrPartySell.interestedParty</td><td>InterestedParty</td><td>Array (AnsiChar)</td><td>8</td><td>3rd party interested in this order or trade.
Optional</td></tr>
<tr><td>tcrPartySell.memo</td><td>Memo</td><td>Array (AnsiChar)</td><td>18</td><td>Free text.
Optional</td></tr>
</table>

7.27. TradeCaptureReportResponse

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty.
tradeReportId	TradeReportId	Array (AnsiChar)	21	Unique identifier of trade capture report.
secondaryTradeReportId	OrderId	Alias (u64)	8	ID of the trade capture report.
status	TcrStatus	Enum	1	Status of the given order. **New** — 1 — New. **Accepted** — 2 — Accepted. **Rejected** — 3 — Rejected. **Cancelled** — 4 — Cancelled.
reason	TcrRejectionReason	Enum	2	Reason for rejecting the given TCR. Please, see ***GPW WATS 2.03 Rejection Codes*** for a full list of rejection codes.

7.28. TradeCaptureReportSingle

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
instrumentId	ElementId	Alias (u32)	4	ID of the instrument included in the order.
tradeReportId	TradeReportId	Array (AnsiChar)	21	Unique identifier of the trade capture report.
secondaryTradeReportId	OrderId	Alias (u64)	8	ID of the trade capture report. **Optional:** required if tradeReportTransType is Replace or Cancel and tradeReportRefId is not set

Name	Type	Kind	Length	Description
tradeId	TradeId	Alias (u32)	4	The unique ID assigned to the trade entity once it is received or matched by the exchange or central counterparty (TVTIC).
tradeReportTransType	TradeReportTransType	Enum	1	Identifies Trade Report message transaction type. **Name** / **Value** / **Description** **New** / 1 / New. **Cancel** / 2 / Cancel. **Replace** / 3 / Replace.
tradeReportType	TradeReportType	Enum	1	Type of Trade Report. **Name** / **Value** / **Description** **Submit** / 1 / Submit **Alleged** / 2 / Alleged **Accept** / 3 / Accept **Decline** / 4 / Decline **TradeReportCancel** / 7 / Trade Report Cancel **TradeBreak** / 8 / Trade Break
tradeType	TradeType	Enum	1	Type of trade. **Name** / **Value** / **Description** **PrivatelyNegotiatedTrade** / 22 / Privately negotiated trade. **BlockTrade** / 38 / Block trade.
algorithmicTradeIndicator	AlgorithmicTradeIndicator	Enum	1	Indicates algorithmic trader. **Name** / **Value** / **Description** **NA** / 1 / Not applicable. **NonAlgorithmicTrade** / 2 / Non-algorithmic trade. **AlgorithmicTrade** / 3 / Algorithmic trade.
execType	ExecType	Enum	1	Type of execution being reported. Uses subset of ExecType for trade capture reports.

<table>
<tr><th>Name</th><th>Type</th><th>Kind</th><th>Length</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td></td><td>
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>NA</td><td>1</td><td>Not applicable.</td></tr>
<tr><td>Rejected</td><td>8</td><td>Rejected.</td></tr>
<tr><td>Expired</td><td>12</td><td>Expired</td></tr>
<tr><td>Trade</td><td>15</td><td>Trade.</td></tr>
<tr><td>TradeCorrect (for future use)</td><td>16</td><td>Trade Correct.</td></tr>
<tr><td>TradeCancel</td><td>17</td><td>Trade Cancel.</td></tr>
</table>
</td></tr>
<tr><td>tradeReportRefId</td><td>TradeReportRefID</td><td>Array (AnsiChar)</td><td>21</td><td>Reference identifier used with Cancel and Replace transaction types. The TradeReportID that is being referenced for trade correction or cancelation.
Optional: required if tradeReportTransType is Cancel and secondaryTradeReportId is not set</td></tr>
<tr><td>lastQty</td><td>Quantity</td><td>Alias (u64)</td><td>8</td><td>Quantity (e.g. shares) bought/sold on this (last) fill.</td></tr>
<tr><td>lastPx</td><td>Price</td><td>Alias (i64)</td><td>8</td><td>Price per unit(smallest portion) of instrument of this (last) fill.</td></tr>
<tr><td>settlementDate</td><td>Date</td><td>Alias (u32)</td><td>4</td><td>Settlement date of the trade is equal to current date plus actual settlement offset calendar days.</td></tr>
<tr><td>side</td><td>OrderSide</td><td>Enum</td><td>1</td><td>Side of order.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>Buy</td><td>1</td><td>Indicates a buy-side order.</td></tr>
<tr><td>Sell</td><td>2</td><td>Indicates a sell-side order.</td></tr>
</table>
</td></tr>
<tr><td>counterpartyCode</td><td>ParticipantCode</td><td>Array (AnsiChar)</td><td>16</td><td>Participant code of the counterparty.</td></tr>
<tr><td>tcrParty.flags</td><td>OrderFlags</td><td>Enum</td><td>1</td><td>Flags raised on an order in compliance with the MiFID directive.
<table>
<tr><th>Name</th><th>Value</th><th>Description</th></tr>
<tr><td>None</td><td>0b0000</td><td></td></tr>
<tr><td>LiquidityProvisionActivity</td><td>0b0001</td><td>Order submitted by any firm providing liquidity to the market.</td></tr>
</table>
</td></tr>
</table>

Name	Type	Kind	Length	Description
				DirectOrSponsoredAccess 0b0010 Order submitted by Sponsored client using Sponsored Access connectivity. **MarketMakerOrSpecialist** 0b0100 Order submitted as part of Liquidity Provision Activity by firm which have an agreement with Market.
tcrParty.mifidFields.client.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrParty.mifidFields.client.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrParty.mifidFields.executingTrader.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrParty.mifidFields.executingTrader.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable. **Algorithm** 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrParty.mifidFields.investmentDecisionMaker.shortCode	ShortCode	Alias (u32)	4	Short code of MiFID participant.
tcrParty.mifidFields.investmentDecisionMaker.qualifier	PartyRoleQualifier	Enum	1	Qualifier of MiFID participant. **NA** 1 Not Applicable.

Name	Type	Kind	Length	Description
				Algorithm 2 Short code of the algorithm used when submitting an order. **FirmOrLegalEntity** 3 Qualifier if an order is submitted by a client being legal entity **NaturalPerson** 4 Qualifier if an order is submitted by a client being natural person
tcrParty.clearingMemberCode	ClearingCode	Array (AnsiChar)	20	Clearing member code. **Optional:** required if clearingMemberClearingIdentifier is different than NotApplicable
tcrParty.clearingMemberClearingIdentifier	ClearingIdentifier	Enum	1	Clearing member's clearing identifier. **Optional:** required if clearingMemberCode is present **NotApplicable** 33 Not Applicable. **Lei** 78 Legal Entity Identifier. **Custom** 68 Custom clearing identifier.
tcrParty.account	Account	Array (AnsiChar)	16	Account number. **Optional**
tcrParty.accountType	AccountType	Enum	1	Type of account associated with the order. **Missing** 1 Account is missing. Account is expected to be filled with 0x00. **Customer** 2 Account is carried on customer side of the books. **House** 3 House trader.
tcrParty.orderCapacity	Capacity	Enum	1	Designates the capacity of the firm placing the order. **Agency** 1 Agency (mapped to AOTC). **Principal** 2 Principal (mapped to DEAL). **RisklessPrincipal** 3 Riskless Principal (mapped to MTCH)

Name	Type	Kind	Length	Description
tcrParty.feeStructureId	u8	Primitive	1	Optional identifier of a fee scheme for billing purposes. **Optional**
tcrParty.interestedParty	InterestedParty	Array (AnsiChar)	8	3rd party interested in this order or trade. **Optional**
tcrParty.memo	Memo	Array (AnsiChar)	18	Free text. **Optional**

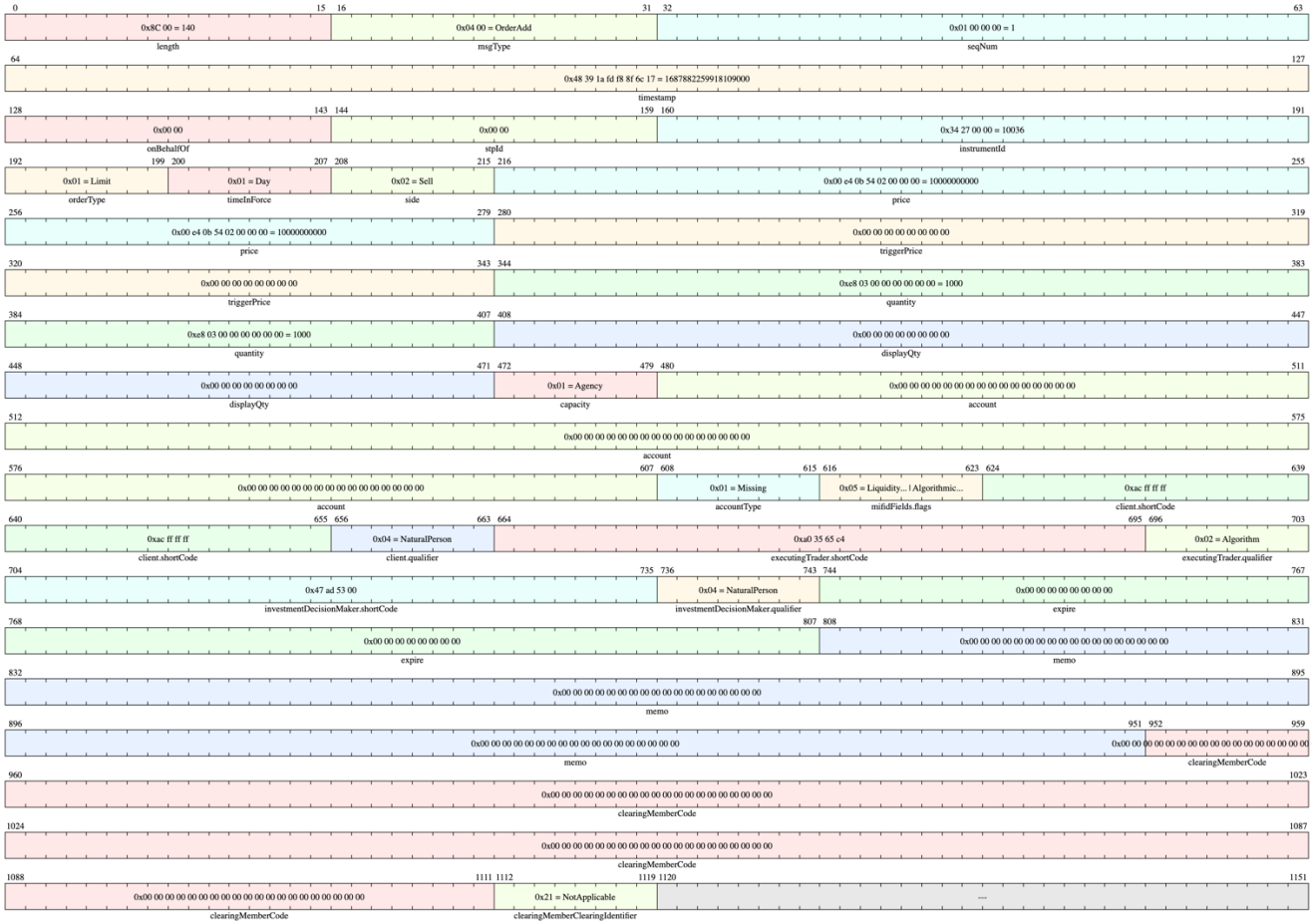
7.29. TradingSessionStatus

Name	Type	Kind	Length	Description
header	Header	Struct	24	Header.
marketId	MicCode	Array (AnsiChar)	4	Market structure's Market Identifier Code (MIC) as specified in ISO 10383.
date	Date	Alias (u32)	4	Date (YYYYMMDD) as integer value. In case of undefined date value of '0' (zero) is used..
TradingSessionEvent	TradingSessionEvent	Alias (u8)	1	Identifies an event related to the status of a trading session. **Name** / **Value** / **Description** **PreviousDayRestateStart** / 1 / Previous day orders restate start. **PreviousDayRestateEnd** / 2 / Previous day orders restate end. **StartOfTradingSessionMIC** / 3 / Start of trading session MIC. **EndOfTradingSessionMIC** / 4 / End of trading session MIC.

7.30. Order Add Example

Below is an example of an OrderAdd message, illustrating sample values for each field along with their binary representation.

Figure 7 OrderAdd message frame



8. Message Kinematics

Scope:

The purpose of this appendix is to present the message flow within the Native communication protocol, as used in GPW WATS. The appendix does not include all available message flows in GPW WATS, but only those selected for presentation purposes.

Remarks:

Due to ensuring transparency of message flow diagrams, the scope of presented fields is limited to basic components and does not include the complete set of all fields that a given message may contain. The scope of the fields for the same messages may also vary depending on the specific example described in the diagram. The complete range of fields in the messages is available in the description of each individual message within the protocol.

Actors:

For the purpose of the presentations, the following actors have been utilized in the context of kinematics:

Native Trading Client Application – Exchange Member authorized to submit buy and sell orders, communicating with System represented by the Native Order Gateway.

Native Order Gateway – a system that communicates with the Native Protocol Client Application.

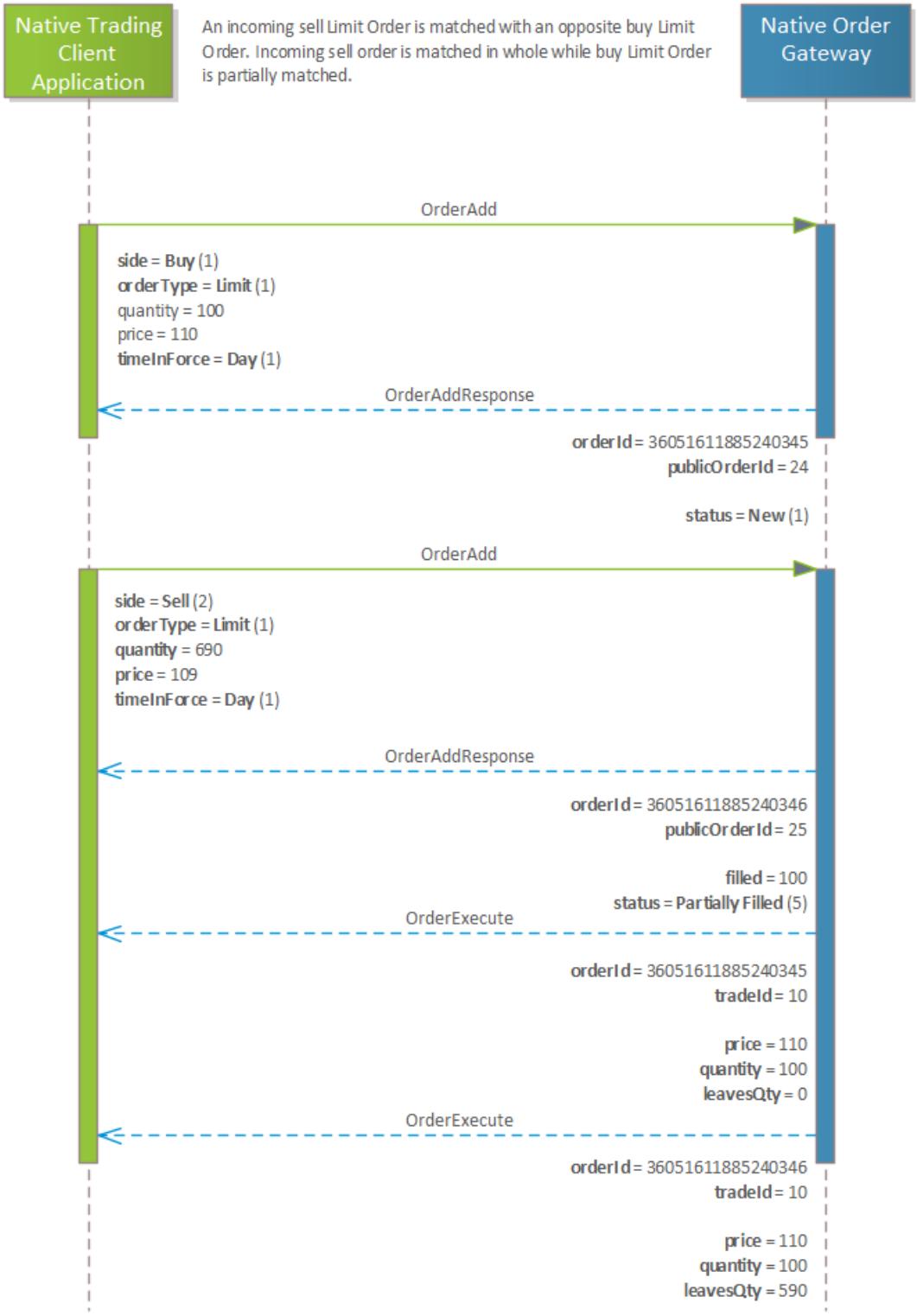
Note: presentation of simultaneous communication between multiple clients and Gateway was omitted. As a result, the Native Trading Client Application can either represent a single Exchange Member or simulate the exchange of messages between two different Exchange Members including buying, selling and conducting transactions.

8.1. CLOB

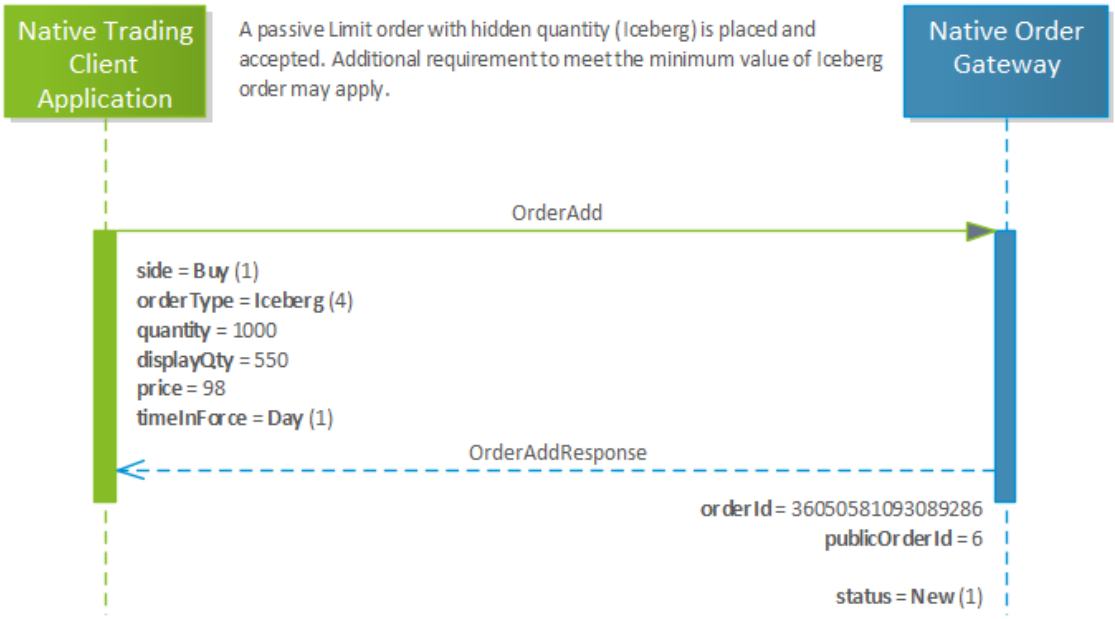
Order - Accepted



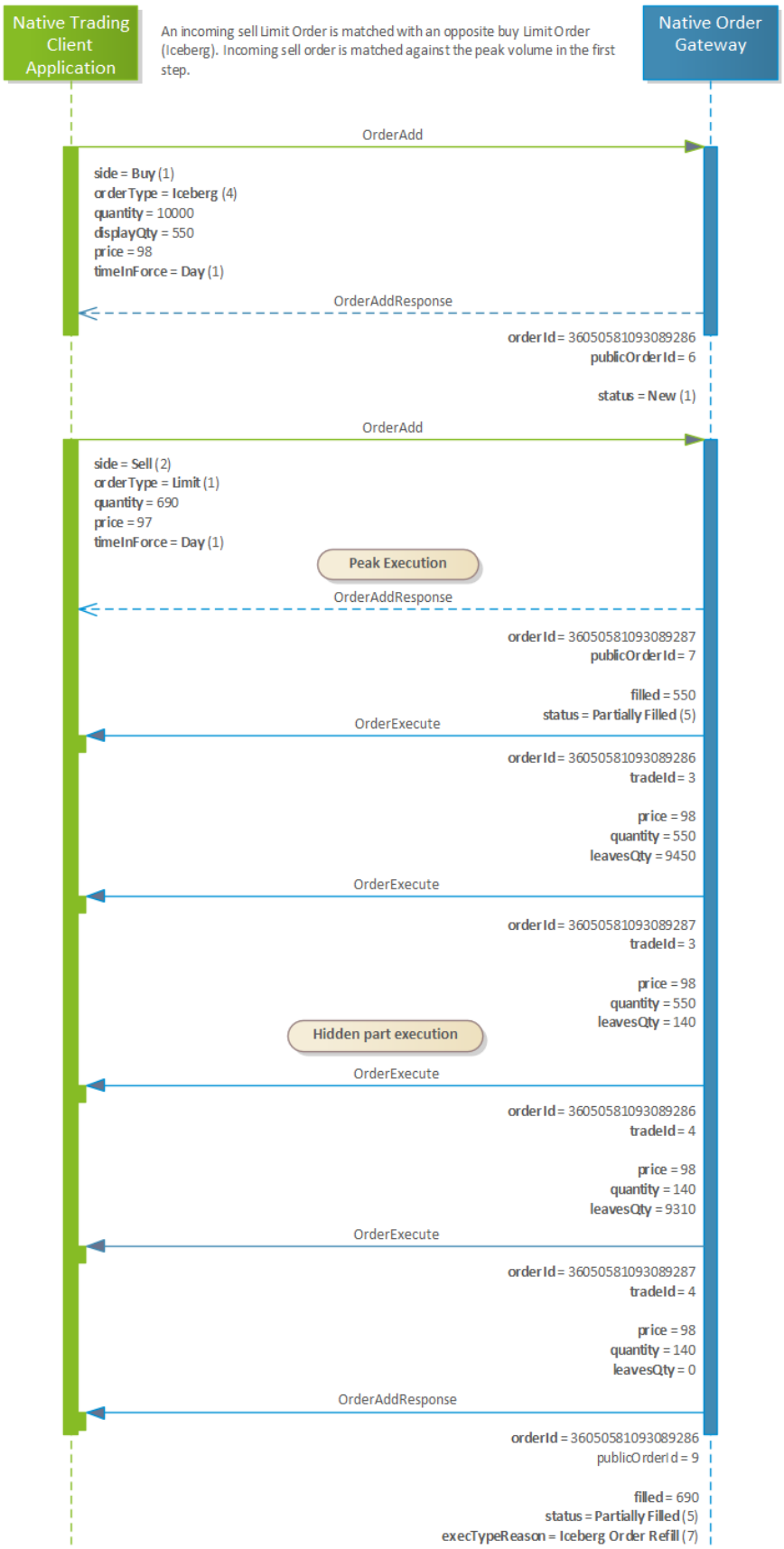
Order – Accepted and Filled – Full and Partially



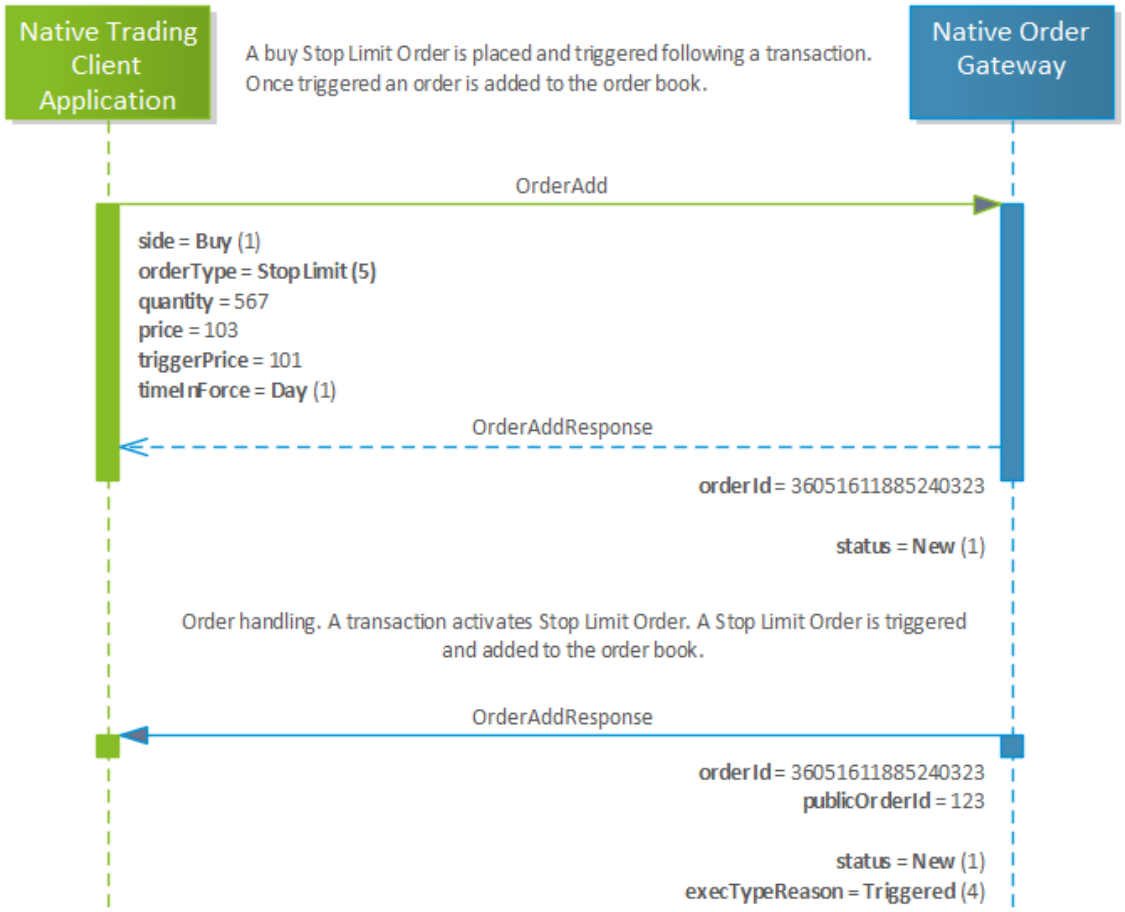
Iceberg Order – Accepted



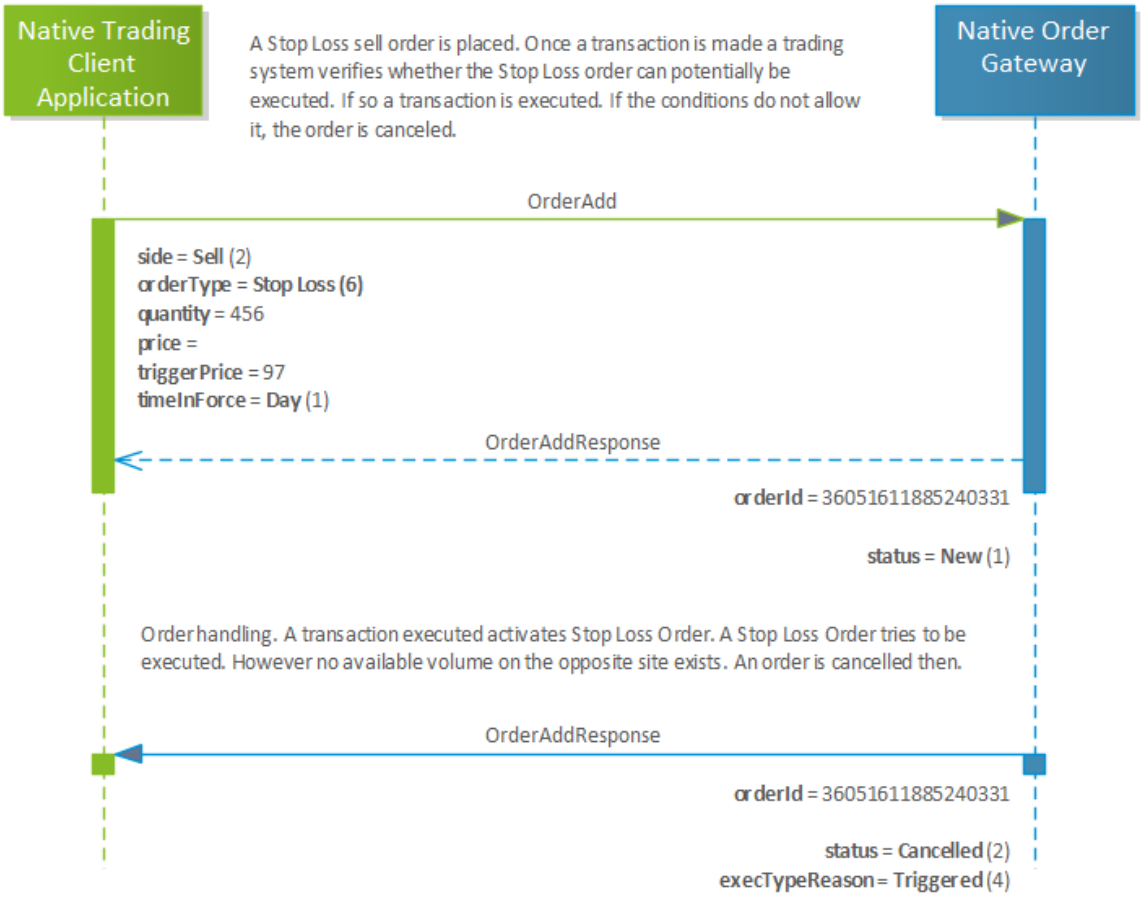
Iceberg Order – Accepted Partially Filled and Restated



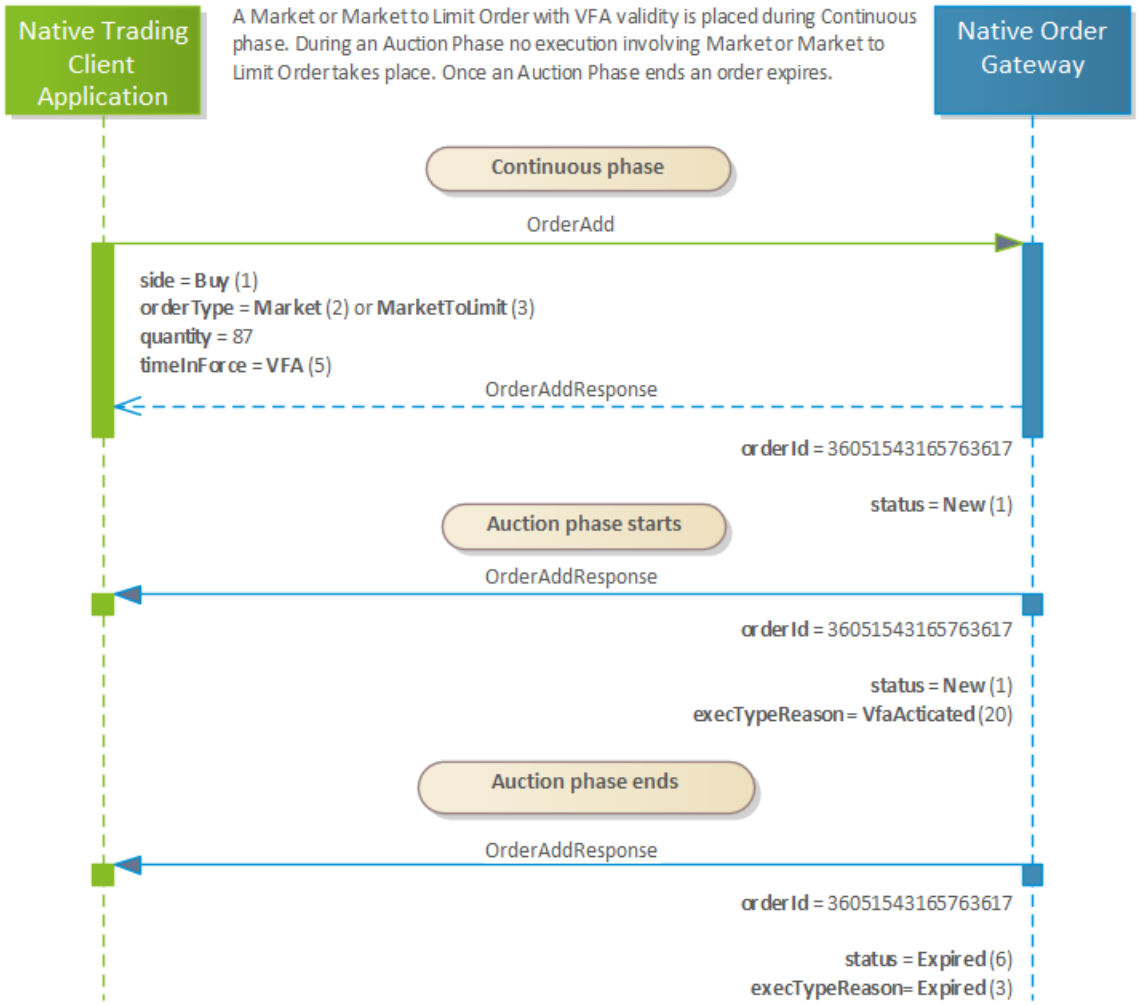
Stop Limit Order – Accepted and Triggered



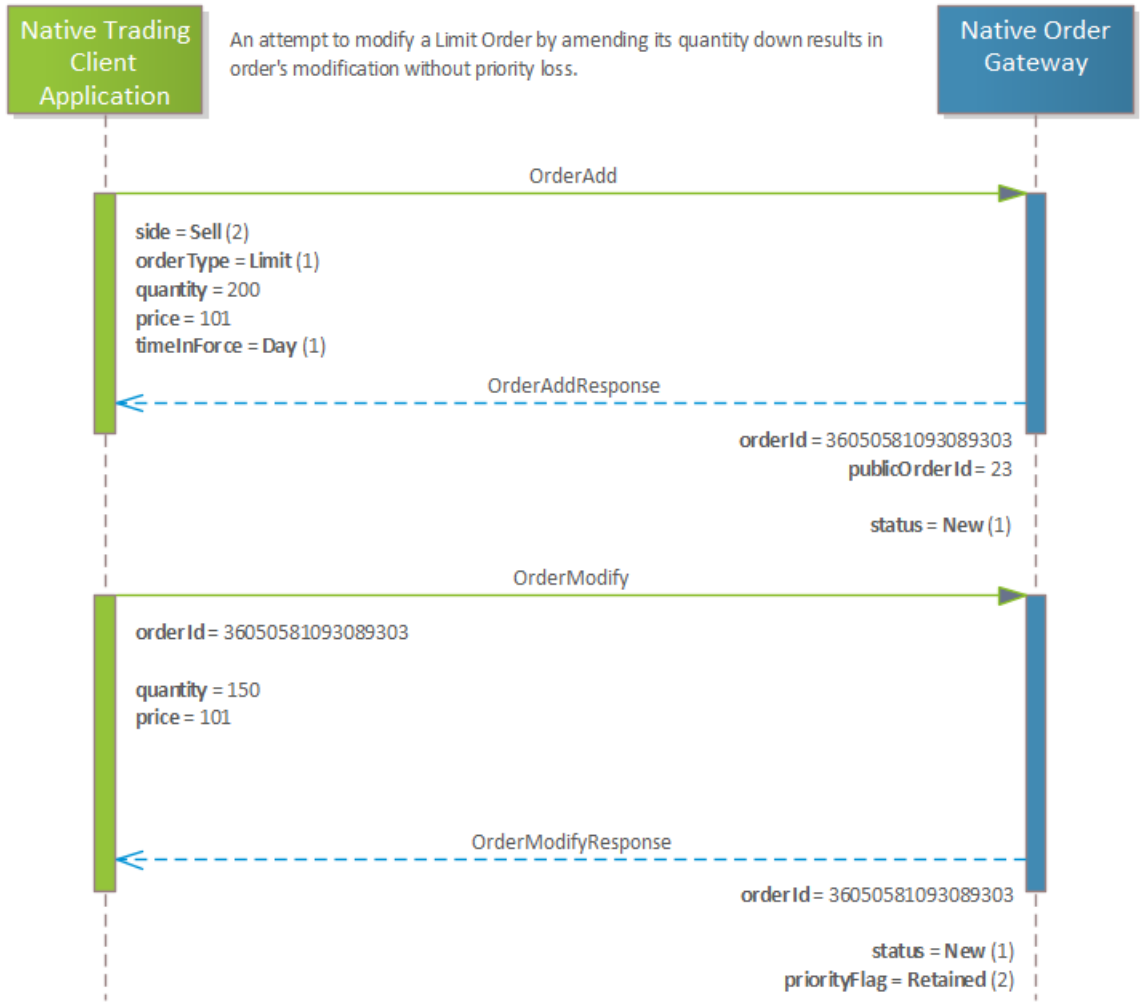
Stop Loss Order – Accepted, Triggered and Cancelled



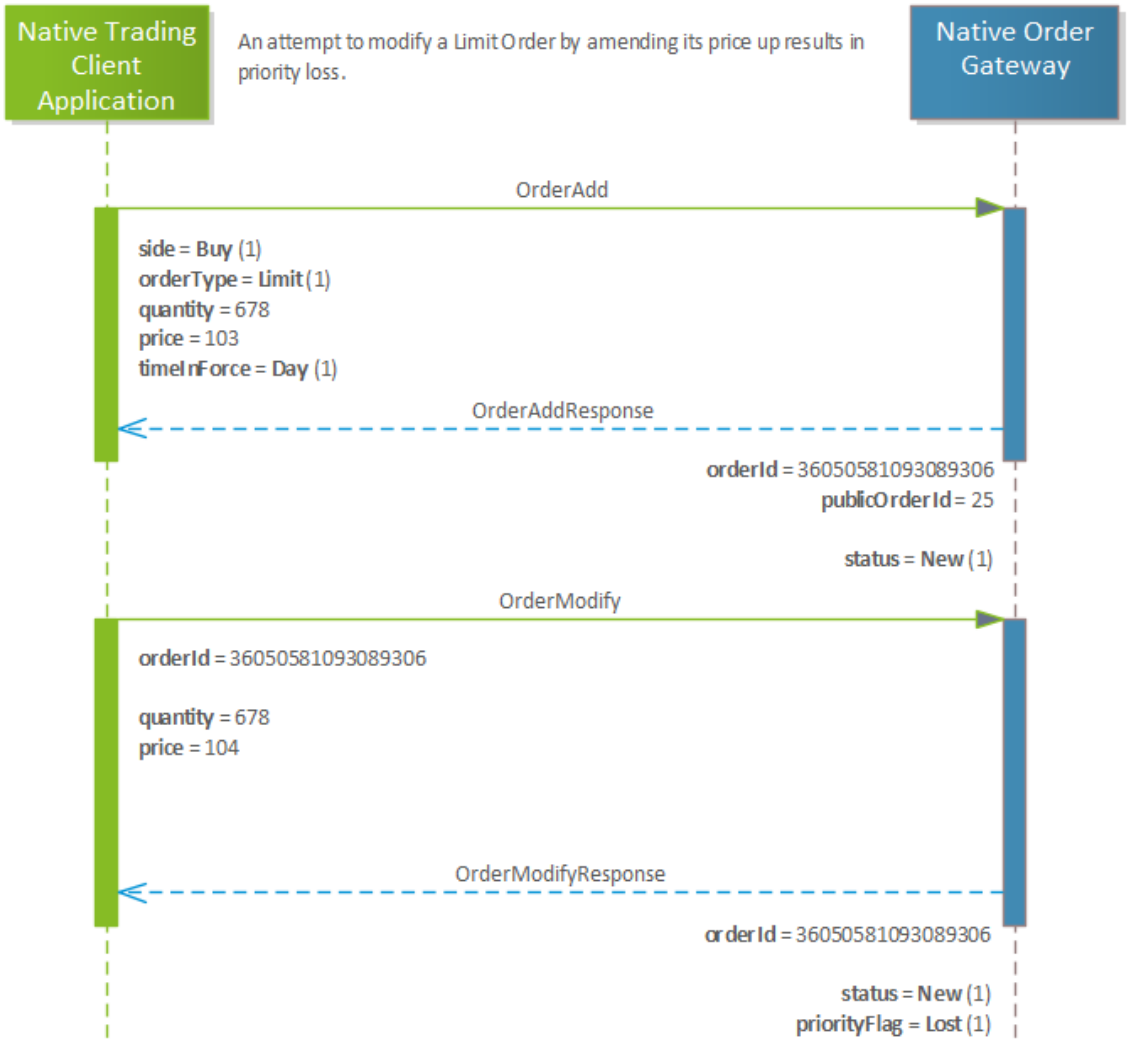
Market Order and Market to Limit Order with TIF VFA – Triggered during Auction



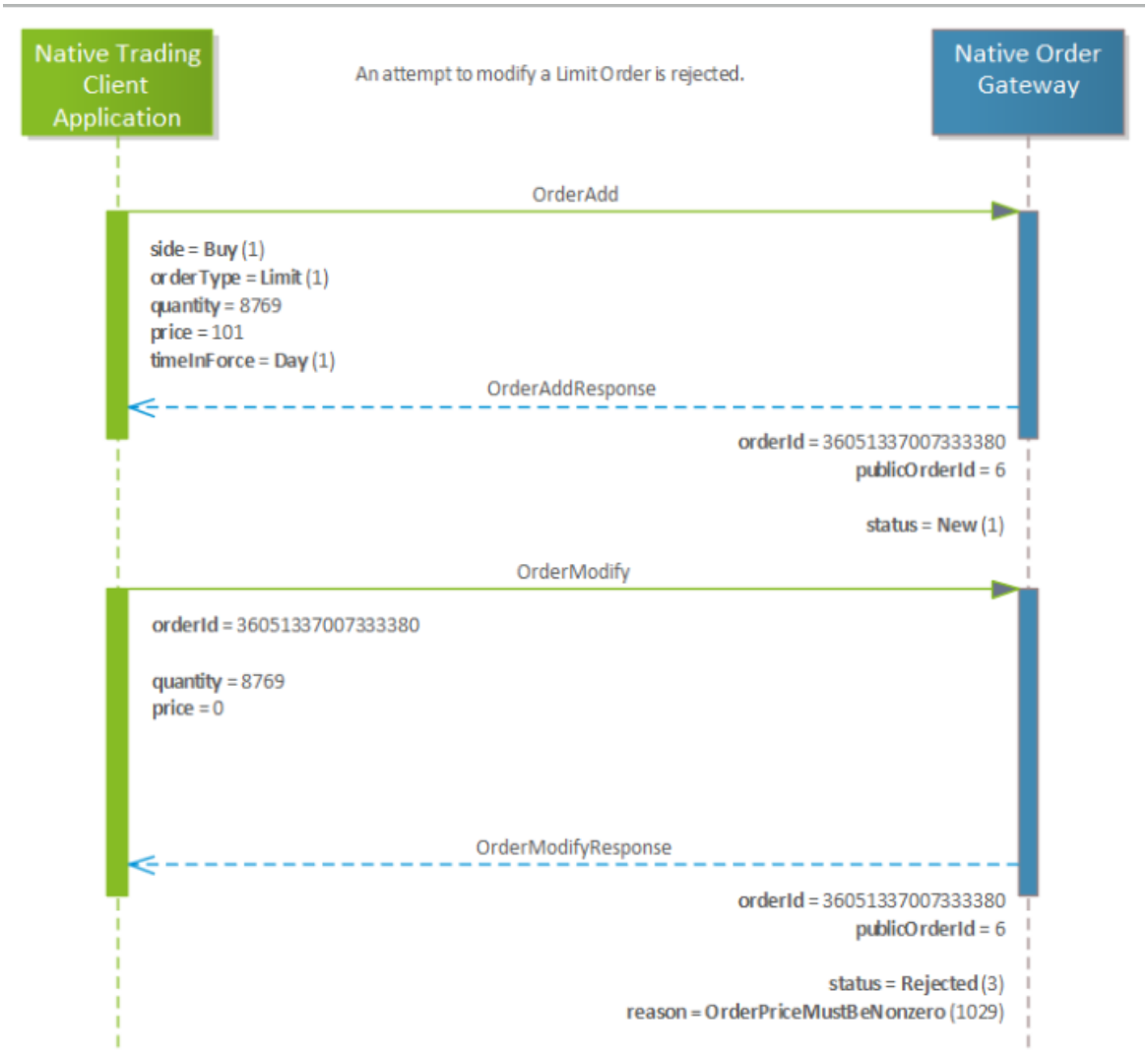
Order Modification without priority loss



Order Modification with priority loss



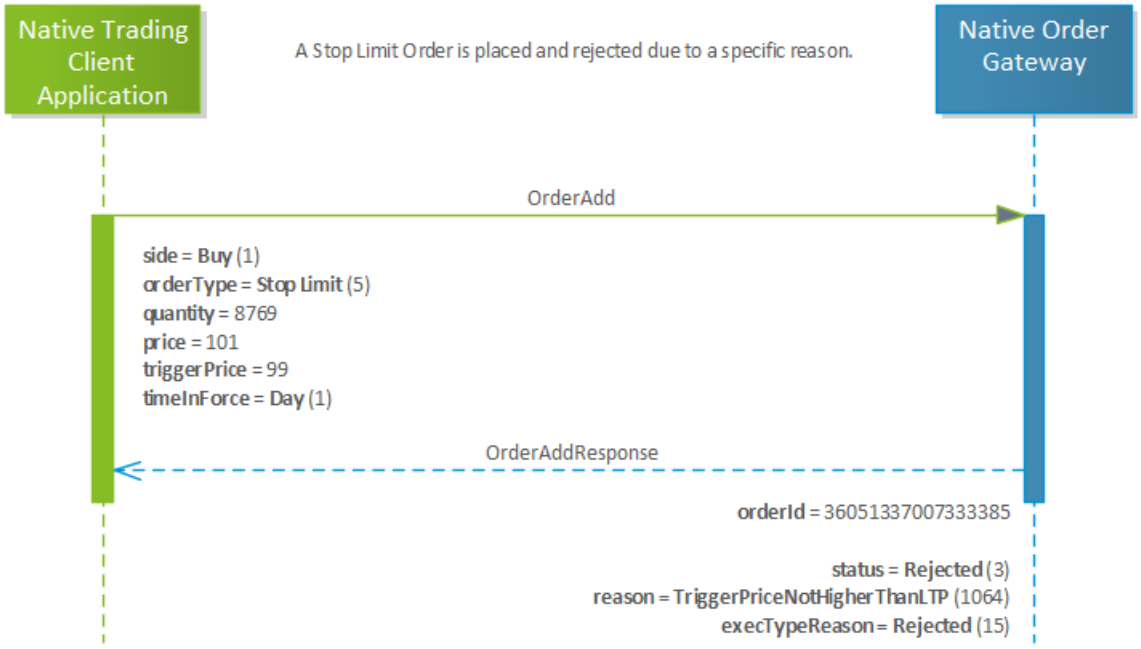
Order Modification – Rejected



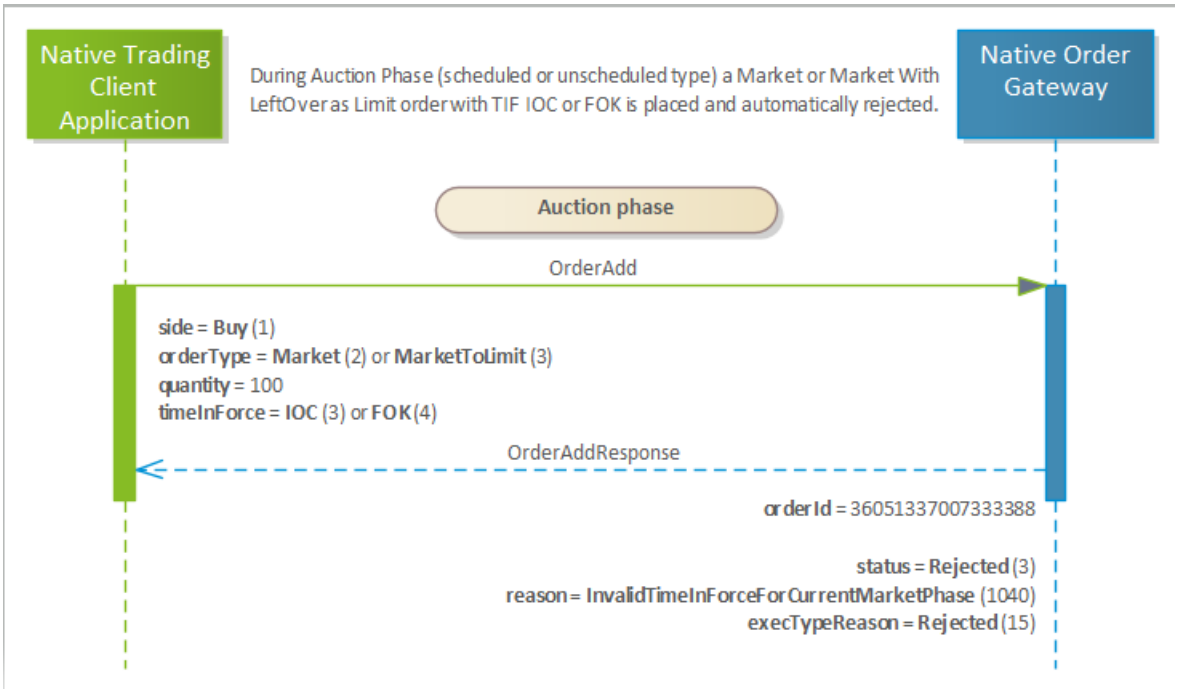
Order – Rejected



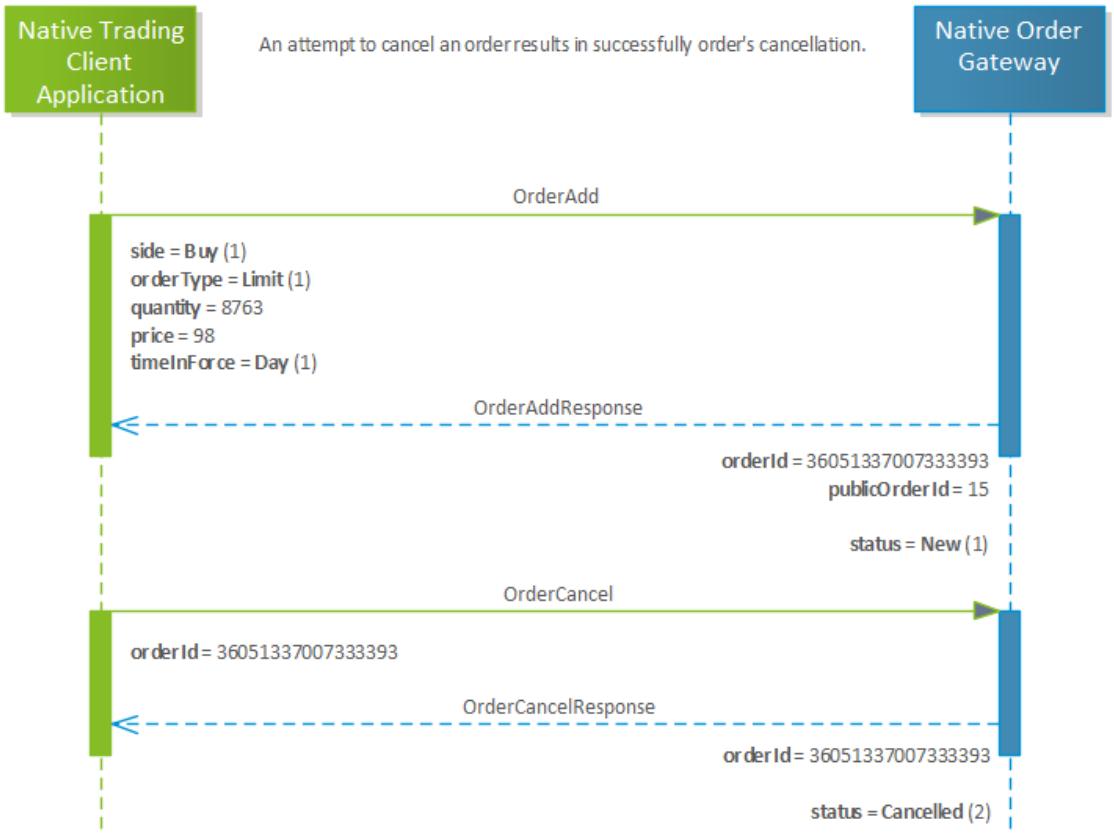
Stop Limit Order – Rejected



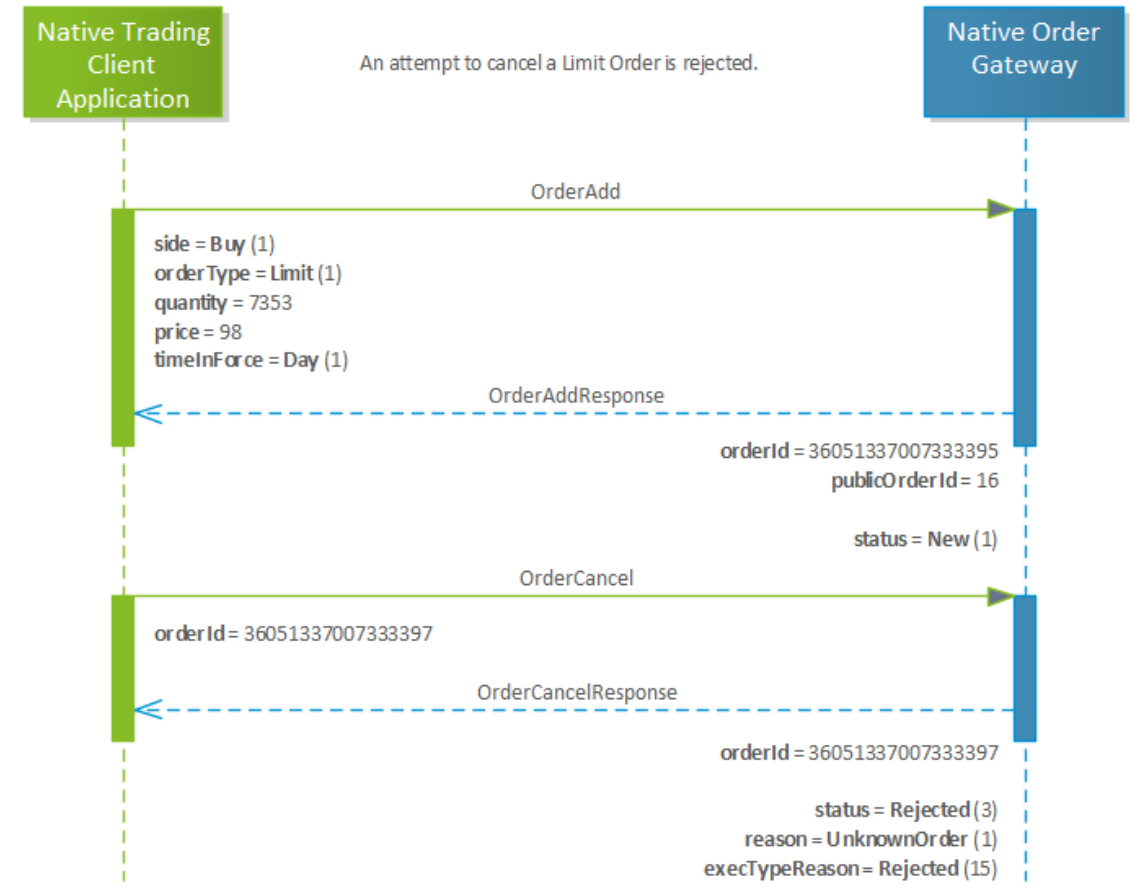
Market Order and Market to Limit Order IOC/FOK – Rejected during Auction



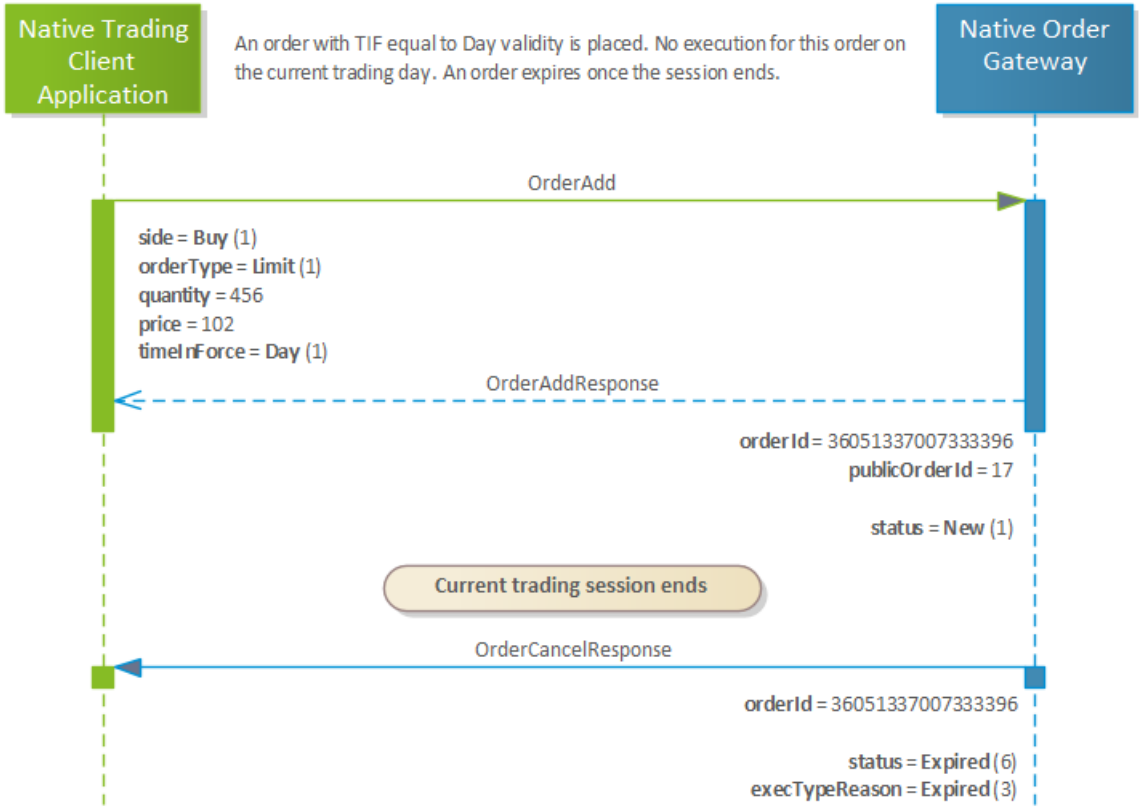
Order Cancellation – Accepted



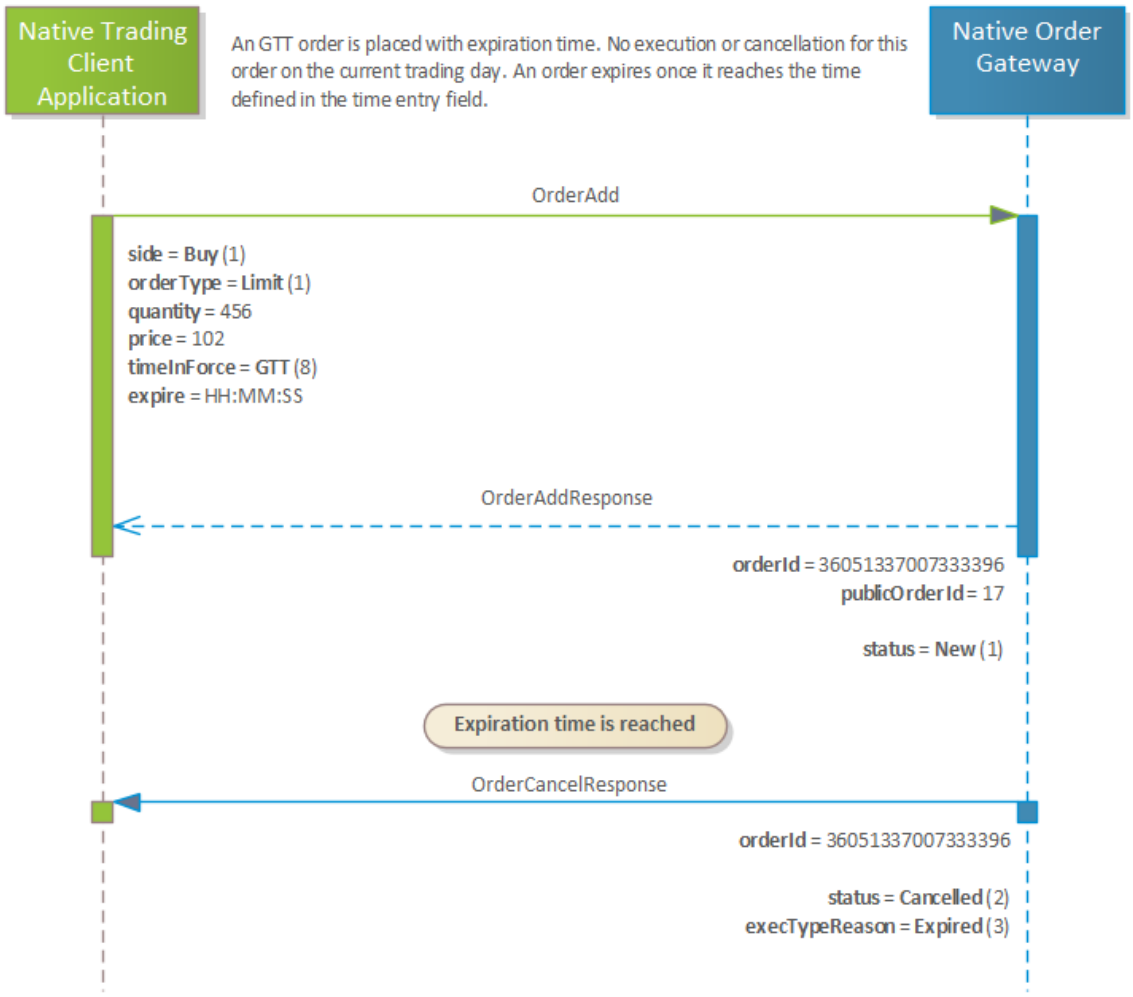
Order Cancellation – Rejected



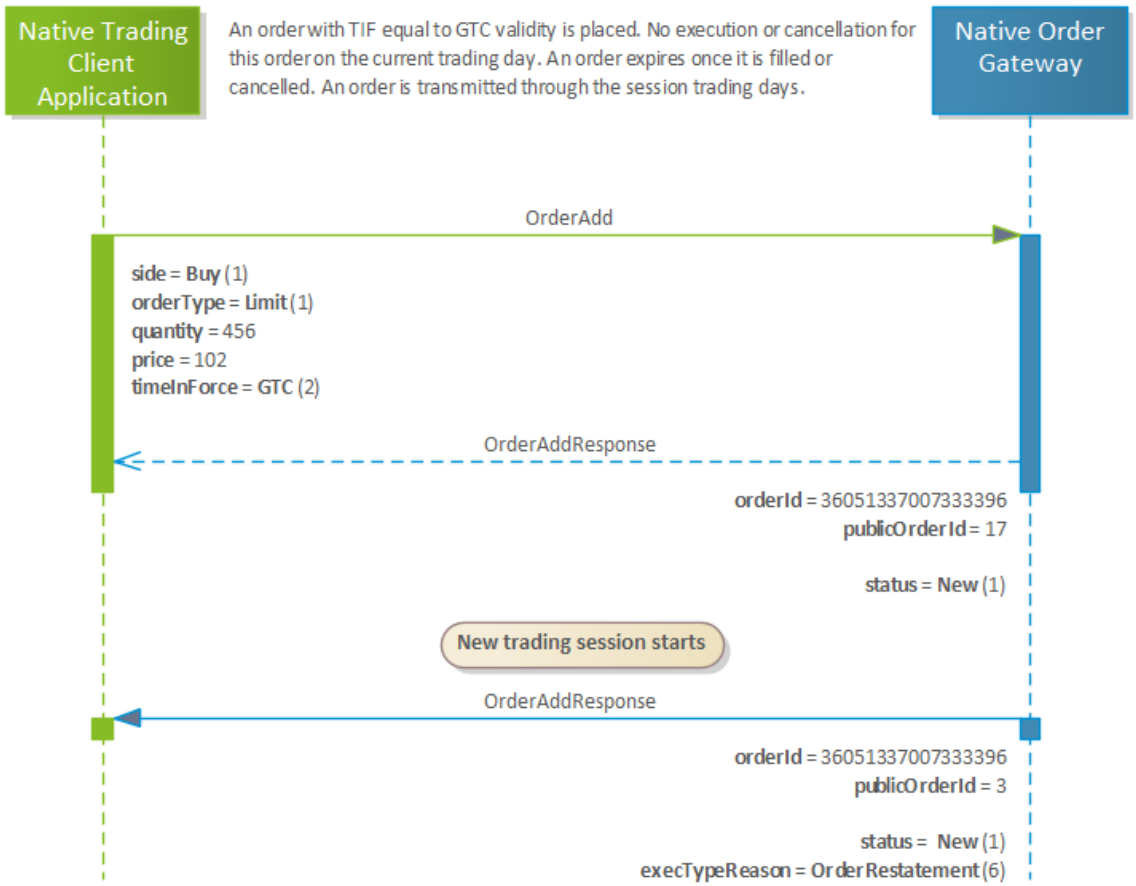
Day Order – Expired



GTT Order – Expired

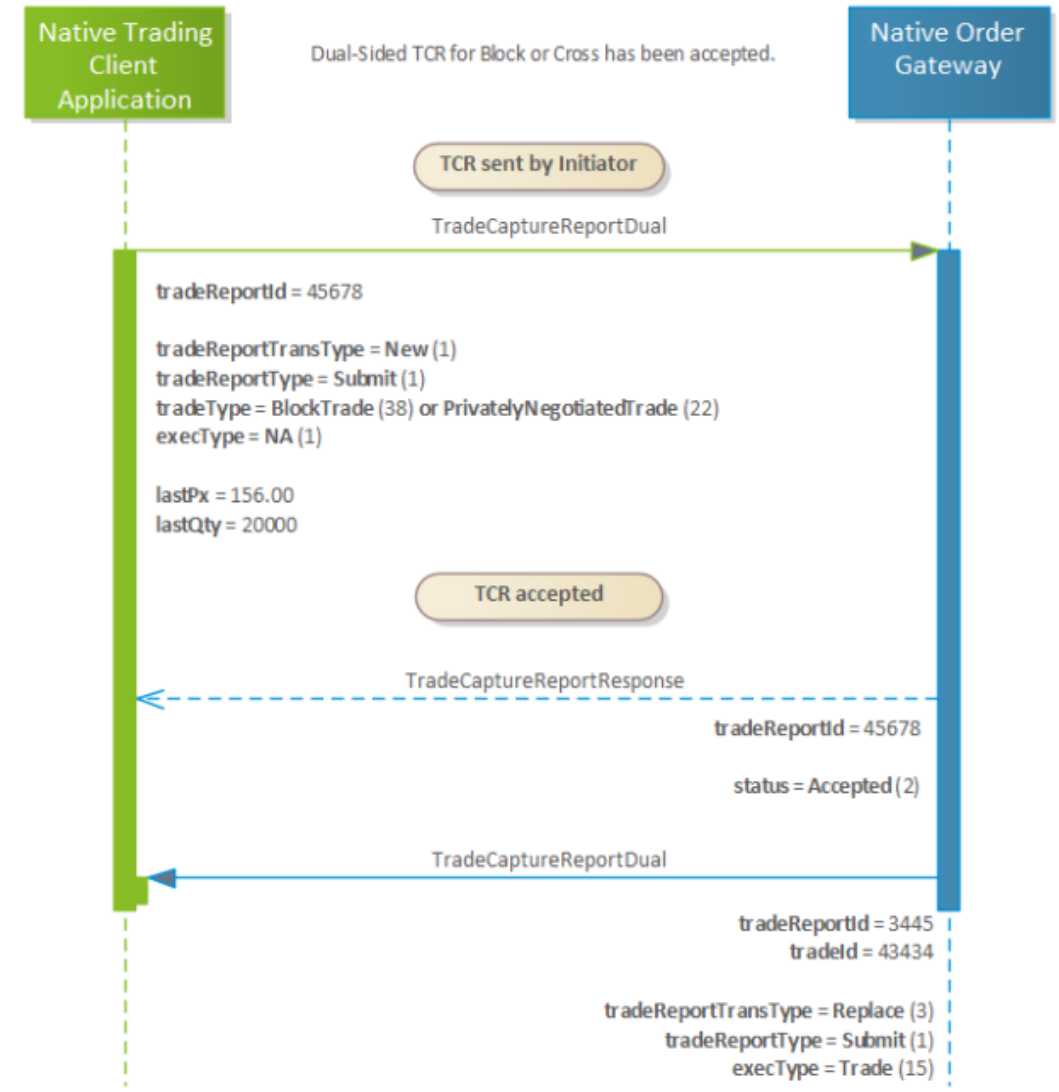


GTC Order – Restated

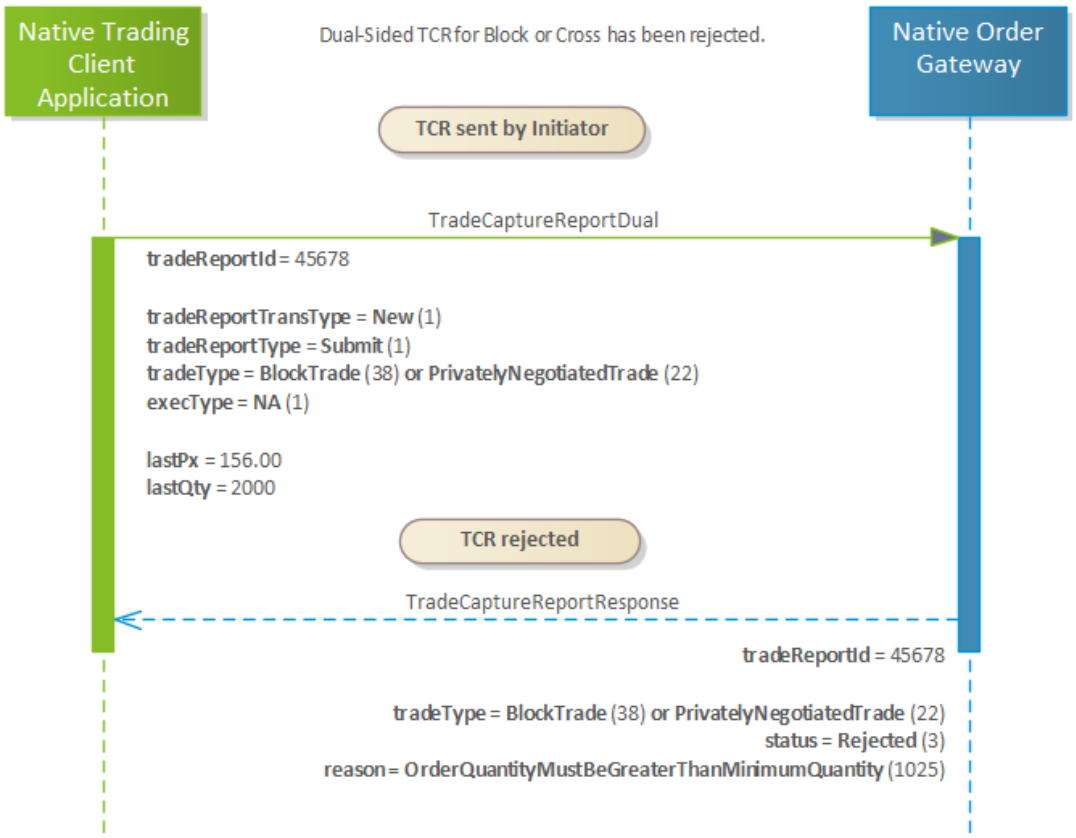


8.2. Off-Book (Block or Cross)

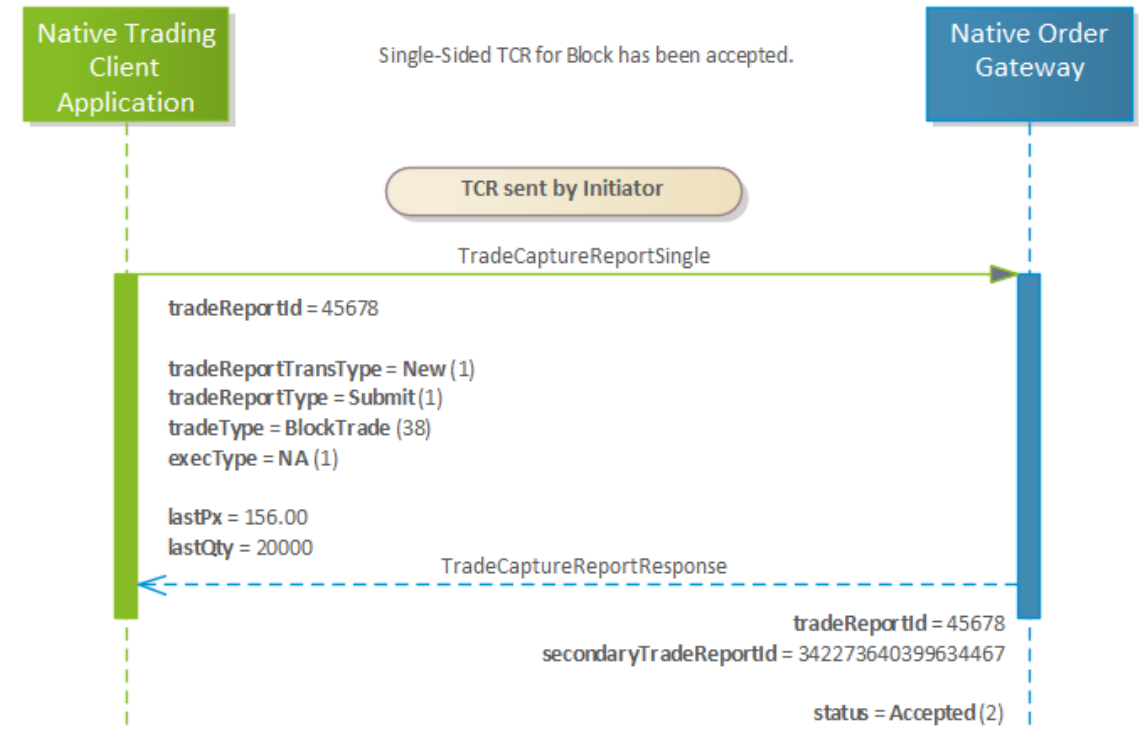
Dual Sided – Accepted



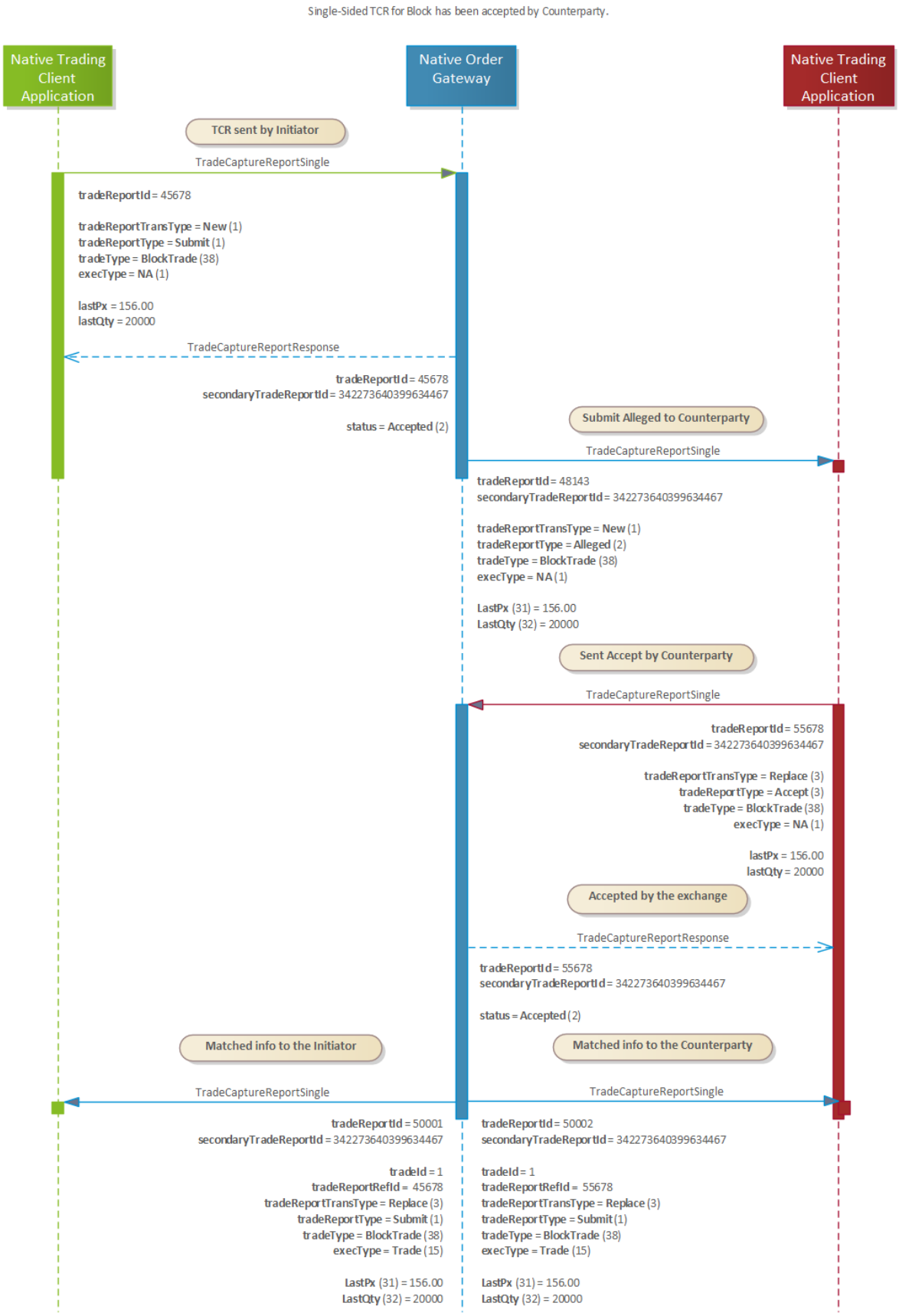
Dual Sided -Rejected



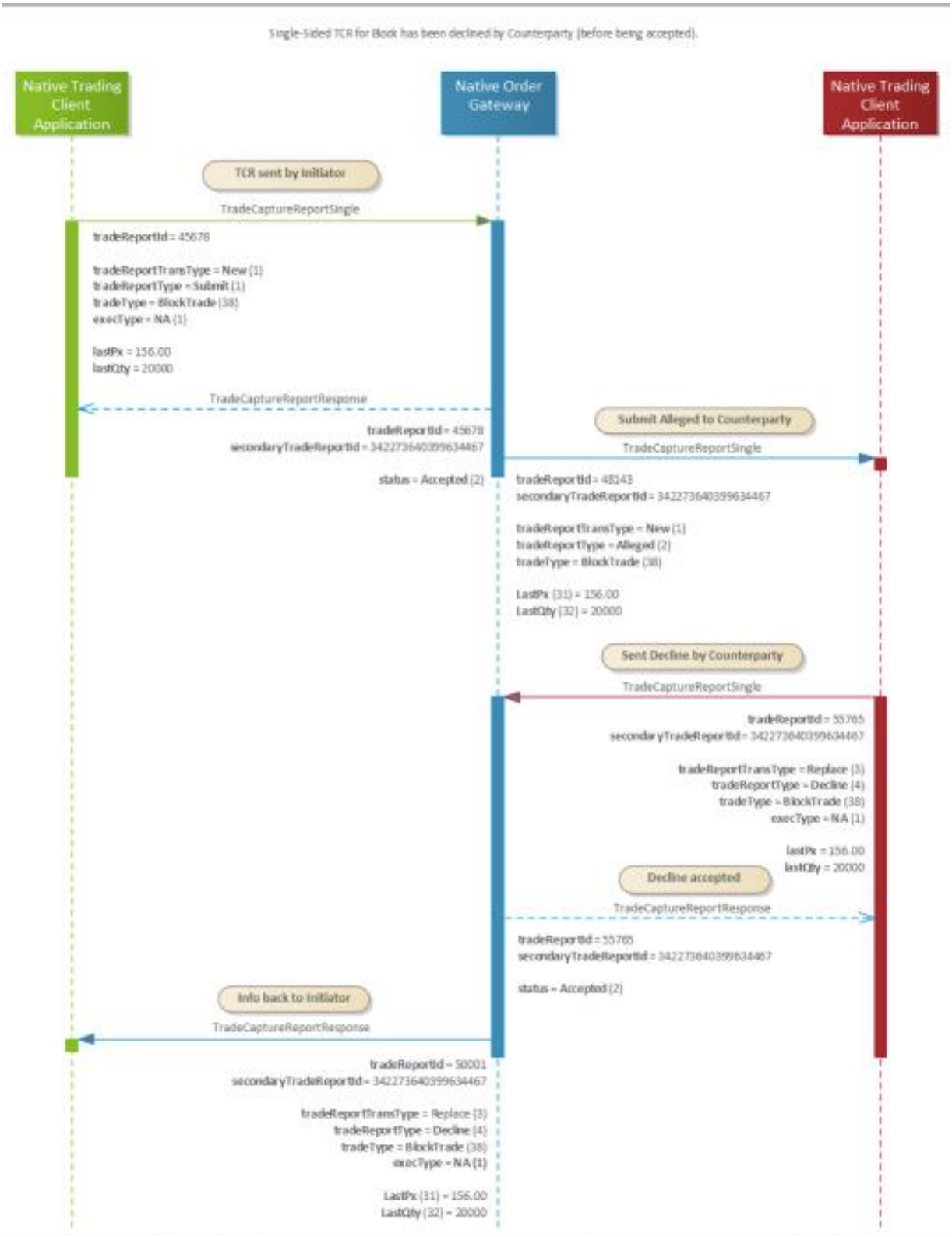
Single Sided – Accepted



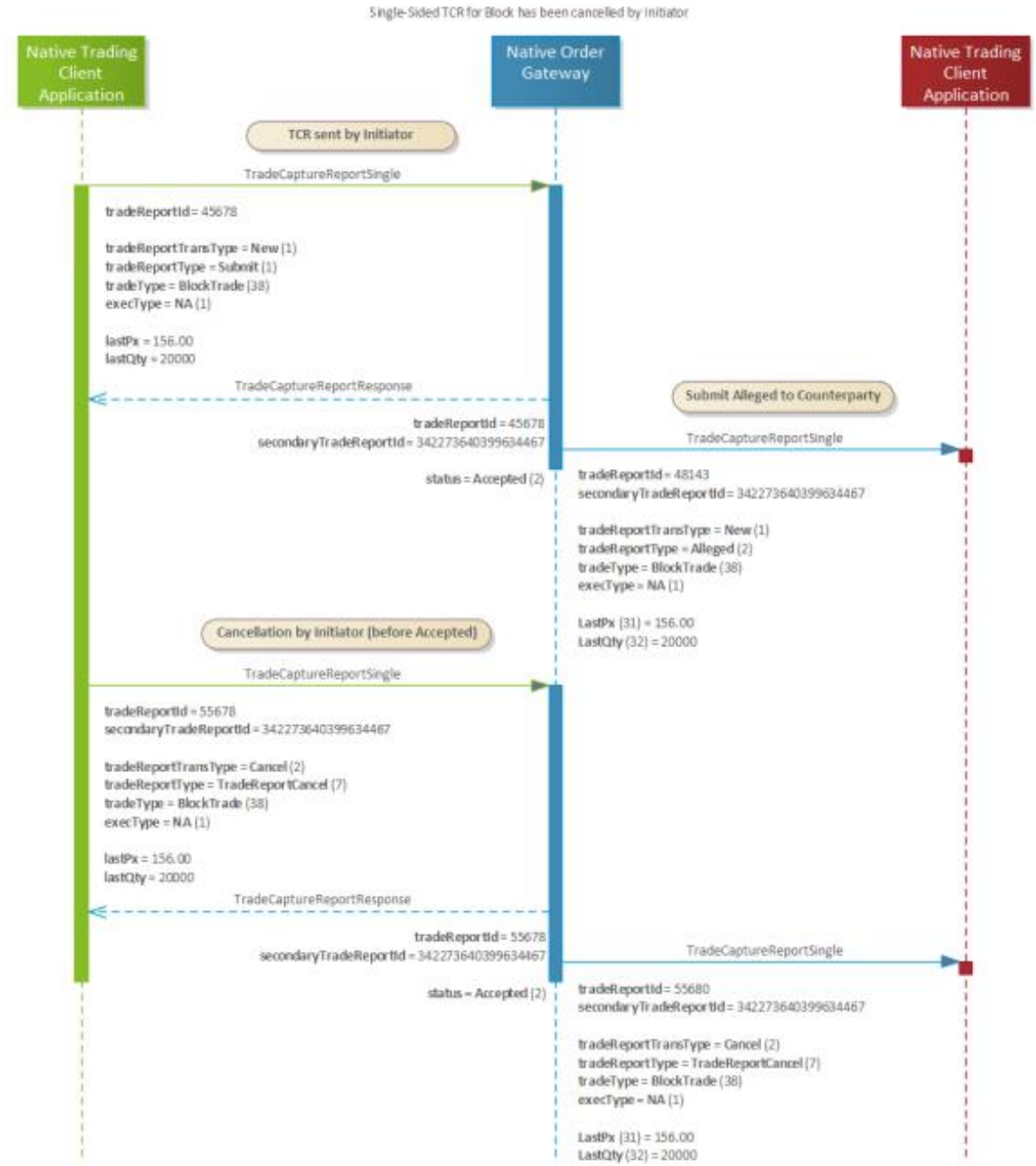
Single Sided – Accepted by Counterparty



Single Sided – Declined by Counterparty



Single Sided – Cancelled by Initiator



Dual Sided – Rejected

